

Beam Angle Selection of Non-Coplanar Fields in Proton Therapy

**Advancing automated treatment planning
techniques of cranial tumours**

by

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Abstract

This study delves into two distinct strategies for addressing the challenge of predicting beam angles of non-coplanar fields in Proton therapy, applied to cranial tumours. The initial approach utilises Convolutional Neural Networks (CNNs), investigated through both supervised and unsupervised learning paradigms. While the former leverages annotated data, the latter relies on a heuristic loss function based on dosimetric prescriptions. These two methods are assessed for their ability to forecast angles for a single field only, but their performances remain unsatisfactory, primarily due to limitations in available patient data. Conversely, the second approach introduces a novel gradient descent variation tailored to angles prediction for three non-coplanar fields. By harnessing the previously mentioned heuristic loss, this approach yields promising outcomes within reasonable computational time, spanning both single and multiple-field dose distributions optimisations. The predicted plans' efficacy is benchmarked against clinical reference plans, albeit without accounting for robustness with respect to patient positioning in this initial implementation.

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1

Introduction

Over 40,000 new cancer cases are diagnosed yearly in Switzerland, and almost 17,000 people die from this disease annually [1]. Cancer has become the second most common cause of death in the country after cardiovascular diseases, responsible for almost a quarter of all deaths. Consequently, providing optimal patient care and pursuing innovative, efficient, and minimally intrusive treatment methodologies gained significant interest in medical research.

The generic term 'cancer' is applied to a large group of diseases that can affect any body segment. They are also referred to as malignant tumours. One of the characteristic features of cancer is the proliferation of abnormal cells that invade (or infiltrate) tissues or organs beyond their usual boundaries or implant themselves in other organs, sometimes far away from the original site (metastatic spread). Four types of cancer are responsible for more than half of the new cases worldwide. Women are mainly affected by breast, lung, or colon cancer, whereas men are more prone to prostate, lung, and intestines tumours. Due to demographic trends (ageing population), the number of new cases and deaths will continue to increase. The Federal Statistical Office (FSO) estimates that within the next 20 years, cancer-related deaths will inflate by about one-third. Simultaneously, the number of 'patients', i.e., patients living with cancer, will also increase significantly. The cancers suffered by children are not the same as those in adults. The most common paediatric cancers are leukemia, brain tumours, lymphomas, and a series of rare tumours arising from immature embryonic tissue, absent in adults. In Switzerland, 350 children and adolescents are diagnosed with cancer every year [1]. In recent decades, the survival chances for children with cancer have significantly improved, and many can recover, despite the high risks of late effects (health problems occurring years after the cancer).

All these improvements have been made possible by continuously improving technologies and techniques, allowing for the treatment of a wide variety of cancer types. The main techniques currently in use are chemotherapy, immunotherapy, surgery (tumour removal), and radiotherapy, either used individually or in combination. For example, a large portion of cancer patients receive radiotherapy as part of their treatment (over half of all cancer patients in the United States [2]). The principle of the latter is to expose tissue, mainly cellular DNA, to "well-designed" high-energy radiations in the form of beams. The nuclear reactions induced, either single or double-strand breaks of the DNA double helix, can result in mutations or the complete functional disruption of the cells, causing apoptosis. This interaction is quantified as "absorbed dose" and expressed in energy (Joules) absorbed per unit mass (kilogram). The unit commonly used instead of J/kg is the Gray (1 Gy = 1 J/kg). The main goal of radiation therapy is to eradicate cancerous tissue while minimising the irradiation of healthy tissues. Unlike chemotherapy, it allows targeting specific parts of the patient's body - a local therapy. How well this goal is achieved describes how "well-designed"

the delivered beams are. Most radiotherapy treatments use photons, produced using linear accelerators, which allow achieving the required radiation specifications for most treatments in an affordable, compact, and relatively inexpensive way. However, the dose deposition might not be ideal for all tumour types and configurations. To address this problem, an alternative particle can be used: protons.

Protons have a finite range of propagation in matter and deposit their energy mainly through electromagnetic (Coulomb) interactions with orbital electrons of atoms, as described by Bethe-Bloch's formalism[3]. The slower the protons become, the more energy they transfer to tissue per track length, causing a maximum dose deposition at a certain depth in the tissue. This results in a peak called the Bragg peak. The depth of the Bragg curve depends on the initial energy of the protons as well as their energy loss via interactions. In practice, several proton beams of different intensities and energies are combined to form a Spread Out Bragg Peak (SOBP), which allows for better target coverage but increases the mean dose at the entry of the tissue. These SOBPs are optimised by tuning their positions and direction of incidence: a technique called Pencil Beam Scanning (PBS). Thanks to these techniques, Proton Therapy (PT) allows for a smaller dose to critical structures, leading to higher doses to the tumour and increased tumour control probability. The current downside of PT is the complex delivery systems, requiring much more advanced particle accelerators and beam lines to achieve the required particle energies. We can cite, for example, COMET, the cyclotron at Paul Scherrer Institut (PSI)'s Center for Proton Therapy (CPT) [4]. PT combines a multitude of technologies and know-how to achieve a complete and successful treatment. Such treatment follows these steps:

- *Imaging of the patient*: Diagnostic and identification of the need for proton therapy;
- *Target and organs delineation*: Tracing the contours of the volumes of interest, namely organs at risk but also the different target volumes.
- *Treatment planning*: Setting and optimising all the characteristics of a treatment most adapted to the patient's pathology;
- *Delivery*: Execution of the treatment planning with the delivery facility. This phase takes into account the evolution of the patient over the course of the treatment (re-planning might be required);
- *Post-treatment follow-up*: Periodic examinations to assess the success of the treatment or to detect potential regrowth or the appearance of second primary cancers .

This work focuses solely on the treatment planning part of the procedure. Treatment planning is the process through which a proton (or radiotherapy) device is programmed to deliver a specific amount of protons (or radiation) to a patient. This process utilises adapted mathematical and physical formalisms to optimise the balance between delivering a high and conformal dose to the target while limiting the doses to critical structures. The therapeutic ratio, which compares the probabilities of tumour eradication and normal tissue complications, is often used to assess this balance. Currently, the "Planning" step involves computer assistance but remains ultimately human-driven, resulting in a complex set of instructions for a delivery facility. Placed at the end of the proton beamline, a typical delivery machine consists of a rotating couch where the patient lies, and a gantry, an apparatus that delivers the proton beams and can rotate around the couch.

Adjusting the variables that govern treatment planning can take hours to several days, particularly for complex cases. While recognising the expertise of proficient clinicians, there is an opportunity to enhance efficiency without compromising treatment quality by incorporating computer-aided predictive modeling of planning variables. The objective of this work is to contribute to reducing the time needed for a solution and assist clinicians in obtaining an initial high-quality plan estimate through automated means. Among potential pathways to automation, this work focuses on the automatic selection of the incidences of the beams delivered to the patient.

The beams outlining the treatment plan, more commonly denoted as **fields**¹, exhibit a range of attributes. These encompass the field centre (position of the beam's iso-centre, i.e. the target point of the field), the positions and energies of PBS spots (with or without absorbers), and the orientations of the gantry and couch. The latter components dictate the angles at which proton beams are directed towards the patient. CPT maintains multiple delivery systems: Gantry 1 (G1), reserved exclusively for research purposes, Gantry 2 (G2), and Gantry 3 (G3). Although G2 and G3 are employed for treating patients, this study solely focuses on clinical plans executed utilising G2.

The research community has already put forth a range of methodologies for beam angles prediction and selection, spanning both photon and proton radiotherapy domains. An overview of existing methodologies is presented in the forthcoming chapter 2. Within this study, two novel and distinct approaches are considered: a convolutional neural network-based Deep Learning (DL) technique and a modified gradient descent method.

¹This terminology finds its roots in photon radiation therapy where X-Rays (i.e., electromagnetic fields) are delivered to the patient. This convention will consistently be adhered to within this document.

2

Background

Before diving into the work conducted towards the automated selection of beam angles, we provide additional insights into proton therapy, precisely Intensity Modulated Proton Therapy. Additionally, a concise literature review of existing solutions is presented, paving the way for the exposition of the first technique exploited for our goal: Convolutional Neural Networks (CNNs).

2.1. Intensity Modulated Proton Therapy

Intensity-modulated proton therapy (IMPT) was initially introduced in the late 1990s and has since emerged as the prevailing paradigm in proton therapy [5]. IMPT involves the meticulous arrangement of multiple proton beams, each characterised by finely tuned angles of incidence. These beams consist of an array of narrow pencil beams, or 'beamlets', following a sequence of distinct energies (recall SOBPs). Advanced magnetic scanning technology is utilised to deliver these beamlets. Beamlets are also commonly called "spots". The optimisation of spots' intensities across all beams is achieved concurrently through intricate mathematical algorithms. This approach aims to generate dose distributions that strike an optimal balance between target coverage and safeguarding of critical organs at risk (OARs). Notably, IMPT benefits from an additional degree of freedom : the range of the Bragg peak, tunable with the protons' energy. This attribute's main advantage is the reduction of the mid-to-low dose levels in comparison to photons used in Intensity Modulated Radiation Therapy (IMRT). Besides, IMPT showcases improved post-treatment quality of life for certain patients (observed for pediatric tumours [6][7][8]).

In current IMPT planning practices, beam count and angles are determined subjectively, drawing upon prior empirical experience and practical constraints (such as treatment delivery efficiency and quality assurance considerations). The number of IMPT beams typically ranges from 2 to 4, rarely exceeding this range. In specific disease contexts, non-coplanar beams might also find application; beams are "coplanar" when all incidences lie within the same plane relative to the patient, often a transverse axial plane. The objective is to amalgamate the distributions delivered by each field to achieve a uniform dose distribution within the target volume while meticulously sparing normal tissues.

Various incarnations of IMPT exist. First, the single-field paradigm, where each individual delivered field's spots are uniformly distributed and their intensities optimised. This is Single Field Optimisation (SFO), also known as Single Field Uniform Dose (SFUD). This approach is straightforward and robust, yet might not yield the most optimal dose conformation to the target. The three-dimensional variant accounts for all delivered fields at a time. Spots are again distributed uniformly within the target volume, but intensities are simultaneously optimised for all fields, en-

suring optimal target coverage. This is Multiple Fields Optimisation, or Multiple Field Uniform Dose (MFUD).

MFO is particularly sensitive to uncertainties, as highly heterogeneous dose distributions of each constituent beam combine to produce the desired uniform dose distribution in the target volume while maximally sparing normal tissues. In the face of uncertainties, this fit may be lost, meaning that the dose distributions delivered to both the target and normal tissues may differ from the planning. To address this problem, in the current practice of proton therapy, the directions most sensitive to anatomic changes or patient positioning uncertainties are avoided. In the case of head tumours, this translates into the avoidance of the oral and nasal cavities, as well as trajectory crossing complex bony structures. Such practice may lead to directions that might be mathematically (and sometimes clinically) sub-optimal. Thus, additional techniques have been developed for robust optimisation of IMPT plans and for the robustness evaluation of proton dose distributions.[9]

Selection of adequate and optimal beam angles is no trivial task to human planners and requires the utmost rigour to craft reliable automated selection pipelines.

2.2. Current Beam Angles Selection Solutions

Despite being a relatively new topic in proton therapy, multiple methods already exist for automatic beam angle selection in PT treatment planning. We present three of them.

Class selection [10]

"Class solution (CS) beam configuration selection" allows the planner to identify predefined beam configurations based on target localisation and on a wish list for the planning optimisations. This wish list is the set of constraints and prioritised objectives, specific to treatment site and protocol. The algorithm then explores all possible beam geometries based on that list. This approach is supervised, with an average time to an acceptable plan of 22 minutes (range 9–29 min.).

Supervised CNN [11] [12]

With this approach, the beam angle selection is treated as a supervised CNN classification task. Only coplanar proton plans for liver tumour cases are considered. The whole reachable angle range $[0, 360^\circ]$ is partitioned into 360 classes (one class per degree increment), and a score is computed for each class. This method requires only geometric information and does not perform fluence optimisation during the selection process (inherent to NN predictions), and a plan with clinically acceptable quality is produced within five minutes. The training time of the neural network was approximately 120 hours, the prediction a matter of seconds. It is unknown how large the training set was (not mentioned).

Bayesian two-steps optimisation [13]

This algorithm can, in principle, handle as many coplanar and non-coplanar beams as desired (more beams, more arduous), but the optimisation is performed for two coplanar, two non-coplanar, and three coplanar beams. The algorithm selects beams out of 36 equally spaced coplanar beams (steps of 10°). For the non-coplanar, the algorithm can choose between 3 couch angles only. ECHO, a fluence map optimiser, is run for some initial beam angle candidates according to hierarchical dose constraints. The resulting treatment plan for each beam configuration is rated using a clinically relevant treatment score function. A Bayesian optimisation process then iteratively predicts the treatment score for not-yet-evaluated candidates to find the best

candidate to be optimised next with ECHO. A single ECHO optimisation for a given beam configuration takes 2–36 minutes (median times for five patients), and acceptable plans are obtained in an average of 23 iterations.

Solutions developed in Intensity Modulated Radiotherapy

In the photon radiotherapy community, more research has been conducted and other approaches developed. Here are some examples: heuristic score optimisation (based on dose objectives) [14], Guided Pattern Search [15], Reinforcement learning approach with Monte Carlo search tree [16].

In this work, we propose and study the capabilities of two novel approaches for beam angle selection, partly inspired by previously described studies. Notably, we make use of CNNs and of a heuristic loss based on dosimetric prescriptions from human planners. Further details on CNNs and such loss are provided in the following sections.

2.3. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) have demonstrated remarkable capabilities in domains like image classification, object detection, and image segmentation [17] [18] [19]—domains that heavily rely on geometry analysis. Considering that our approach exclusively employs patient geometries as inputs (chapter 3), CNNs present a fitting avenue for addressing our problem. This section offers a succinct overview of the foundational principles and techniques that underpin CNNs [20] [21].

2.3.1. Definition

CNNs are constructed based on neuron models that emulate the functionality of human brain neurons. This behaviour is achieved by assigning activation functions to each neuron, introducing non-linearities into these models. This mechanism facilitates the learning of intricate decision boundaries.

CNNs differentiate themselves from other models through a novel foundational unit: convolutional operations. Drawing inspiration from the receptive fields of visual neurons, these operations involve learnable filters that convolve with input data, extracting local features while maintaining spatial relationships. By employing shared weights and local connectivity (ensuring that each neuron is linked to a limited input region), CNNs efficiently capture hierarchical representations. Activation layers apply activation functions element-wise to the outcomes of convolutional layers, introducing non-linearities and bolstering the network's expressive capabilities. Convolutions and activation layers are often complemented by pooling layers, which assume a pivotal role in downsampling feature maps, diminishing computational complexity, and introducing translational invariance. The sequence of these layers forms an **Encoder**.

Sequential to the encoder is the **Predictor**, comprising one or more fully connected layers. A fully connected layer embodies the fundamental element of Perceptrons. Within this layer type, each neuron in one layer connects to every neuron in the subsequent layer, facilitating high-level pattern recognition.

2.3.2. Training of CNNs

Multiple components contribute to train a CNN effectively. Backpropagation is a foundational operation for all neural networks and plays an essential role in updating CNNs. This process involves propagating loss function gradients backward through the network to adjust the weights of the neurons. Gradient descent techniques optimise network parameters by iteratively fine-tuning them in the direction of the steepest decline. These techniques, often referred to as Optimisers,

are also chosen for better performances in the training. Largely popular optimisers are Gradient Descent, Stochastic Gradient Descent and Adam [22] and its variants (NAdam, AdaDelta, etc...).

After establishing the model and selecting the optimisation technique, input data must be handled. Data preprocessing procedures, like normalisation and resizing, ensure that input data conforms to a suitable standard format for the network. Additionally, data augmentation techniques, encompassing operations such as rotation, scaling, and flipping, enhance the diversity and resilience of the training data. It is imperative, however, to exercise caution and avoid introducing unintended biases in the dataset through augmentation.

With the data prepared, the next step involves hyperparameter tuning. Hyperparameters are values that characterise the model and its training process. Key hyperparameters include the learning rate governing weight optimisation, the batch size for input data, regularisation strength, and more, sometimes even comprising network depth and width. Accurate configuration of these values significantly influences CNN performance, rendering precise tuning vital for attaining optimal outcomes.

Following hyperparameter tuning, another prevalent practice is implementing regularisation to curb model overfitting. Regularisation imposes constraints on network weights to encourage simpler and more generalisable models. Various forms of regularisation exist, with the most common including dropout, L_1 or L_2 regularisation, all used in this study.

IMPT and CNNs having been defined and the task of beam angles selection presented. We shall transition to the setting of the study and the different methods implemented.

3

Methods

As mentioned in the preceding chapter, the initial strategy for addressing the beam angles prediction challenge involves the construction and training of a Convolutional Neural Network (CNN). Prior to creating this network, we introduce the dataset available for analysis, present the formatting of neural network inputs, and detail the diverse strategies implemented to enhance outcomes. Additionally, we will provide an overview of the hardware utilised and the software developed for conducting this experiment. It is worth mentioning that, before transferring to CNNs, a supervised Multiple Layers Perceptron was constructed for beam angles selection and trained using hand-crafted features. The results of this preliminary study are available in appendix A.

3.1. Dataset

3.1.1. General Information

The dataset used in this study comprises 94 patients, segregated into two distinct sets: the training set composed of the first 75 patients, while the test set comprises the remaining 19 patients. To each patient in each set corresponds a unique identifier, denoted ID.

All patients are diagnosed with intracranial and extracranial cancers, exclusively involving tumours within the head region. These patients exhibit varied and complicated cases judged untreatable with conventional radiotherapy, making the dataset non-homogeneous and increasing the task's complexity. These patients underwent successful treatment between 2013 and 2020 at G2 within CPT. Details of test patients' diseases and prescribed doses to the target volumes are given in appendix B. The assemblage of this dataset was carried out by Renato Bellotti, as part of the development of an Automatic Treatment Planning library in Julia language named **JulianA** [23]. It is briefly over-viewed in section 3.4.

For every patient within the dataset, several essential components are available. These encompass the prescribed dose in Gray Relative Biological Effectiveness (Gy RBE) – a unit quantifying the biological impact of radiation on living tissue, shortened to Gy from here onwards – the original planning Computed Tomography (CT) scan, the human planner-derived treatment plan (also referred to as the reference plan), the delineated contours of the Volumes Of Interest (VOIs) in the patient, and the approved dose distribution utilised for clinical treatment (termed reference dose distribution). The reference plans encapsulate comprehensive details concerning the Planning Target Volumes (PTVs), patient-specific dosimetric prescriptions allocated to OARs, and the ultimate fields employed for treatment.

3.1.2. Fields and Angles

As highlighted earlier, the patients included in the dataset come with reference treatment plans approved and delivered by CPT clinics. These plans encompass various pieces of information, primarily pertaining to the attributes of the fields to be administered. Each field within a treatment plan is associated with the following attributes:

- ☐ Field identifier;
- ☐ Gantry angle (in degrees);
- ☐ Couch angle (in degrees);
- ☐ Gantry nozzle extraction (in centimetres);
- ☐ Preabsorber setting, which might be required for the delivery of low energy beamlets;
- ☐ Target structure name, i.e the target of the field;
- ☐ Weight of the field, i.e. a fraction representing the importance of the field in the treatment.

Traditionally, a treatment plan consists of two to four fields, each representing a distinct physical beam directed at the patient. Among the various attributes, the most crucial for our study are the two angles. These angles define the incident trajectory of the proton beam – the precise quantity we aim to predict. In the subsequent sections of this document, the gantry angle will be represented as α , the couch angle as β . The comprehensive representation of angles for all fields is denoted as $\vec{\alpha}$.

For G2, the permissible ranges for these two angles are $\alpha \in [-30^\circ, 180^\circ]$ for the gantry angle, and $\beta \in [-180^\circ, 180^\circ]$ for the couch angle. These angles and Gantry 2 are represented in fig. 3.1. In practical applications, clinicians do not use the full range of the couch angle and rarely exploit the complete range of gantry angles. This tendency becomes evident when analysing the composite distribution of angle combinations employed across all fields administered to train and test patients, depicted in fig. 3.2. Notably, only one field has gantry angle below 0° , while all couch angles fall within the range of $[0^\circ, 180^\circ]$. Consequently, the predictions of our methods will exclusively cover the intervals $\alpha \in [0^\circ, 180^\circ]$ and $\beta \in [0^\circ, 180^\circ]$.

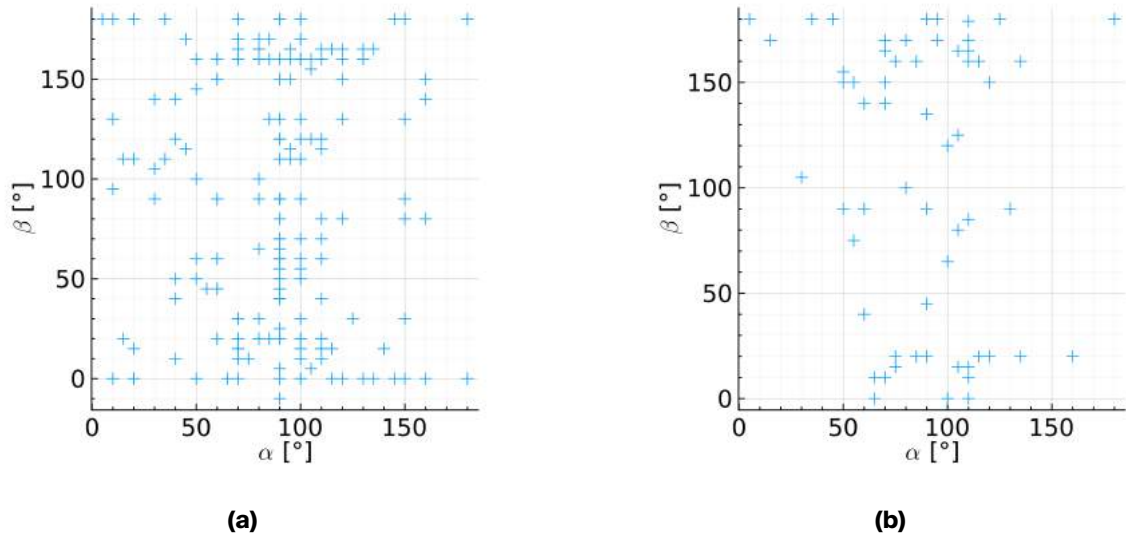


Figure 3.2: Distributions of (α, β) pairs of all the fields delivered to the patients reconciled in the data set at hand. α is the gantry angle, β is the couch angle.
(a) Training patients, **(b)** Test patients.

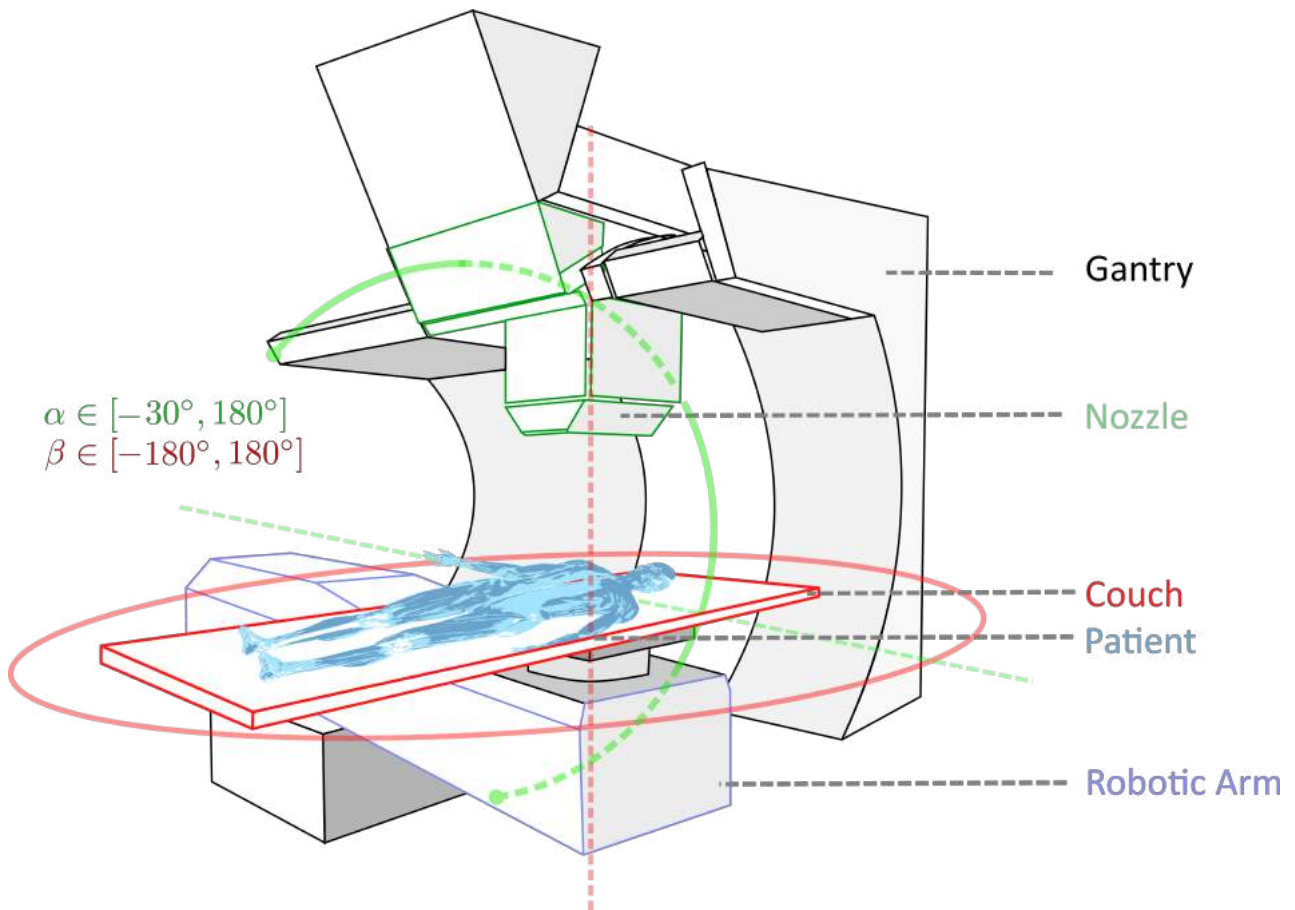


Figure 3.1: Diagram of Gantry 2 at PSI. The angle ranges are indicated and represented. The patient position represented is $(\alpha, \beta) = (0^\circ, 90^\circ)$ in an Head First Supine configuration. The positive range of the couch angle is explored via trigonometric rotation, and the negative range via anti-trigonometric rotation. The robotic arm's disposition is governed by the couch angle, the patient's configuration and the target structure. It is not a degree of freedom.

One could wonder why patients are not grouped by pathology type for this angles analysis. While this would be pertinent in the case of a large dataset, the set at hand only counts 94 patients with a large variety of pathologies. Splitting the dataset any more would result in unusable and small datasets. Hence, the entire population is considered for both this analysis and the approaches attempted for the beam angles selection problem.

No clear pattern emerged from the distributions of angles. With the lack thereof, no pre-processing could be achieved on the dataset to make the angles selection process more educated.

3.2. CNNs and Inputs Processing

3.2.1. CNNs for Beam Angles Selection

As described in chapter 2, Convolutional Neural Networks conventionally excel at tasks that involve visual representations. In the context of this research, the dataset contains three-dimensional (3D) details regarding patients, materialised as contours of VOIs. From these delineations, binary masks and distance masks can be generated. Further elaboration on these objects and their application within our context is provided below.

Beyond the 3D data, the dataset also incorporates comprehensive dosimetric prescriptions assigned to each Organ at Risk (OAR) by medical professionals for each patient. Examples are accessible in appendix F.1. From these two sources of information, two distinct learning approaches are studied in this investigation with CNNs: a supervised paradigm and an unsupervised paradigm. Importantly, only the latter approach relies on prescriptions. Detailed insights into these methodologies and their results are presented in chapter 4 and chapter 5.

3.2.2. Binary Masks

As previously mentioned, the geometric attributes of the patients serve as the foundation for predicting beam angles with CNNs. For this purpose, we shall construct a standardised three-dimensional object that comprises all pertinent 3D data about each patient.

Within the dataset, every patient is supplied with hand-traced contours of visible anatomical structures, an endeavour undertaken by clinical experts. The JulianA library is employed to convert these contours to binary masks on the CT grid. A binary mask is a tensor characterised by specified dimensions, housing binary values of 0 or 1, thereby delivering spatial information. In our case, these tensors are three-dimensional. An elemental component of such a tensor is named a **voxel** and represented as v_{ijk} for tensor indices (i, j, k) . The formulation for an element of a binary mask M with dimensions (W, D, H) that corresponds to the contours of structure s is expressed as follows:

$$M_{i,j,k}^{(s)} = \begin{cases} 1 & v_{ijk} \text{ enclosed by delineated contour} \\ 0 & \text{elsewhere} \end{cases} \quad (3.1)$$

To enable the CNN to differentiate between OARs and PTV, the inputs consist of two channels: $(M^{(OARs)}, M^{(PTV)})$. The first channel amalgamates all the OARs to produce a unified mask. Meanwhile, the second channel represents the PTV mask as a distinct entity. Furthermore, a prerequisite for CNN's functionality is that these channels adhere to a standard format: their dimensions must remain consistent across all patients. Consequently, an extra pre-processing stage is essential to identify the most fitting standardised input size.

Upon scrutinising the CT grids of all patients, all their dimensions are uniformly set to $(256, 256, x)$, where x denotes a variable height unique to each patient. To obviate any necessity for cropping the current masks and to facilitate the potential augmentation of the dataset in the future, the

maximum height is selected and supplemented with an additional margin. The definitive standard height is 150. In cases where x is less than 150, zero-padding is appended on both ends of the final masks. A mathematical definition of these channels would be the following :

$$M_{i,j,k}^{(OARs)} = \begin{cases} 1 & v_{ijk} \text{ enclosed by any OAR contour} \\ 0 & \text{elsewhere} \end{cases} \quad (3.2)$$

$$M_{i,j,k}^{(PTV)} = \begin{cases} 1 & v_{ijk} \text{ enclosed by PTV contour} \\ 0 & \text{elsewhere} \end{cases} \quad (3.3)$$

A depiction of a standardised input is presented in fig. 3.3.

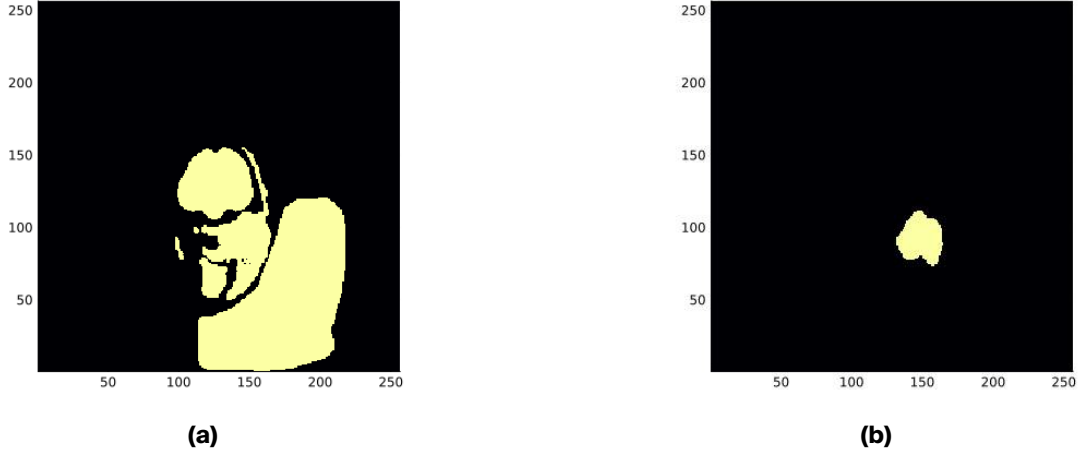


Figure 3.3: Binary masks for training patient with ID=25. These binary masks of size (256,256,150) are a first attempt at building an input to the CNN. Here, slice 50 is represented.

(a) Channel 1 : All the OARs $M^{(OARs)}$, **(b)** Channel 2 : PTV $M^{(PTV)}$

These channels may contain irrelevant information. Indeed, not all structures traced by clinicians are assigned dosimetric prescriptions. This becomes apparent when examining fig. 3.3a, where the structure in the lower-right corner corresponds to the patient's couch. Consequently, it appears natural to consider structures subject to prescriptions exclusively. This alignment between the inputs and the loss function is crucial for consistency, especially in the unsupervised paradigm where dosimetric prescriptions are key to the approach. More details are given in chapter 5.

Additionally, these channels often manifest significant sparsity. To address these issues, a different standardised input must be devised. To achieve this, an analysis is performed on the ranges along all three dimensions of the VOIs' masks obtained with JulianA from the contours. This analysis involves identifying the smallest bounding frame (cube) capable of containing all organs (OARs and PTVs) within the binary masks for each patient. Based on the collected frame sizes, a standardised frame applicable to all patients is determined: (130, 130, 85). This strategic approach shrinks the previous masks to match the frame without truncating any patient's organs. This adjustment reduces the sparsity of inputs and the overall element count per input, subsequently leading to a relative increase in pertinent information.

The definitions of the two input channels are thus slightly revised, as given in eq. (3.4) and eq. (3.5). Notation-wise, m replaces M to signify the alteration in channel size.

$$m_{i,j,k}^{(OARs)} = \begin{cases} 1 & v_{ijk} \text{ enclosed in an OAR appearing in the prescriptions} \\ 0 & \text{elsewhere} \end{cases} \quad (3.4)$$

$$m_{i,j,k}^{(PTV)} = \begin{cases} 1 & v_{ijk} \text{ enclosed by PTV contour} \\ 0 & \text{elsewhere} \end{cases} \quad (3.5)$$

fig. 3.4 shows the new masks for patient 25.

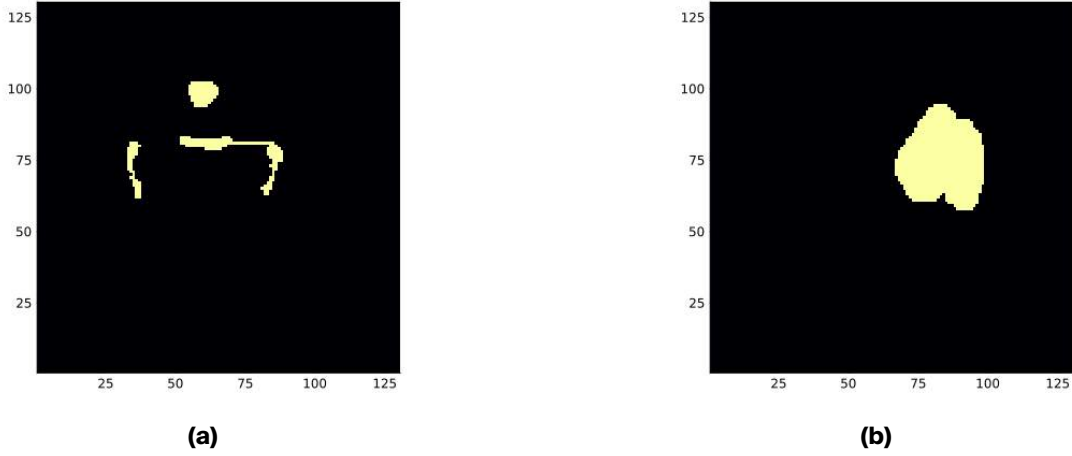


Figure 3.4: Binary masks for training patient with ID=25. With size (130,130,85), they are inputs to the CNN. Here, slice 40 is represented.

(a) Channel 1 : All the OARs $m^{(OARs)}$, (b) Channel 2 : PTV $m^{(PTV)}$.

3.2.3. Distance Masks

The masks formulated in the previous section retain a degree of sparsity and could potentially lack spatial insight. Indeed, the network might not deduce the distance of any voxel to any other structure. A possible solution is the generation of distance masks. JulianA enables the computation of such masks for an organ based on its contour points. For every voxel, the in-slice¹ distance (d_{slice}) in centimetres to the closest in-slice contour point is determined. When the voxel is positioned within the contours (inside the structure), the distance value is assigned as negative. Conversely, when outside, the value is positive. The contour points are confined within a "bounding box" (not to be confused with the "bounding frame" mentioned earlier). Beyond this box, with an additional margin², the values are set to $+\infty$. A "vanilla" distance mask element corresponding to a structure s can be defined as described in eq. (3.6):

$$D_{i,j,k}^{(s)} = \begin{cases} (-1) \cdot d_{\text{slice}}(v_{ijk}) & v_{ijk} \text{ inside of contour} \\ d_{\text{slice}}(v_{ijk}) & v_{ijk} \text{ outside of contour but inside bounding box and margin} \\ +\infty & v_{ijk} \text{ outside of bounding box} + 2.2\text{cm margin} \end{cases} \quad (3.6)$$

Starting from the individual structures' masks, the OARs and PTV channels can be computed by determining the minimum in-slice distance to an OAR corresponding to each voxel. Similarly to binary masks, the inputs are standardised to dimensions (130,130,85) using the same bounding frames established earlier. Furthermore, the infinite values that persist within the bounding frame are assigned the maximum distance value found within that frame. This modification enables the neural network to process the input. The minimum distance to an OAR is defined in eq. (3.7):

$$d_{\min}(v_{ijk}) = \min_{s \in \text{OARs}} (\{D_{i,j,k}^{(s)}\}) \quad (3.7)$$

¹In this context, in-slice distance corresponds to the distance calculations that involve only the contour points located within the same axial transverse slice as the voxel under consideration.

²2.2 cm, because JulianA defaults to a 2 cm margin for placing optimisation points, i.e. the points in space where the dose distribution is calculated.

The definition of the distance masks element at voxel v_{ijk} inside the bounding frame is thus newly defined for the OARs and the PTV :

$$d_{i,j,k}^{(OARs)} = \begin{cases} d_{\min}(v_{ijk}) & \text{if not infinite} \\ \max_{v_{ijk} \in \text{bounding frame}} (\{d_{\min}(v_{ijk})\}) & \text{else} \end{cases} \quad (3.8)$$

$$d_{i,j,k}^{(PTV)} = \begin{cases} D_{i,j,k}^{(PTV)} & \text{if not infinite} \\ \max_{v_{ijk} \in \text{bounding frame}} (\{D_{i,j,k}^{(PTV)}\}) & \text{else} \end{cases} \quad (3.9)$$

An example of such masks is presented in fig. 3.5.

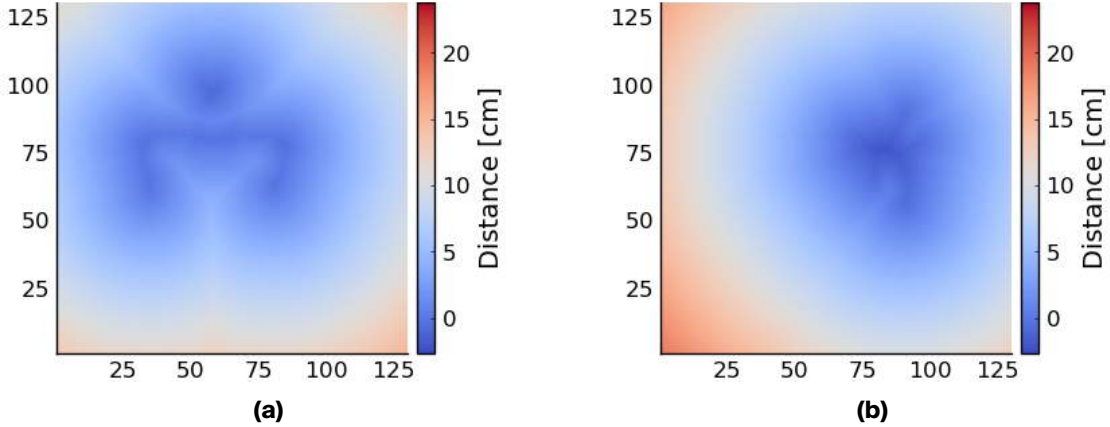


Figure 3.5: Distance masks for training patient 25. With size (130,130,85), they are inputs to the CNN.

Here, slice 50 is represented.

(a) Channel 1 : All OARs $d^{(OARs)}$, **(b)** Channel 2 : PTV $d^{(PTV)}$.

Both of these input types, binary and distance, are used in the training of the supervised and unsupervised CNN approaches.

3.3. Data augmentation

3.3.1. Dataset limitations

As previously explained, the dataset available for this study counts 75 training patients, a (very) limited number for a DL approach. To address such data scarcity, a conventional solution involves dataset augmentation. With this process, relevant synthetic data is generated from the existing set. When dealing with CNNs, data augmentation frequently consists of alterations to the input data, such as translations, rotations, blurs, elastic deformations, mirroring, and more. However, in the context of this dataset, the process is less straightforward.

The masks created are derived from contours traced over standardised CT images. All head CTs adhere to a consistent pattern: the head is positioned at the centre of the image and maintains a uniform configuration. Left, right, anterior, posterior, inferior, and superior orientations are identical across patients. Any augmentation efforts must uphold this organisational structure. Moreover, if deformations are introduced, they must be organic, since a deformation could potentially impact multiple organs simultaneously, and should reflect in the patients' contours. This decision is complex as it would necessitate medical approval to validate the "naturalness" of a patient's state after transformation.

Beyond image transformations, our unsupervised approach also relies on prescriptions provided by doctors for the VOIs within the patient. augmentation that influences the geometry could necessitate a reevaluation of prescriptions by doctors, akin to creating a whole new treatment plan.

An effort in this direction would require collaboration with doctors to establish acceptable protocols for such augmentation. This option was not feasible in the course of this study.

3.3.2. augmentation technique

The human head has a sagittal symmetry: any structure on the left side has a symmetric counterpart on the right side. It is not the case when considering other body segments, namely the digestive system or the upper part of the thorax. With this property, an augmentation is possible without facing the aforementioned limitations. A representation of this transformation is provided in fig. 3.6.

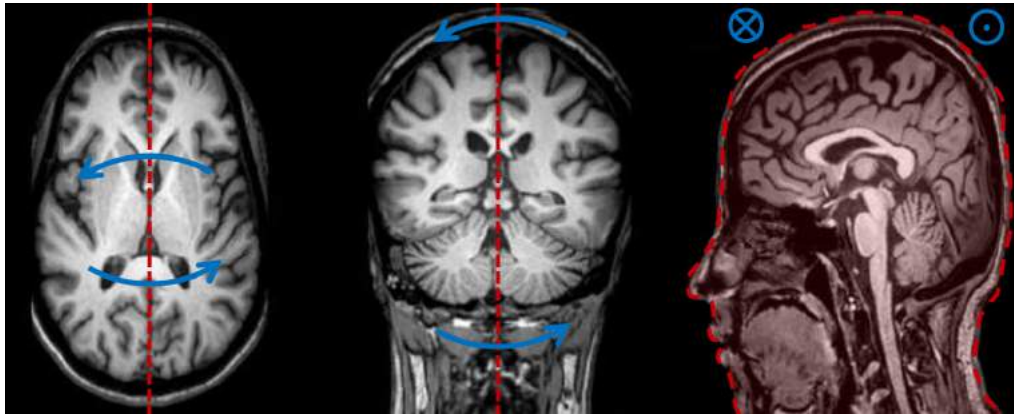


Figure 3.6: Data augmentation technique of the patient based on the sagittal symmetry of the human head. From left to right, axial, coronal, and sagittal views of the sagittal plane (dotted red). The blue arrows represent the mirroring of the voxels with respect to the sagittal plane. MRI images extracted from figure 2 of [24].

The augmentation requires the following steps and transformations with respect to the sagittal plane:

- ☐ Reversal of the CT;
- ☐ Reversal of the contours;
- ☐ Reversal of the prescriptions when a side is mentioned. For example, a prescription for the left eye becomes a prescription for the right eye;
- ☐ Transformation of the fields angles.

The transformation for the fields' angles, used in the supervised training pipeline, is as follows:

$$(\alpha, \beta) \rightarrow (\alpha, 180^\circ - \beta) \quad (3.10)$$

With this technique, a new patient emerges, referred to as the **symmetric** patient throughout the study. This patient possesses a valid CT, contours, and prescriptions. Consequently, the new dataset is doubled compared to the previous one and maintains clinical coherence. However, one limitation persists with this augmentation. One of the validation steps for its usability is to confirm that FlonA's, CPT's treatment planning system, dose calculation engine generates symmetrical dose distributions for both the original and symmetric patients. After this verification, a noticeable difference arises when subtracting the reference dose distribution from the mirrored symmetric dose distribution, as demonstrated in fig. 3.7.

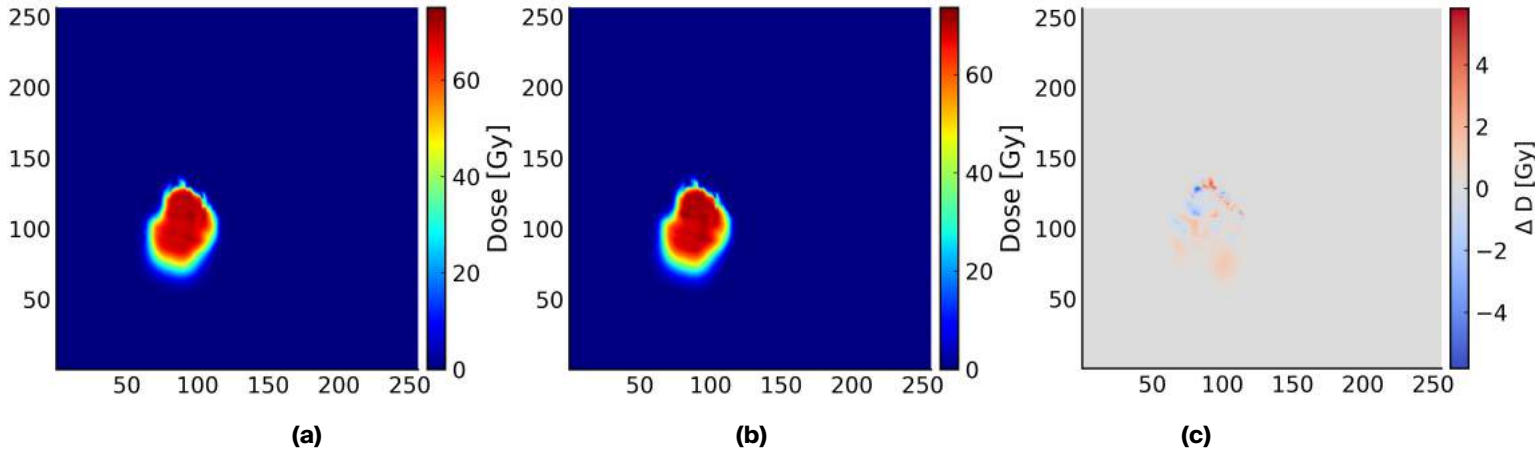


Figure 3.7: Reference Clinical dose distributions for training patient with ID=25 at slice 50 of the CT grid:
(a) Reversed normal patient
(b) Symmetric patient
(c) Difference between both distributions.

The observed dose disparity results from the asymmetry in both the optimisation points and the spots' configurations computed by FlonA. The dose discrepancy is deemed insignificant within the scope of this study, leading to the acceptance of the augmentation.

3.4. Framework

The whole framework developed for this experiment is named "**A**utomatic and **R**apid tool for **O**ptimal **B**eam **A**ngles **S**election", or in short AROBASE (the French name for character "@"). It is written in **Julia 1.8.1** [25] and makes use of several libraries aside from its core tools. The most central in the development is **Flux.jl** [26][27], a library for machine learning in the Julia language. It comes with out-of-the-box solutions and proposes simple yet extensible and efficient building blocks to build ML and DL pipelines. Flux is high level without sacrificing efficiency, for both computation time and memory usage, and proved suitable for this work. Another library used is the automated treatment planning library **JulianA** [23]. Apart from spot weight optimisation and dose distribution calculation, this library offers a number of utility functions essential for handling radiotherapy-related data structures, such as CT, contours, optimisation points, and dosimetric prescriptions. It also provides dose volume histogram computations, along various treatment plan quality metrics.

AROBASE is divided into two parts, corresponding to the two approaches presented in this study. The architecture of the code developed specifically for the development of the CNN approaches and its abilities are detailed in fig. 3.8

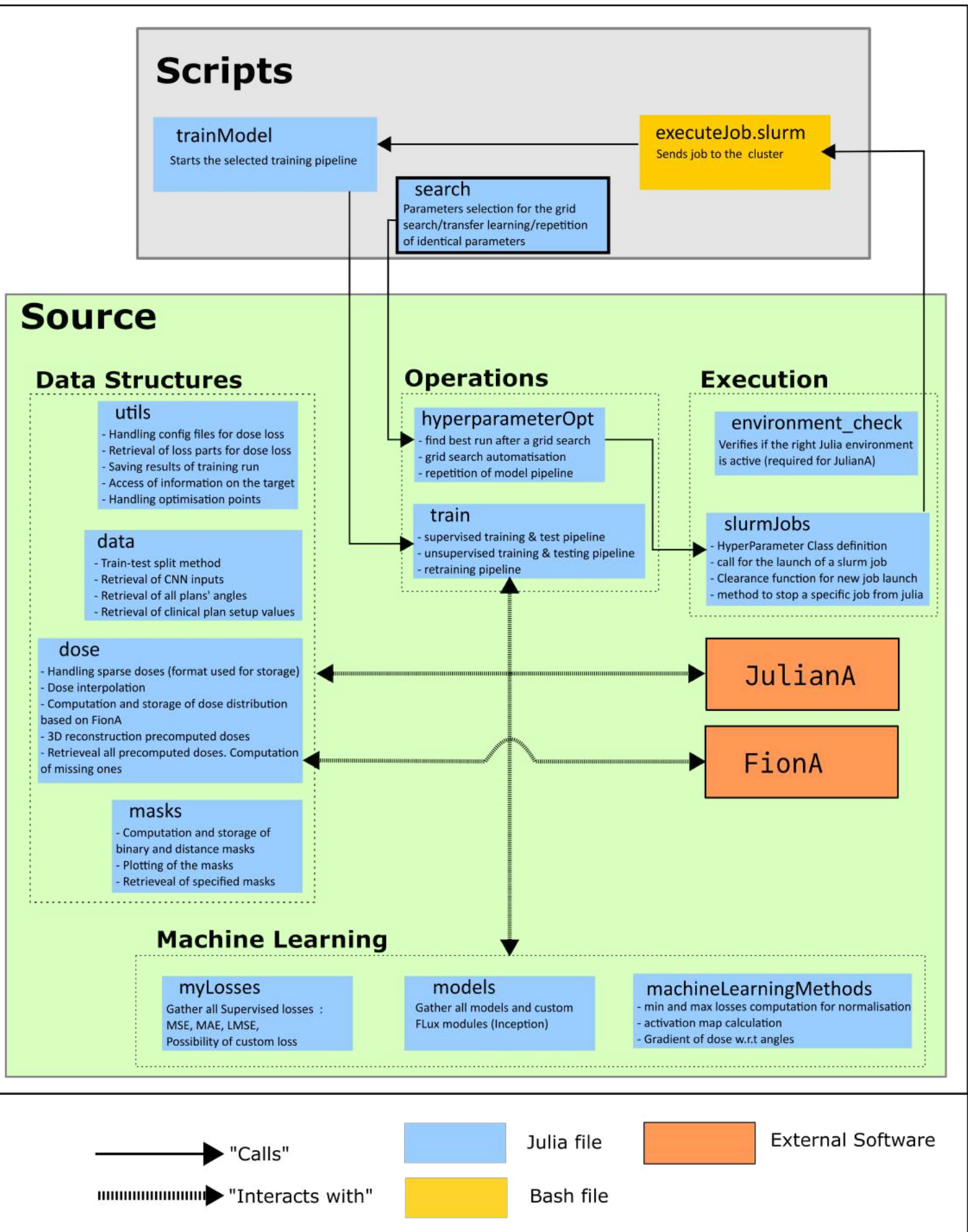


Figure 3.8: Architecture of the code developed for the experiment of supervised and unsupervised CNNs for single non-coplanar field beam angles selection in proton therapy.

AROBASE has the following capabilities (non-exhaustive) related to the CNN approach:

- **Data Structure Handling:** The code provides utility functions to manage data structures like dose distributions, masks, angles.
- **Selection of Learning Approach:** It permits users to opt for either supervised or unsupervised learning approaches from the outset.
- **SLURM Automation:** The code streamlines SLURM job management and automation exclusively through Julia, relieving users of manual handling.
- **Experiment Tracking:** It enables the tracking of experiments by logging data such as losses, saved models, final angle predictions, and hyper-parameters;
- **Grid Search Automation:** It provides automatic grid search functionality for designated hyper parameter ranges;
- **Automated Best Run Selection:** The code automates the selection of the best run, tailored to the specific supervised or unsupervised context;
- **Flexible Customisation:** The code facilitates effortless and adaptable customisation of machine learning components, encompassing model, optimiser, and supervised loss type;
- **Optimisation Incorporation:** It allows integration of various optimisation techniques mentioned earlier, including regularisation strength and type, as well as batch size;
- **Retraining and Transfer Learning:** The code supports model retraining and transfer learning, for instance, from supervised to unsupervised scenarios.

Besides this source code directed to CNN training, AROBASE contains several Jupyter notebooks for data exploration, dose distribution production, and the entire Three Fields Dose Descent approach presented in chapter 6. Version control is done using PSI's in-house Gitlab infrastructure. More details about this framework and the requirements to run it are available in a repository³. The scripts and source code handling the CNN models are automated using Simple Linux Utility for Resource Management (SLURM) [28] on the Merlin/Gwendolen (GPU) cluster of PSI. They are run on an NVIDIA DGX A100 system, providing eight A100 GPUs with 320GB cumulated GPU memory (40GB each) [29].

³Available with PSI credentials at https://gitlab.psi.ch/bellotti_r/bastien_golomer_msc.

4

Supervised Learning

The initial strategy for beam angles selection exploits supervised learning. In this paradigm, a model is trained to establish a mapping between input data and corresponding target labels, relying on a labeled dataset. For this supervised training approach, the angles characterising from the fields in the reference plans, as described in section 3.1.2, serve as the labels. Prescriptions are not utilised in this context. The rationale behind this choice lies in the role of the supervised approach as a preliminary attempt at constructing a foundational machine learning pipeline that can be extended and refined in the future. The objective is to establish an initial network architecture, select suitable activation functions and weight initialisation schemes, determine an appropriate optimiser, and undertake a hyper-parameter exploration. This results in an initial implementation that serves as a fundamental framework to evolve into a more intricate, unsupervised, and prescription-informed pipeline.

Three to four fields are employed for the treatment of each patient. In this preliminary configuration, only a **single non-coplanar field** is predicted by the networks, for simplicity. They shall be extended to predict additional field angles only if they yield acceptable results. Consequently, a single field is selected across all fields in patient's reference plan to serve as labels. The label field has lowest gantry and couch angles values among the fields list. A visual representation of this supervised approach is given in fig. 4.1.

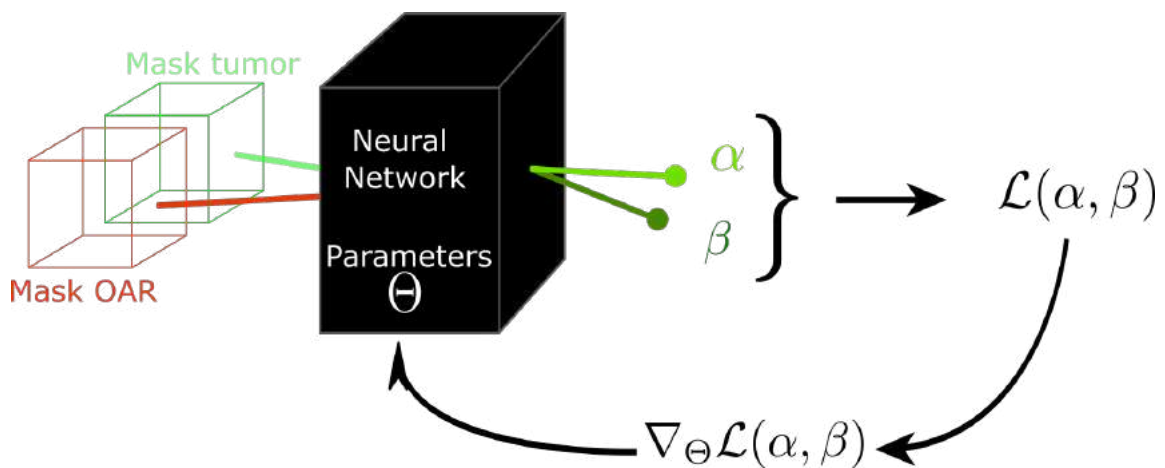
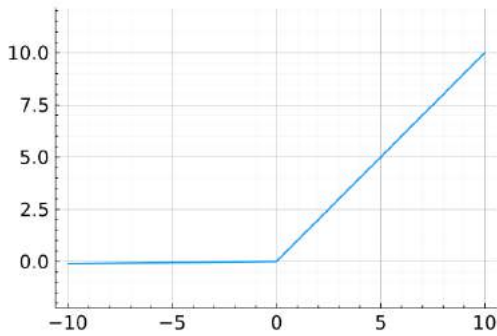


Figure 4.1: Supervised approach to the single field beam angles prediction problem. $\mathcal{L}$ is a loss function comparing outputs to labels, (α, β) are the predicted gantry angle and couch angle respectively, Θ is the set of all the learnable network parameters. The gradient $\nabla_{\Theta} \mathcal{L}(\alpha, \beta)$ is calculated in the backpropagation process of the weights update.

As described earlier, the inputs consist of two channels: the mask aggregating the OARs, and the PTV mask. These inputs are passed inside the NN with set of learnable parameters Θ . A single learnable parameter is noted θ . The network predicts a pair of real values, the gantry and couch angles (α, β) . In the following sections, the ingredients to the pipeline are detailed, namely the network architectures and the loss function.

4.1. Network Architectures

Several networks are crafted for this experiment. The goal of this multiplicity of models is to test different degrees of architecture complexity and assess if one's performance stands out among others'. For all the networks constructed, the weights of each layer are initialised using a Glorot uniform initialisation with gain of 1 [30].



The activation function σ selected for all the layers in all the presented networks is Leaky ReLU, a variation of ReLU (Rectified Linear Unit). It showcases non-zero output and gradient values for negative input values, allowing in principle for more network expressivity. A posteriori, it showed better results than conventional ReLU for all the attempted networks.

Figure 4.2: Leaky ReLU activation function on the range $[-10, 10]$.

4.1.1. Full Model

The network we start with is a chain of all the basic operations of CNNs listed in chapter 2. An additional Global Mean Pooling layer transforms the final 512 3D-tensors into 512 scalars by taking their individual mean values. These values are inputs for the encoder. The neural network is represented in fig. 4.3. This "vanilla" network is denoted as **"Full" model** in the rest of this document.

4.1.2. Reduced Model

The first additional model constructed is a reduced version of the Full network presented, as illustrated in fig. 4.4. This network's purpose is to verify whether the Full model is too deep. Indeed, a neural network too deep for a task captures the variances of the input training data, thus losing generalisation ability and inducing over-fitting. Minimizing the complexity of the network allows in principle to reduce this effect and balance back with the learning of the variances with the overall bias. This network is denoted **"Reduced" model** in the rest of the document.

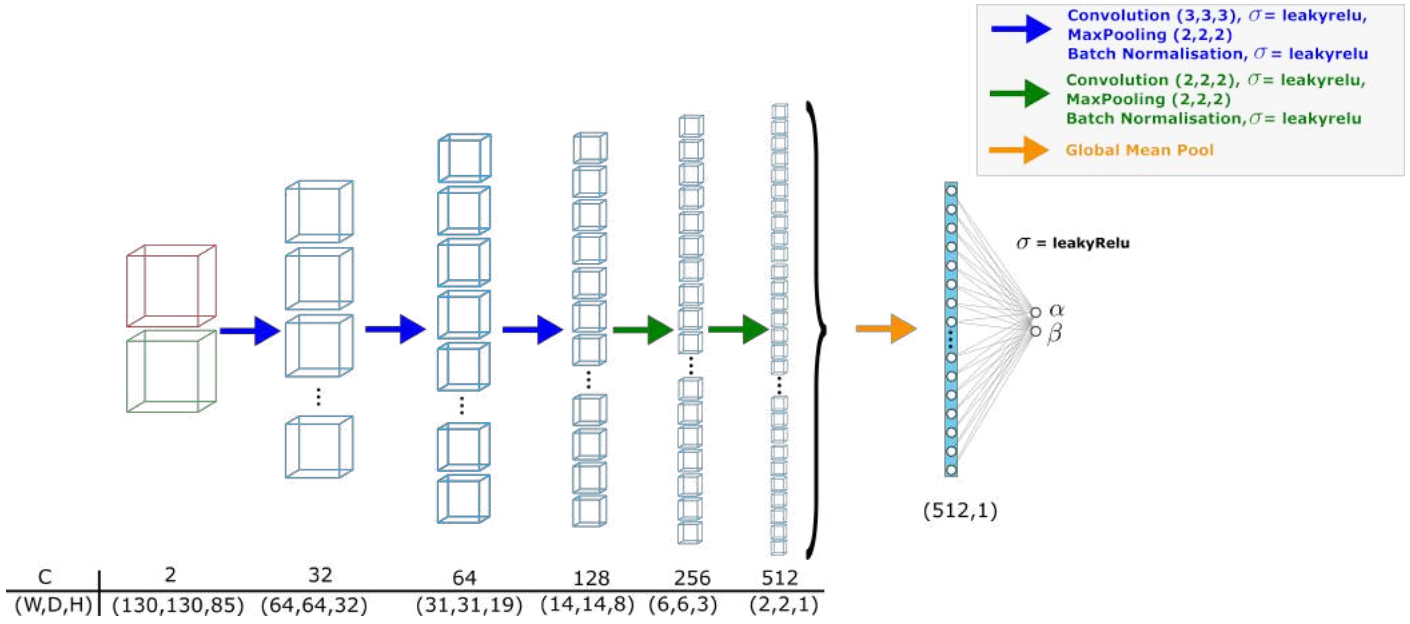


Figure 4.3: Full model architecture for the 1-field beam angles selection task. "C" denotes the number of channels of the current layer, (W, D, H) represent (Width, Depth, Height). The legend indicates what operations are applied. The convolution layers and max pooling layers have no padding, and strides of 2. Each transformation's kernel size is indicated by a triplet of integers.

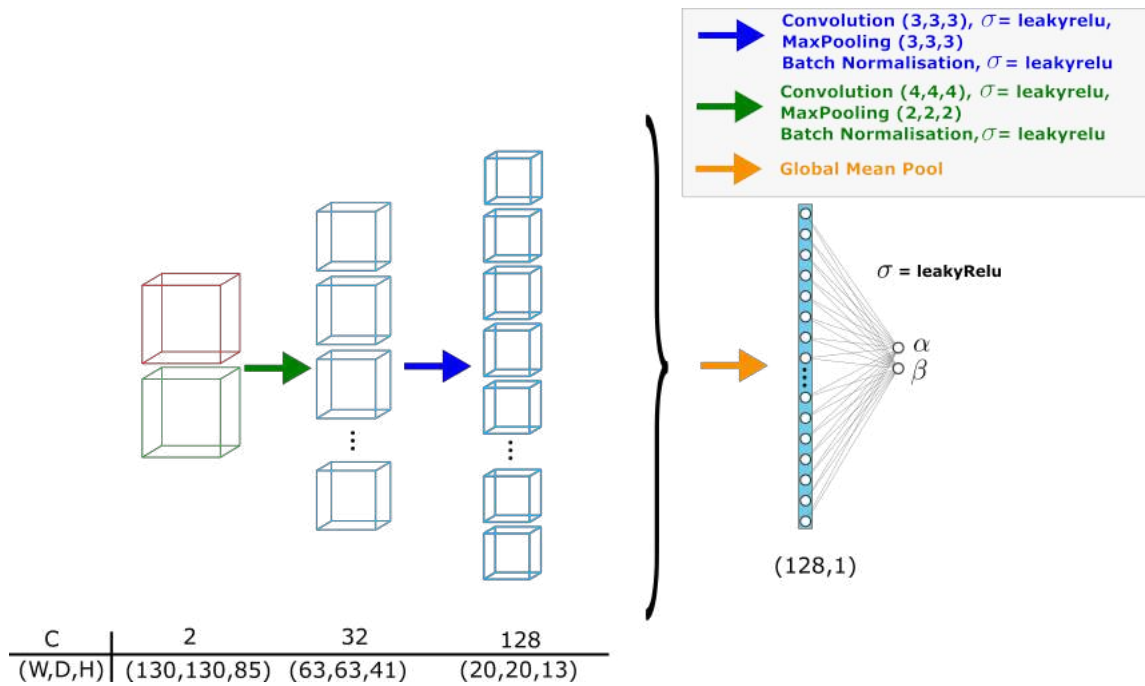


Figure 4.4: Reduced model architecture for the 1-field beam angles selection task. "C" denotes the number of channels of the current layer, the triplet (W, D, H) represent (Width, Depth, Height). The legend indicates what operations are applied. The convolution layers and Max Pooling layers have no padding, and strides of 2. Each transformation's kernel size is indicated by a triplet of integers.

4.1.3. Inception Model

The second model variation attempted is based on the Inception module developed in [31] in the context of high-performance computer vision. Such module is presented in fig. 4.5. Citing the abstract of the paper describing it: *"The main hallmark of this architecture is the improved utilisation of the computing resources inside the network. By a carefully crafted design, we increased the depth and width of the network while keeping the computational budget."*[31].

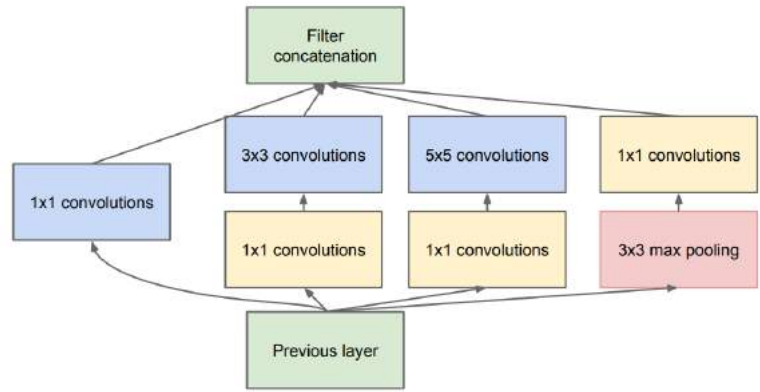


Figure 4.5: Inception Module with dimensionality reduction [31].

It appeared an interesting addition to this study to potentially improve performance while keeping low training costs. The final model, denoted **"Inception" model**, is illustrated in fig. 4.6

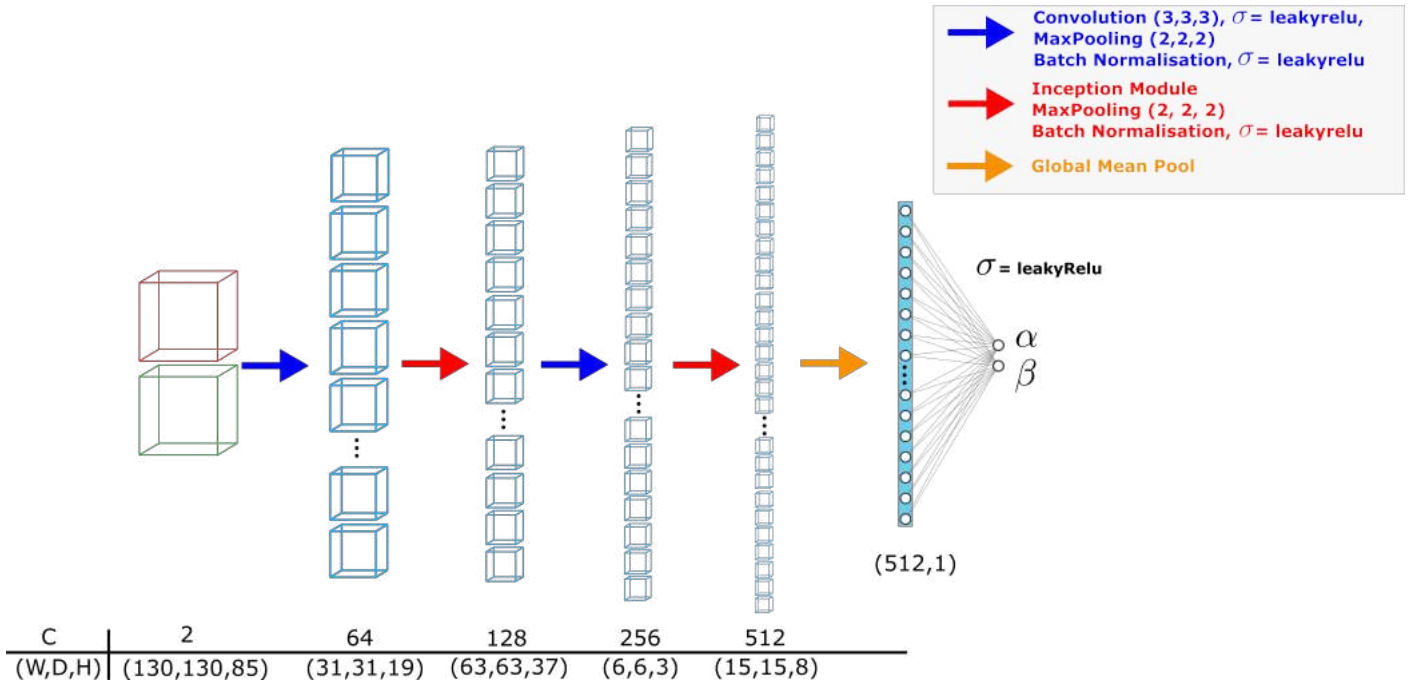


Figure 4.6: Inception model constructed with Inception Modules for the 1-field beam angle selection task. The legend indicates what operations are applied. The triplet (W, D, H) represent (Width, Depth, Height). The convolution layers and the Max Pooling layers have no padding, and strides of 2 and 3. Each transformation's kernel size is indicated by a triplet of integers.

4.2. Loss Function

As presented in fig. 4.1, the networks' predictions are directly input into the loss function $\mathcal{L}(\alpha, \beta)$ and can be contrasted with the target values (α^*, β^*) using a Mean Square Error (MSE) as a loss function. To mitigate overfitting concerns, a regularisation term with a tunable strength $\lambda_{\text{reg}} \in \mathbb{R}^+$ is incorporated. The final loss function for the supervised training is formulated as shown in

eq. (4.1).

$$\mathcal{L}(\alpha, \beta) = \mathcal{L}_{MSE}(\alpha, \beta) + \mathcal{L}_{\lambda}(\alpha, \beta) = \underbrace{\frac{1}{2} ((\alpha - \alpha^*)^2 + (\beta - \beta^*)^2)}_{\text{MSE}} + \underbrace{\lambda_{\text{reg}} L_{\text{reg}}}_{\text{Regularisation}} \quad (4.1)$$

The regularisation L_{reg} can be selected between L_1 and L_2 regularisations :

$$L_1 = \sum_{\theta \in \Theta} |\theta| \quad L_2 = \sum_{\theta \in \Theta} \theta^2$$

4.3. Training Modalities

The training of the networks was first set to 300 epochs with the binary masks and later limited to 150 epochs for the distance masks, as no improvement was seen past this number and for sparing of computational resources. Adam optimiser [22] was used for the update rule of the network weights, with the learning rate denoted η . To construct the best-performing networks for each architecture and mask combination, grid searches are performed across specified hyperparameter ranges. For the supervised learning approach, these ranges are $\eta \in \{10^{-4}, 5 \cdot 10^{-5}, 10^{-5}\}$, $\lambda_{\text{reg}} \in \{10^{-4}, 5 \cdot 10^{-5}, 10^{-5}, 5 \cdot 10^{-6}\}$, and batch sizes $b \in \{1, 2, 4\}$. These ranges were determined based on initial empirical attempts. The choice of relatively small batch sizes is the result of the limited size of the dataset. Both the L_1 and L_2 regularisation were attempted. The training and test loss evaluations for each of the best networks obtained through the grid searches are presented in fig. 4.7. The qualification "best" is based on the overall best mean value of the test losses of the network across all epochs. This implies that the final losses might present higher values. The best network is selected after the grid search is completed.

To facilitate outcomes interpretation, the results of the median predictor are visualised. A median predictor computes the median of the training labels and uses it as its prediction for any new test input. If the performance of the median predictor surpasses that of the network, it implies that the network failed to output meaningful predictions.

4.4. Results

Initial Dataset

We start this analysis with the initial dataset (no augmentation). The results are presented in fig. 4.7. The parameters of each best networks are given in table 4.1

Model Mask Type	Full Binary	Inception Binary	Reduced Binary	Full Distance	Inception Distance	Reduced Distance
η	$1.0 \cdot 10^{-4}$	$5.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$
Regularisation	L_2	L_1	L_2	L_2	L_1	L_1
λ_{reg}	$5.0 \cdot 10^{-6}$	$5.0 \cdot 10^{-6}$	$5.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$	$5.0 \cdot 10^{-5}$	$5.0 \cdot 10^{-6}$
b	1	2	4	2	4	1

Table 4.1: Best hyperparameter combinations obtained after grid search for the supervised learning paradigm. The symmetric patients are not considered.

The best pairing of model and input type are the Full model with binary mask, Inception with binary and distance masks, and Reduced model with binary masks. There is a noticeable over-fitting in all cases, where both the training loss and the test loss decrease until the test loss rises up again while the training loss keeps on decreasing. This typical behaviour is illustrated in fig. 4.8.

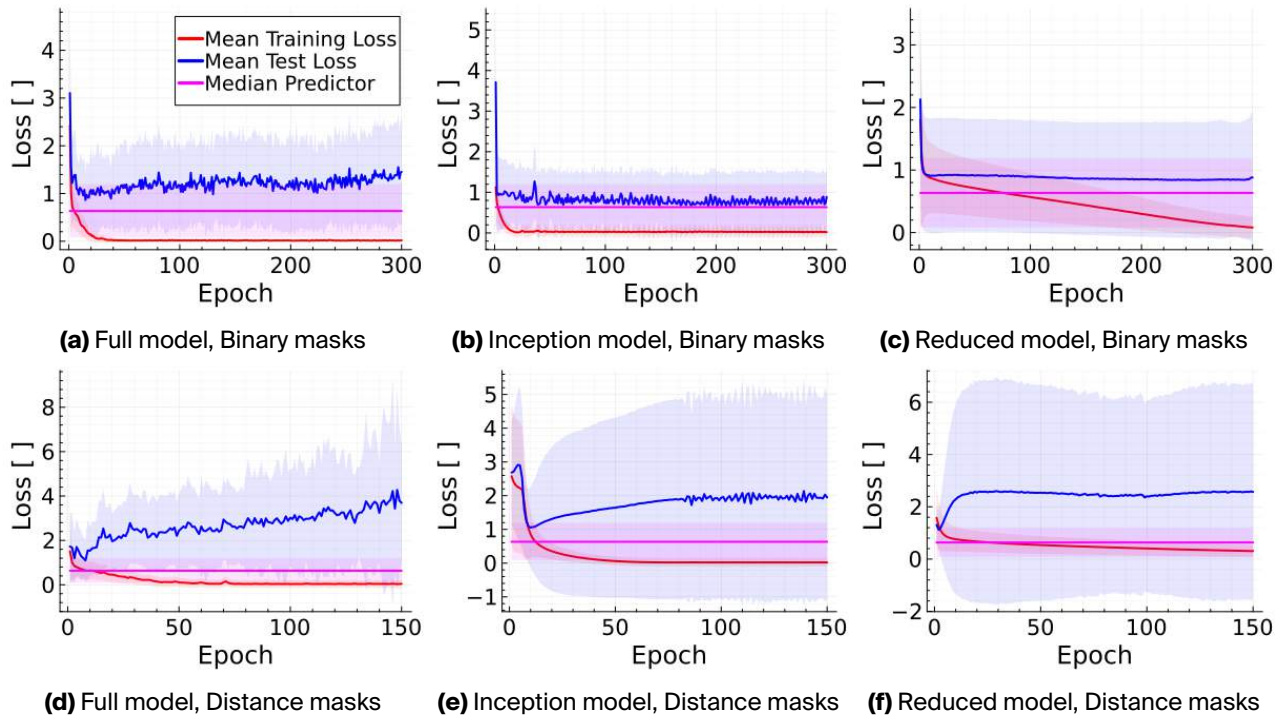


Figure 4.7: Training loss and Test loss evolutions across all the epochs in the supervised training, for each model and input combination. The solid lines represent the mean loss value across all patients in each set. The lesser saturated areas around the mean curves show the standard deviations from the mean for each set.

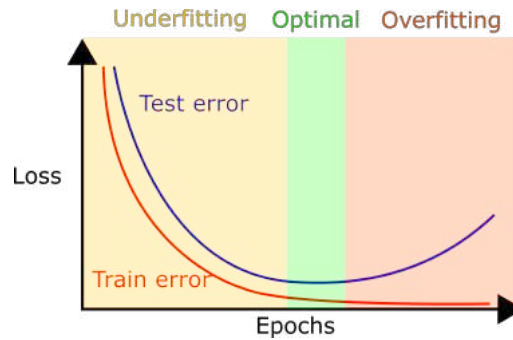


Figure 4.8: Typical behaviour of the loss functions (errors) of a neural network in training. Underfitting corresponds to the learning of the biases of the training data only. Overfitting demonstrates the learning of the variances of the training data, which yields a divergence of the test error and a further convergence of the training error.

Augmented Dataset

We extend the analysis to the augmented dataset to assess if this manoeuvre has an impact on the quality of the results. A new hyperparameter scan is performed using the same ranges. The new results are presented in fig. 4.9.

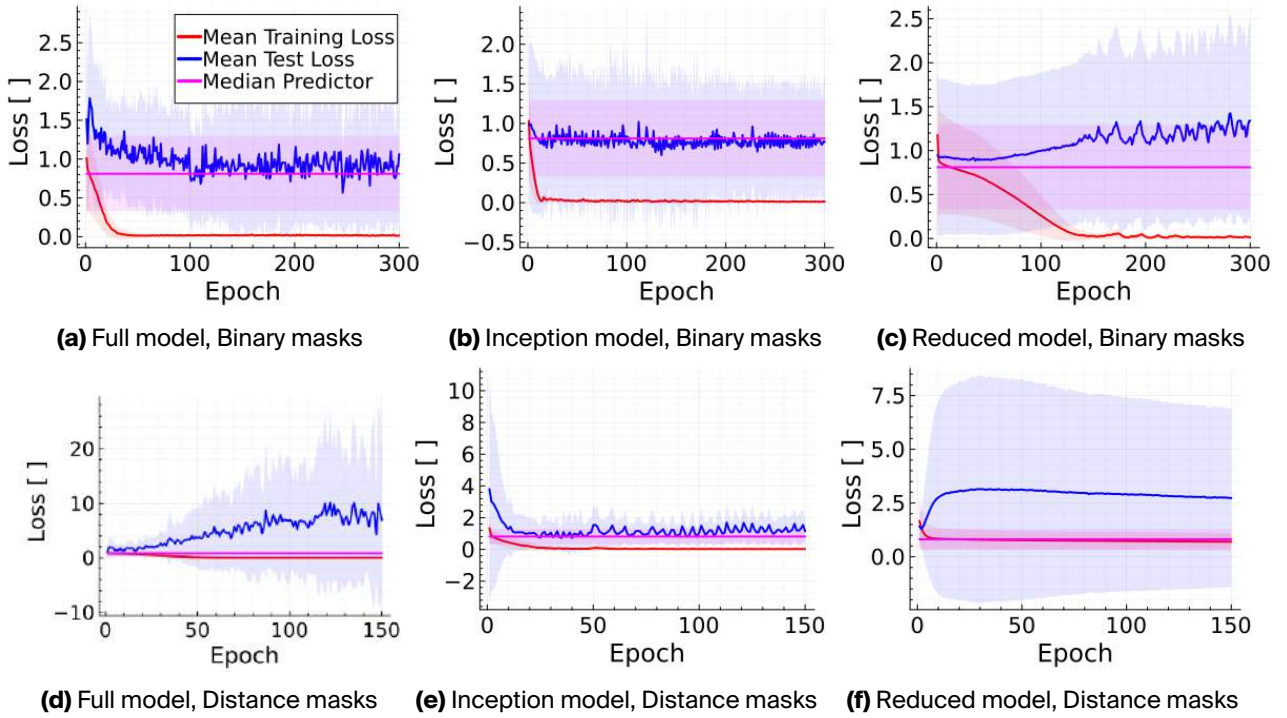


Figure 4.9: Training loss and Test loss evolutions across all the epochs in the supervised training with the augmented dataset. The solid lines represent the mean loss value across all patients in each set. The lesser saturated areas around the mean curves show the standard deviations from the mean for each set.

The parameters of each best networks are given in table 4.2

Model	Full	Inception	Reduced	Full	Inception	Reduced
Mask Type	Binary	Binary	Binary	Distance	Distance	Distance
η	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$
Regularisation	L_2	L_1	L_1	L_1	L_2	L_2
λ_{reg}	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$5.0 \cdot 10^{-6}$	$5.0 \cdot 10^{-6}$	$1.0 \cdot 10^{-4}$
b	1	2	2	1	4	1

Table 4.2: Best hyperparameter combinations obtained after grid search for the supervised learning paradigm. The symmetric patients are considered.

Three models reach the median predictor : the Full model with the binary masks, and the Inception model with the binary and distance masks. None clearly beats it, which again demonstrates the lack of performance of the networks. From the illustrations, it seems the augmentation allows for slightly better results in these three cases, but a clear conclusion cannot be drawn. However, no improvement is achieved concerning the reduced model.

The combinations with distance masks showcase early and large divergences. This is due to the definition of "best" run. Indeed, when analysing the individual experiments, some of them display lower final losses but, overall, do not reach test errors as low as those presented in the fig. 4.7 and fig. 4.9.

4.5. Discussion

4.5.1. Gradient-based Activation Maps

With this lack of performance from the networks, a valid assumption is that they do not comprehend the underlying geometry in the masks or target the regions of interest. In order to investigate these hypotheses, gradient-based activation maps are evaluated.

Definition

These maps, denoted A , demonstrate the value of the derivative of the output with respect to each individual input voxel. In other words, it showcases which small variations in the inputs affect the outputs. The exact quantities that are assessed for a selected input $(m^{(\text{OAR})}, m^{(\text{PTV})})$ are the following:

$$\begin{aligned} A_{ijk}^{(\alpha, \text{OAR})} &= |\partial_{v_{ijk} \in m^{(\text{OAR})}} \alpha|, & A_{ijk}^{(\beta, \text{OAR})} &= |\partial_{v_{ijk} \in m^{(\text{OAR})}} \beta| \\ A_{ijk}^{(\alpha, \text{PTV})} &= |\partial_{v_{ijk} \in m^{(\text{PTV})}} \alpha|, & A_{ijk}^{(\beta, \text{PTV})} &= |\partial_{v_{ijk} \in m^{(\text{PTV})}} \beta| \end{aligned} \quad (4.2)$$

where $(\alpha, \beta) = NN(m^{(s)})$ with NN the trained neural network, $m^{(s)}$ the binary masks or distance mask for structure (s) .

Patient Masks

The masks of each patients used to construct these activation maps are presented in fig. 4.10 and fig. 4.11

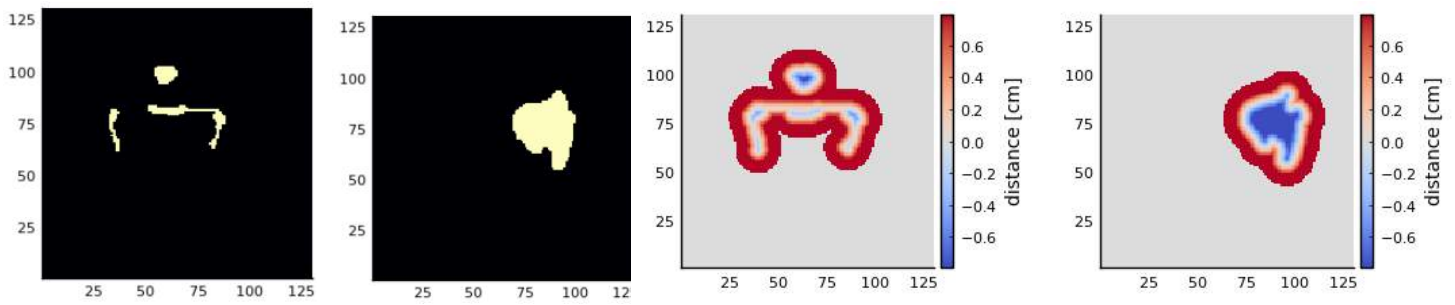


Figure 4.10: Binary and Distance masks of training patient with ID = 25. Slice 50 for binary, slice 51 for distance. The distance masks were cropped at the absolute value of the minimum distance in the masks, for easier visualisation.

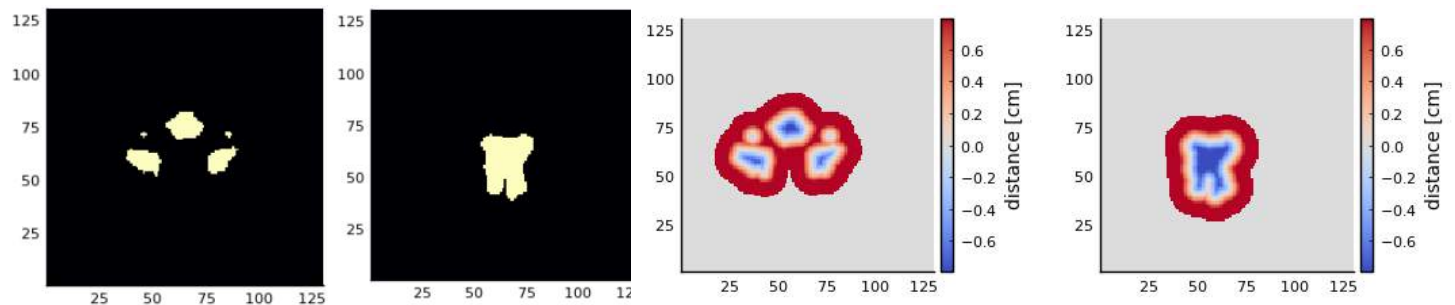


Figure 4.11: Binary and Distance masks of test patient with ID = 00. Slice 50 for binary, slice 45 for distance. The distance masks values were cropped at the absolute value of the minimum distance in the masks, for easier visualisation.

Maps

The maps presented correspond to training patient with ID=0. The maps of the remaining patients considered are gathered in appendix C. The colour bars' ticks are shared by all the representations.

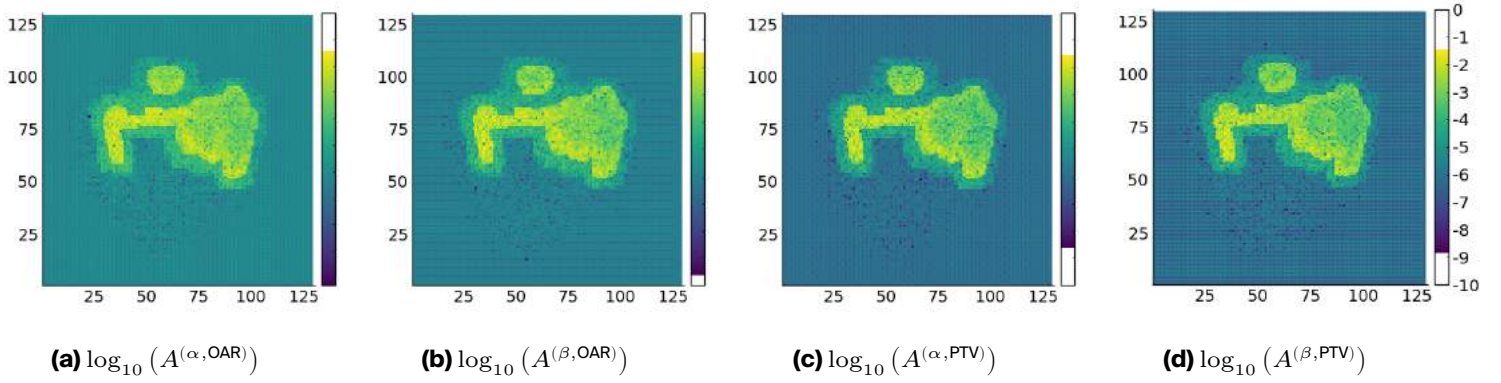


Figure 4.12: Gradient-based activation maps when predicting the angles for training patient with ID=25. The model selected is the supervised Full model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs. Slice 50 is represented.

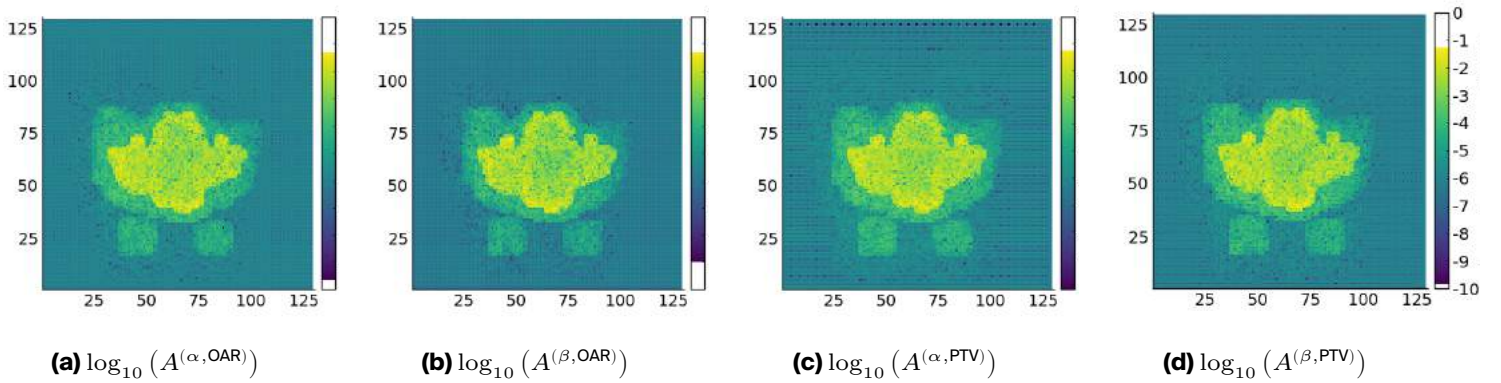


Figure 4.13: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the supervised Full model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs. Slice 50 is represented.

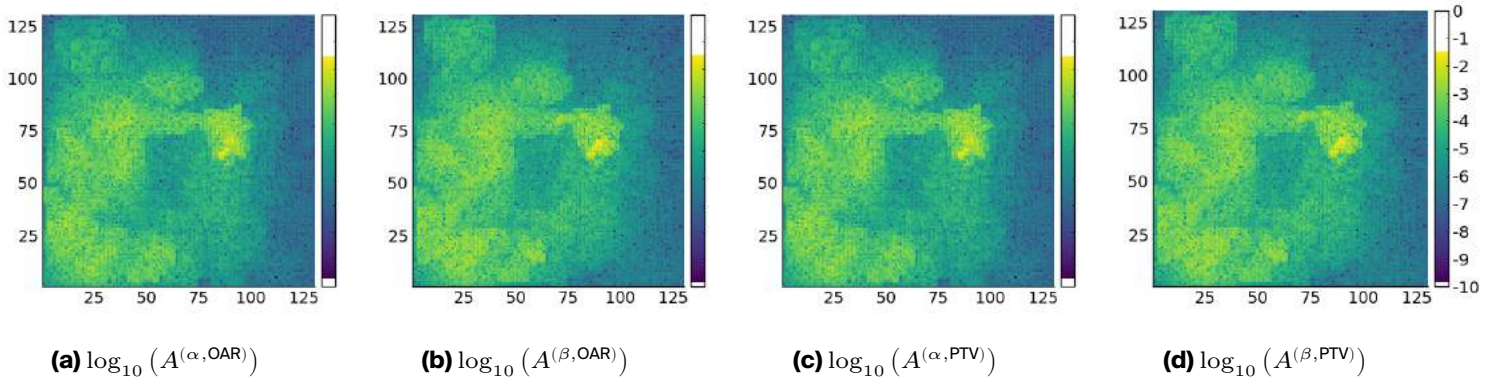


Figure 4.14: Gradient-based activation maps when predicting the angles for training patient with ID=25. The model selected is the supervised Full model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs. Slice 50 is represented.

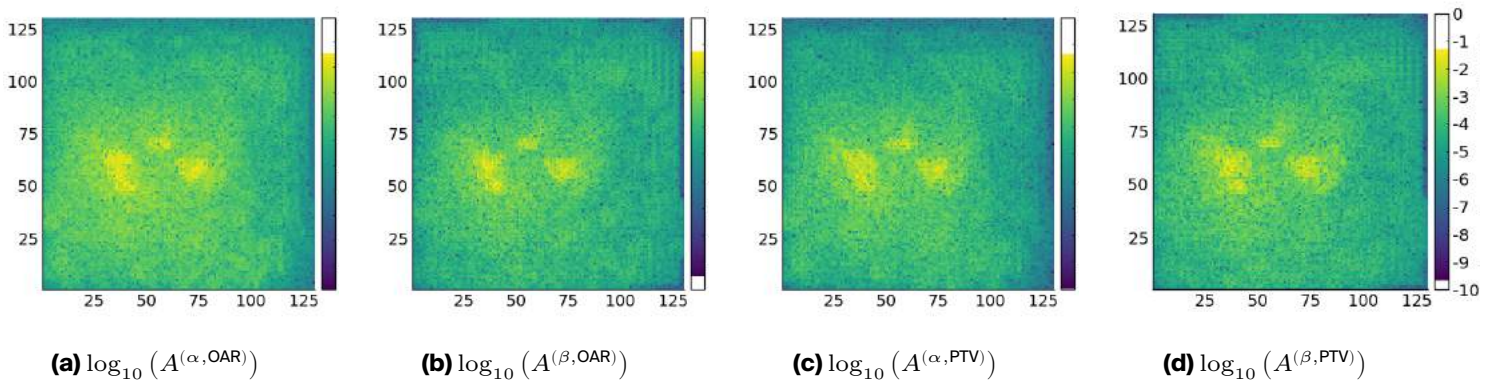


Figure 4.15: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the supervised Full model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs. Slice 50 is represented.

Discussion

The activation maps yield several valuable insights. First, the networks demonstrate a distinct emphasis on the portions of the masks that carry the most informative content. This is notably prominent when considering the binary masks, where a distinct boundary outlining the organs is evident. When examining the distance masks, discerning this demarcation becomes arduous, possibly because the variations of values across these masks are smooth and continuous. Consequently, the network tends to focus its attention on less pertinent regions, as indicated by the left section of the activation maps in fig. 4.14 and the surrounding regions in fig. 4.15. Despite this, the internal segments of the organs still seem to be recognised, standing out amidst the other voxels.

Another noticeable phenomenon is that the network doesn't appear to distinguish OARs from PTV. When observing the derivative of α or β with respect to a voxel in the OAR mask that aligns with a tumor voxel in the PTV mask, this derivative showcases a high value. Thus, the network appears to seek PTV voxels within the OAR mask. This is noticeable the other way around, where voxels in the PTV mask showcase high derivative magnitude when they align with OAR voxels. This observation suggests that the current strategy of segregating OARs and the PTV into separate channels might not be optimal for the network to distinguish between these two types of structures.

Lastly, in the case of the test patient featuring a binary mask and all employed models, the net-

work's attention is directed towards voxels that typically correspond to the locations of eyes and temporal lobes. This behaviour signifies that the network puts special attention to typical organs located in the cranial region; it probes these regions of the input to formulate its predictions. A similar behaviour is witnessed in the distance masks, although Conversely, such behaviour is absent in the training patient. This is expected from the network's tendency to overfit the training set: it doesn't need to explore these regions to yield a prediction.

This final observation regarding the network's capability to look for certain structures suggests that augmenting the dataset could provide the networks with an even deeper perception of the pertinent structures across all patients. This augmentation could facilitate the network in establishing relationships between angle values, the nature and position of represented OARs, and PTV characteristics (placement and geometry).

4.5.2. Underlying Trends

Despite the poor performance of individual outcomes, the apparent recognition of organ placement in the body of some networks is encouraging. Underlying trends that could suggest superior performance for a particular combination of model and mask type, if additional data was provided, might appear. To explore this possibility, histograms depicting the best mean loss values for each experiment across the entire grid searches for each model and input type pairing are presented in fig. 4.16 and fig. 4.17. This analysis aims to provide a broader perspective and identify a potential optimal combination. The results for the augmented dataset are also presented.

In these figures, the Inception model consistently demonstrates superior performance compared to the Full and Reduced models, irrespective of whether binary or distance masks are employed. Indeed, the overall histograms depict lower loss values, and the best-performing runs (represented by the leftmost bars on the histograms) exhibit reduced values compared to those of the Full and Reduced models.

The Distance masks present overall worse results for every model, despite the higher density of information they contain compared to binary masks. This might be related to the previous analysis regarding the ease with which the networks trained with binary masks identify organ boundaries.

Evaluating the impact of data augmentation across all networks is a complex task. A discernible enhancement is evident in the case of the Inception model when utilising distance masks, along with a slight improvement observed for the Full model with both mask types (medians of lower values). Conversely, the Reduced model exhibits worse training losses when employing distance masks, and testing losses worsen for the Reduced model with binary masks. In other scenarios, performance levels remain equivalent. A larger increase in the dataset would allow for improved analysis and a net distinction in model and mask-type performances.

4.5.3. Regularisations

Fully Convolutional Network

Numerous strategies are explored to reduce overfitting and enhance results. First, a fully convolutional neural network (FCN) [32] is constructed as an alternative to traditional max pool layers. FCNs are engineered for tasks necessitating the preservation of spatial information and generating dense pixel-level predictions. They exclusively employ convolutional layers, omitting fully connected layers and pooling layers. This configuration proves particularly useful in tasks like semantic segmentation, which revolves around classifying each pixel in an input image into a specific category. Contrariwise, max-pooling layers are favoured in tasks like image classification, where spatial information holds less significance compared to capturing higher-level features. Essentially, this attempt marks a shift in the paradigm of model philosophy. Unfortunately, this alteration did not result in a performance improvement, as illustrated in fig. 4.18.

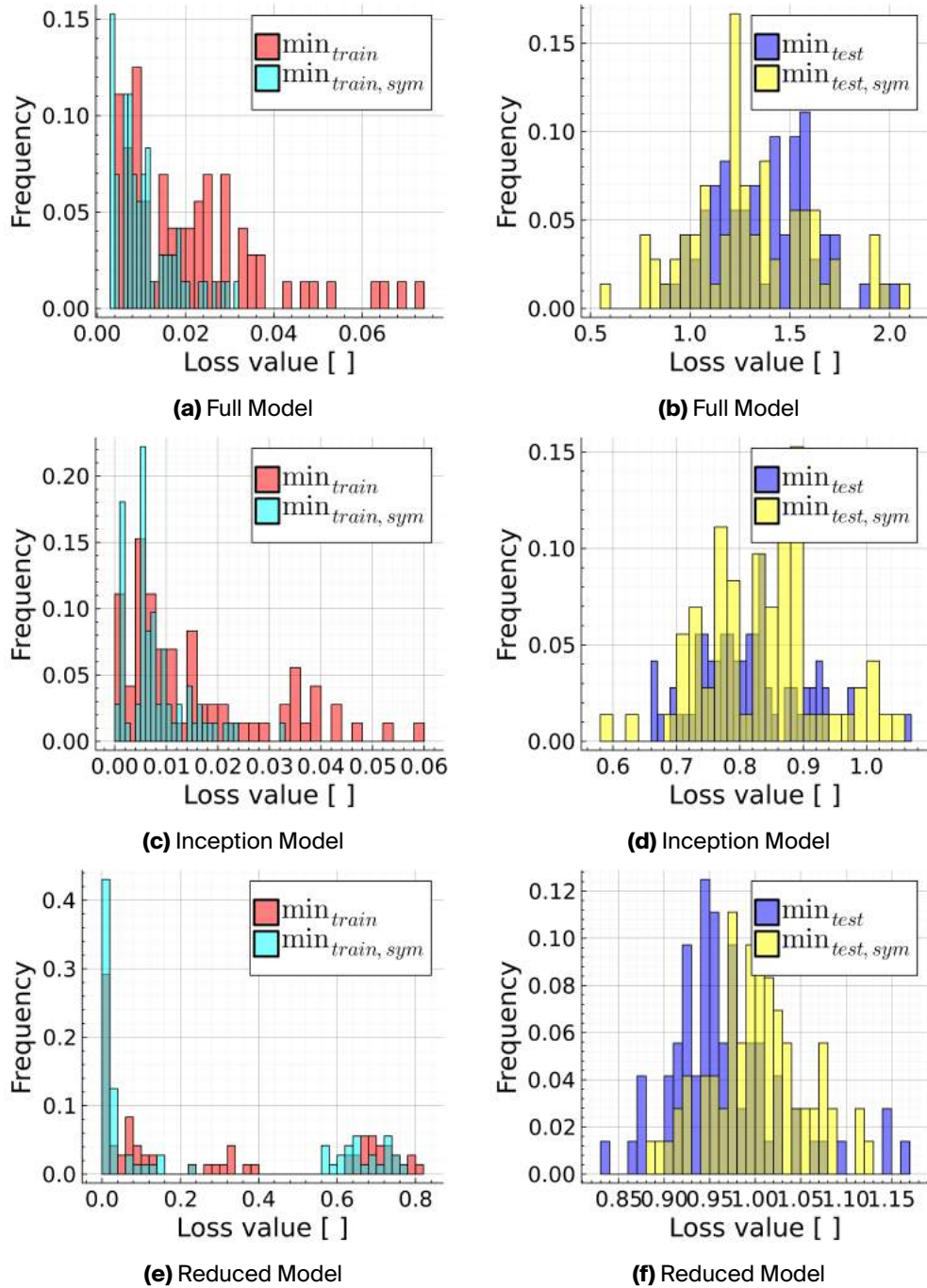


Figure 4.16: Histograms of best mean test loss values of each model trained in a supervised fashion. The models were trained with both the initial and augmented dataset. Both results are compared. The input masks are of type **Binary**.

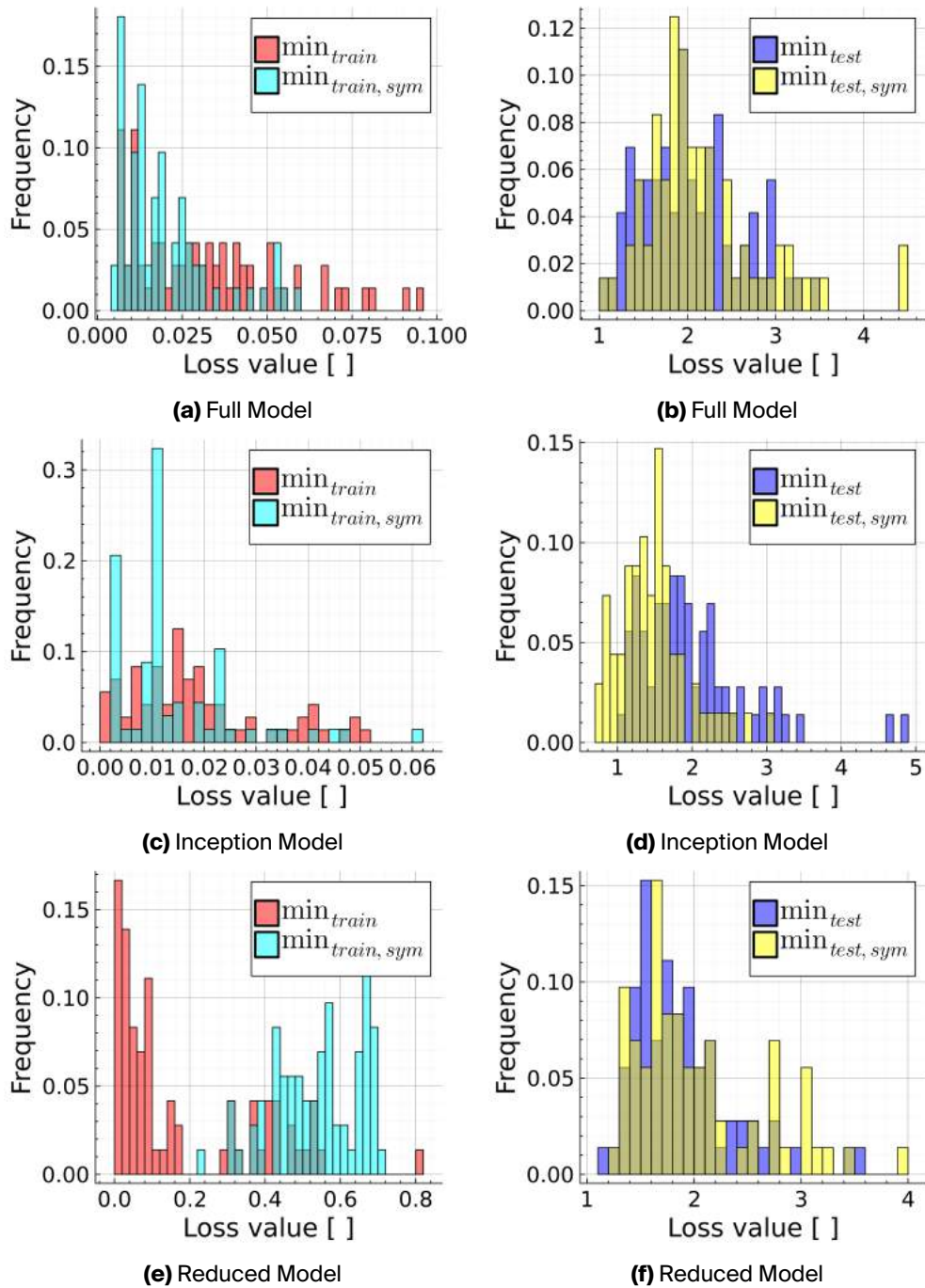


Figure 4.17: Histograms of best mean test loss values of each model trained in a supervised fashion. The models were trained with both the initial and augmented dataset. Both results are compared. The input masks are of type **Distance**.

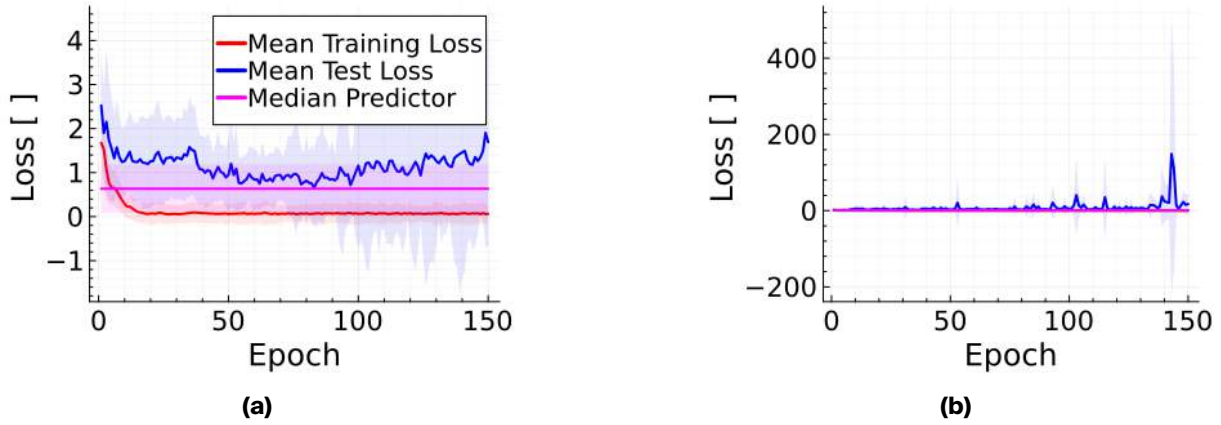


Figure 4.18: Training and Test losses evolution across all the epochs in the supervised training with the initial dataset with FCN. The solid lines represent the mean loss value across all patients in each set. The lesser saturated areas around the mean curves show the standard deviations from the mean for each set. Both mask types are used : **(a)** : Binary Masks, **(b)** : Distance Masks

The histograms of best mean test losses over the entire grid searches (identical ranges to other models) are presented in fig. 4.19. No improvement is to be noted compared to the three initial network architectures.

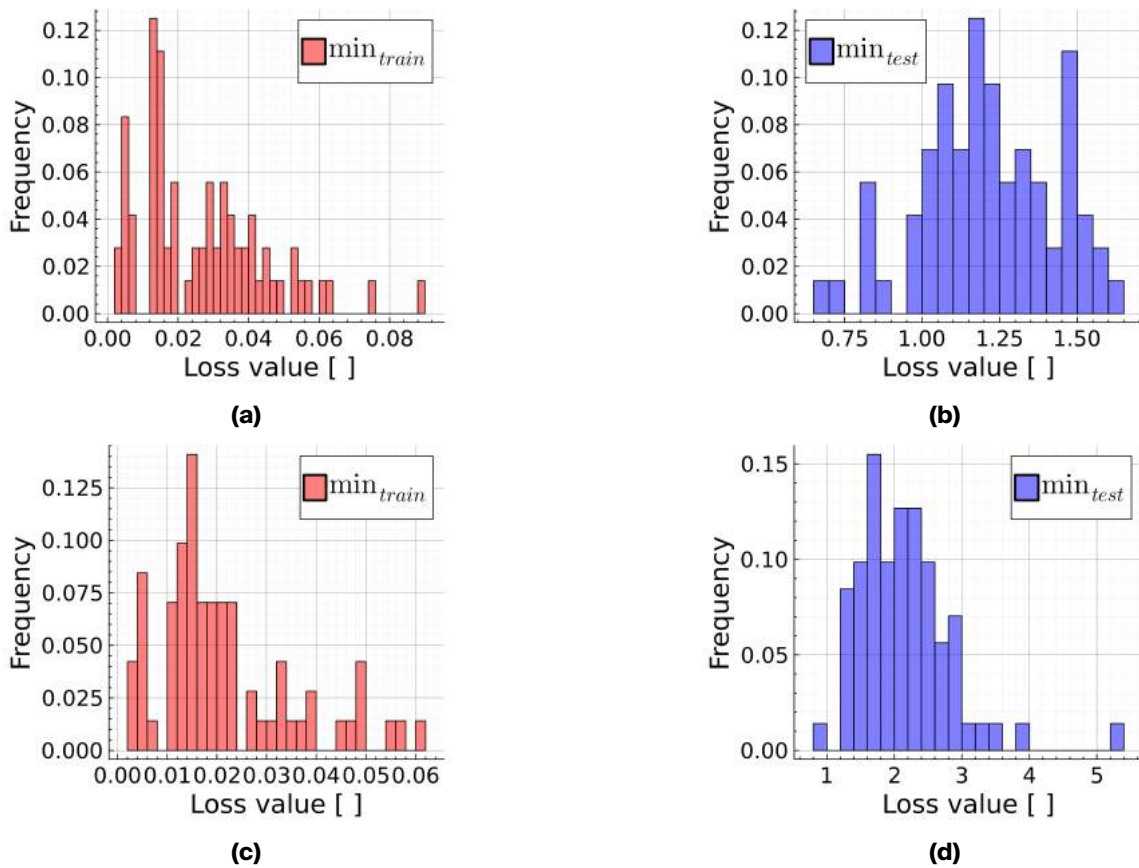


Figure 4.19: Histograms of final mean loss values of FCN model for training and test patients. The models were trained with the initial dataset, and both results can be compared.

(a) (b) : Binary Masks
(c) (d) : Distance Masks

Dropouts

Besides this attempt with FCN, an additional regularisation is tested : Dropouts. Dropouts involve randomly "dropping out" (deactivating) a portion of neurons or units within a neural network layer during each iteration of training. Consequently, the deactivated neurons do not contribute to computations for that specific iteration. Dropouts effectively introduce variability into the network, compelling it to grasp more robust and general features by discouraging reliance on specific neurons. Various dropout rates are tried and positioned at different layers within the networks. While this implementation did lower the convergence rate of the training losses, it did not yield improvements in overall best test losses.

Traditional ways to reduce overfitting are constructing a network with reduced complexity, using regularisations, batches and batch normalisations, Dropouts, and increasing the training set's size. All techniques have been attempted but the last, for which a small augmentation only was performed. From these statements, we infer that additional data is required for better performance of the CNN trained in a supervised way.

More data or more information ?

In addition to the already showcased paucity of data points, the networks' lack of performance could potentially stem from an insufficient perception of the actual objective they are meant to accomplish. Within the supervised approach, the singular quantity to be optimised is the fitting to the labels. It has been observed that the networks are able to perceive the geometric attributes, especially evident with binary masks. However, there remains a potential gap in comprehending the intricate relationship between the geometries and simple labels. By incorporating the dosimetric prescriptions within an unsupervised approach, there arises a prospect of achieving improved outcomes without the need for supplementary data. This idea is developed in the subsequent section of this study, relying heavily on the groundwork established in this first phase.

5

Unsupervised Learning

Following the establishment of a basic model through the supervised pipeline and the conclusion of a lack of information, the implementation of an unsupervised model becomes considerable. With this methodology, training does not explicitly rely on prior medical experiences, focusing solely on optimising the plan's score for any given patient rather than fitting labels. Further details about his plan score are given in the next sections.

For this new unsupervised pipeline, the network with parameters Θ first predicts angles, as illustrated in fig. 5.1. These angles are then employed to compute a dose distribution, passed to a novel loss function.

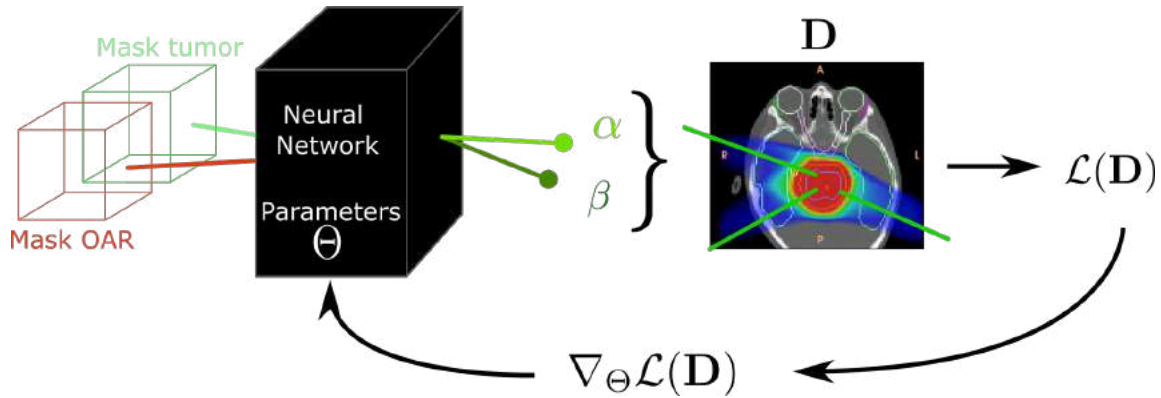


Figure 5.1: Unsupervised approach to the single field beam angles prediction problem. The loss function $\mathcal{L}$ is a combination of several terms, which most important term relies on dosimetry prescriptions of planners for each specific patient. (α, β) are the gantry angles and couch angles respectively, Θ is the set of all the learnable network parameters. The gradient $\nabla_{\Theta} \mathcal{L}(\alpha, \beta)$ is calculated in the backpropagation process of the weights update.

5.1. Methods

5.1.1. Dose Distribution Calculation

As illustrated in fig. 5.1, an on-the-fly dose distribution calculation is essential to evaluate the dose loss $\mathcal{L}_D$ value and compute gradients for network weights backpropagation. This computation, performed by CPT's proprietary software FionA, takes an average time of approximately 25 seconds if done on the fly. This duration varies based on hardware and patient's geometry complexity. However, this process is not adapted to rapid neural network learning. To circumvent this drawback and achieve acceptable training times, an alternative approach must be devised.

The new strategy relies on pre-computed dose distributions stored for each patient on selected angle combinations. In our case, we have the following set for both angles $\{\alpha_{\text{itp}}\}, \{\beta_{\text{itp}}\} \in \{10 \cdot k^\circ | k \in [0, 18] \subset \mathbb{N}\}$ (i.e. angles from 0 to 180 with increments of 10 degrees). From these dose distributions, a dose distribution corresponding to the output angles is interpolated as described in eq. (5.1).

$$\mathbf{D}(\alpha, \beta) = \frac{\beta_2 - \beta}{\beta_2 - \beta_1} \left(\frac{\alpha_2 - \alpha}{\alpha_2 - \alpha_1} \mathbf{D}_{11} + \frac{\alpha - \alpha_1}{\alpha_2 - \alpha_1} \mathbf{D}_{21} \right) + \frac{\beta - \beta_1}{\beta_2 - \beta_1} \left(\frac{\alpha_2 - \alpha}{\alpha_2 - \alpha_1} \mathbf{D}_{12} + \frac{\alpha - \alpha_1}{\alpha_2 - \alpha_1} \mathbf{D}_{22} \right) \quad (5.1)$$

where $\alpha_1 = \lfloor \alpha/10 \rfloor \cdot 10$ (resp. β_1) and $\alpha_2 = \lceil \alpha/10 \rceil \cdot 10$ (resp. β_2). $\alpha_1, \alpha_2 \in \{\alpha_{\text{itp}}\}$ (resp. $\{\beta_{\text{itp}}\}$). Thus $\alpha_2 - \alpha_1 = 10^\circ$. $\mathbf{D}_{ij}$ $i, j \in \{1, 2\}$ is the precomputed dose distribution at angles (α_i, β_j) ¹. If the prediction of the network exceeds the acceptable ranges, then they are rounded to the closest boundary of the interval (either 0 or 180), and the interpolation is performed using this value for the angle. Non-physical values are penalised as described in section 5.1.3

The dose distributions calculated during training are not deliverable because they are calculated using an interpolation of precalculated dose distributions, without running the usual ray casting algorithm for a given set of optimised spots. The precomputed spot lists for each beam angle are calculated using FlonA's default spot optimiser without any OAR sparing because it has the shortest runtime. However, our unsupervised learning approach is agnostic of the underlying spot weight optimisation algorithm and more powerful ones such as the one implemented in JulianA [23] could be used.

The computation of each patient's set of dose distributions demands approximately 2 hours and 30 minutes per patient. This constitutes a notable overhead, especially in the context of the current dataset (comprising 75 training patients). However, it is important to note that this computation is a one-time occurrence and can be parallelised across available computational resources (AROBASE parallelises over several patients). Storing these dose distributions is essential. Each standard dose distribution is in a 256x256x120 format, output by FlonA in a single-precision floating-point format (float 32), resulting in 30 MB files. Considering the need to compute 19x19x75 dose distributions, the storage requirement amounts to nearly 800 GB. To reduce memory use, a series of manipulations are performed. Initially, the type conversion to Half-precision floating-point format (float 16) occurs, which, in the context of dose distributions, does not yield significant information loss. Subsequently, doses below 0.5Gy are set to zero, a viable approach in line with CPT experts recommendations. This sets the stage for the final step: serialising dose distributions as 2D sparse arrays. The transfer from 3D to 2D array is achieved by flattening each axial slice (slices along the z-dimension) of the dose distribution into a 1D array and concatenating them. Since all dose distributions share the same size along the x and y dimensions, reversing this operation is possible. The object obtained is then stored as a sparse array, a functionality of the standard library in Julia [25]. This last manoeuvre can potentially reduce the total size by a factor of three or more, depending on the patient. For instance, considering training patient with ID=00, the regular dose distribution scales down from 29 MB to a mere 6.5 MB when serialised. Extrapolating to encompass all patients, this shift translates the total storage requirement from close to 800 GB to approximately 175 GB. This is doubled when considering the symmetric patients.

When it comes to training, the initial approach for the machine learning pipeline entailed loading all distributions, input masks, loss components, and configuration objects at the beginning of the procedure. However, this approach imposed a notable strain on the system's available RAM, particularly upon the inclusion of symmetric patients. Consequently, an alternative strategy was

¹Not to be mistaken for the D_{ij} in-fluence matrix used in most spot weight optimisation engines.

adopted, wherein dose distributions are accessed on the fly just prior to interpolation computations. While this introduces a marginal increase in computation time due to the additional I/O performed, it significantly conserves RAM resources.

5.1.2. Network Architectures

The models developed in chapter 4 are extended to the unsupervised approach. The purpose remains to consider different architectural complexities. It also allows comparison between both paradigms for the single field beam angles selection.

5.1.3. Loss function

Dose Loss $\mathcal{L}_D$

Our new algorithm optimises the plan by minimising $\mathcal{L}_D \in \mathbb{R}$, a scalar loss function built upon dosimetry constraints on VOIs set by clinicians and established for JulianA [23]. The equation of this loss is given in eq. (5.2).

$$\mathcal{L}_D = \alpha_{\text{ideal}} \mathcal{L}_{\text{ideal}} + \alpha_{\text{hom}} \mathcal{L}_{\text{hom}} + \alpha_{\text{min}} \mathcal{L}_{\text{min}} + \alpha_{\text{max}} \mathcal{L}_{\text{max}} + \sum_o \alpha_{\text{oarmean}} \mathcal{L}_{\text{mean}}(o) + \alpha_{\text{oarmax}} \mathcal{L}_{\text{max}}(o) \quad (5.2)$$

Citing the original publication [23]:

$\mathcal{L}_{\text{ideal}}$ [penalises] deviations from an ideal dose distribution that is inspired by the physics of pencil beams. $\mathcal{L}_{\text{hom}}$ [penalises] variance within the targets. $\mathcal{L}_{\text{min}}$ ensures a sufficient target coverage. $\mathcal{L}_{\text{max}}$ [ensures] that no optimisation point receives more dose than 105% of the prescribed dose. $\mathcal{L}_{\text{mean}}$ and $\mathcal{L}_{\text{max}}(o)$ [...] penalise violations of OAR constraints.

The strength of this loss function lies in its independence from prior plans and ensures that each plan's quality is judged in a consistent way. As stated earlier, human planners are at the risk of making biased decisions relying on their own personal experience. A plan might thus be optimised differently from one facility to another. The objective weights in the loss function have default values that do not need change for the patients in our dataset. However, they could be modified for fine-tuning by the planners. The terms of the loss as well as the weights are parsed from a constraints file by JulianA, which makes the loss patient-specific.

Due to the technical limitations of Flux with the backpropagation of the network weights, a variant of this dose loss is used in the experiment. This modified loss is the following :

$$\mathcal{L}_D^*(\alpha, \beta) = \partial_\alpha \mathcal{L}_D \cdot \alpha + \partial_\beta \mathcal{L}_D \cdot \beta \quad (5.3)$$

The complete explanation of this manipulation is explained in appendix D. This loss solves the problem of the computation of the derivatives of the loss with respect to the network weights, performed using reverse mode automatic differentiation in Flux.jl. A subtlety about this loss function is that it is convex in the dose distributions, but highly non-convex with respect to the field angles².

Additional terms in the loss

The sole dose loss is not enough to ensure acceptable results.

²The composition of a convex function with a non-convex function does not necessarily yield a convex function.

Penalty on angles

First, we need to ensure the physicality of the angles predicted. Indeed, with $\mathcal{L}_D$ as the sole contribution to the loss function, the networks output real scalars, which might not fit the ranges described in fig. 3.2. A penalty term for predictions outside of these ranges must thus be added to enforce physicality of the results. The penalty used is given in eq. (5.4) and illustrated in fig. 5.2

$$\begin{aligned}\mathcal{L}_\pi(\alpha, \beta) &= \mathcal{L}_\pi(\alpha) + \mathcal{L}_\pi(\beta) \\ &= \exp(\gamma(\alpha - \pi - \varepsilon)) + \exp(-\gamma(\alpha + \varepsilon)) \\ &\quad + \exp(\gamma(\beta - \pi - \varepsilon)) + \exp(-\gamma(\beta + \varepsilon))\end{aligned}\quad (5.4)$$

where $\gamma = 3$ and $\varepsilon = 0.1$. A visual representation is given in fig. 5.2.

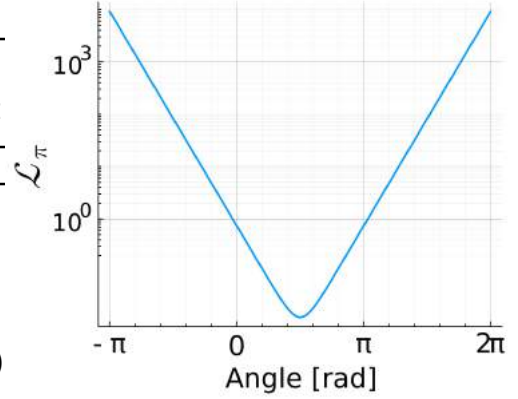


Figure 5.2: $\mathcal{L}_\pi(x)$ loss function. $\gamma = 3$ and $\varepsilon = 0.1$. This term ensures the physicality of the output angles of the CNNs trained in an unsupervised paradigm.

Regularisation

Similarly to the supervised approach, L_1 and L_2 regularisation terms are available and tested.

Total Loss Function

The final loss function used for the training of the network assembles all three components mentioned previously :

$$\mathcal{L}_{\text{total}}(\alpha, \beta) = [\mathcal{L}_D^* + \lambda_\pi \mathcal{L}_\pi + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}](\alpha, \beta) \quad (5.5)$$

5.1.4. Normalisation of the Loss

When representing the performance of the training, we are more interested in the $\mathcal{L}_D$ loss than the combination of all the terms in eq. (5.5). Indeed, this term embodies how effectively the network solves the beam angle selection problem. However, due to the difference in prescriptions for each patient, the $\mathcal{L}_D$ loss values' orders of magnitude can largely vary from one patient to another. This is seen in fig. 5.3. To compensate for this effect in the visualisations and allow for a better representation of the learning over the course of the training, each patient's loss undergoes a min-max normalisation. To this effect, minimum and maximum losses are computed for each individual patient by evaluating $\mathcal{L}_D$ at all their corresponding precomputed doses. The patient-wise normalisation is thus the following eq. (5.6).

$$\mathcal{L}_D' = \frac{\mathcal{L}_D - \min(\mathcal{L}_D)}{\max(\mathcal{L}_D) - \min(\mathcal{L}_D)} \quad (5.6)$$

5.2. Training Modalities

In a similar way to the supervised approach, the plots of the evolution of the normalised training and validation losses for each pairing of model and input type are presented in fig. 5.4. Again, grid searches are performed across specified hyperparameter ranges. The Adam optimiser [22] was used for the update rule of the network weights. For the unsupervised learning approach, these ranges are $\eta \in \{10^{-4}, 5 \cdot 10^{-5}, 10^{-5}\}$, $\lambda_{\text{reg}} \in \{10^{-4}, 5 \cdot 10^{-5}, 10^{-5}\}$, $\lambda_\pi \in \{10^{-4}, 5 \cdot 10^{-5}, 10^{-5}\}$, and batch sizes $b \in \{1, 2, 4\}$. These ranges were determined based on initial empirical attempts. The choice of relatively small batch sizes is the result of the limited size of the dataset. The training was limited to 150 epochs.

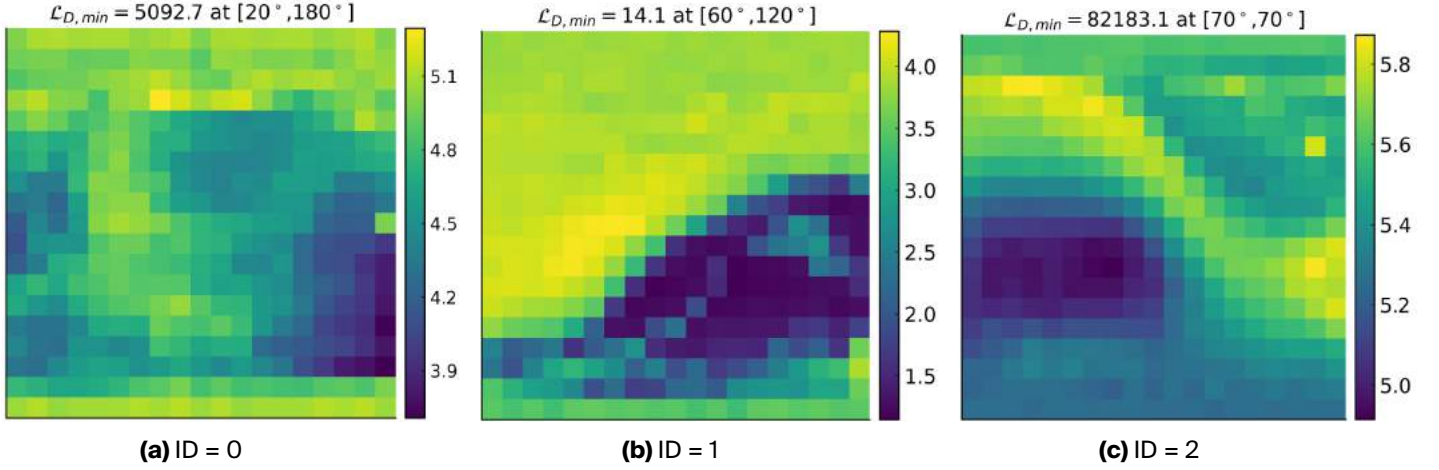


Figure 5.3: Dose loss $\mathcal{L}_D$ landscape for single field dose distributions of training patients. The colorbar scale is logarithmic. The y-axis represents $\alpha \in [0, 180^\circ]$ and x-axis $\beta \in [0, 180^\circ]$. Each coloured tile corresponds to a range of 10° for both α and β , corresponding to where the dose distributions are calculated for assessing the loss. The minima are indicated in the titles and showcase the variation in loss values across patients.

In the time frame of this study, only the initial dataset could be used due to the computational burden of each grid search. Nonetheless, grid searches on the augmented data are possible thanks to the framework developed for the experiment and can be performed in future studies.

5.3. Results

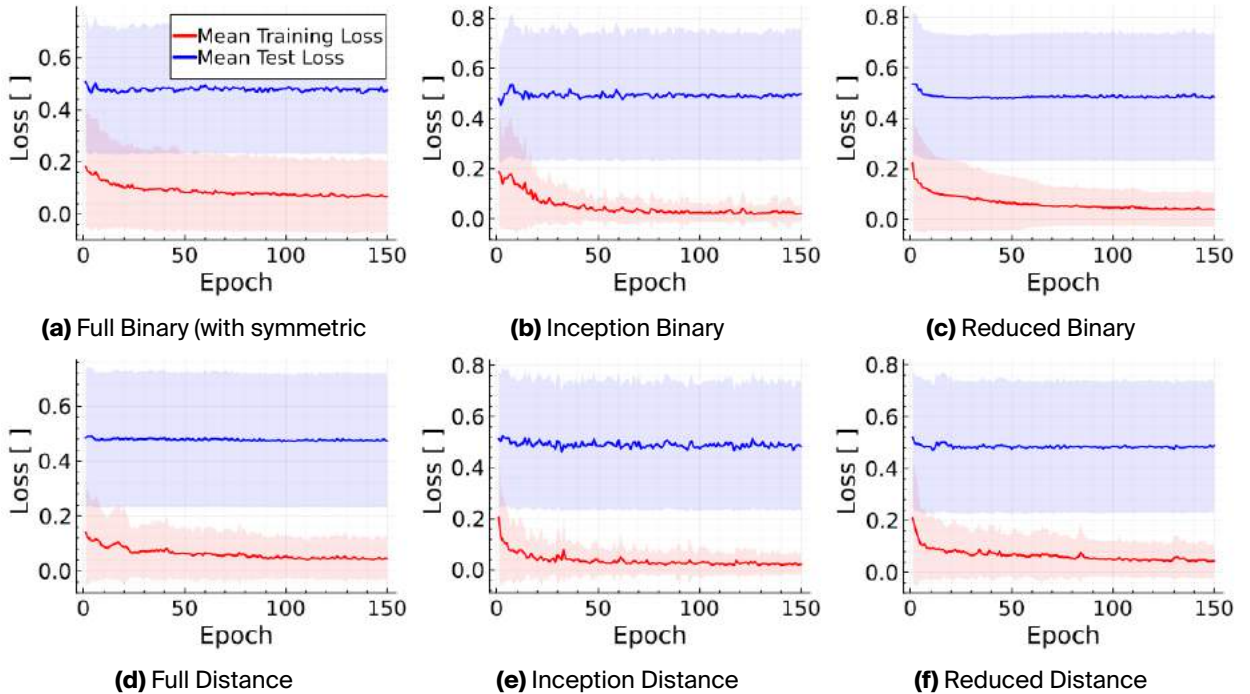


Figure 5.4: Normalised Training loss and Test loss $\mathcal{L}_D'$ evolutions across all the epochs in the supervised training. The lesser saturated areas around the mean curves show the standard deviations from the mean for each set.

In these illustrations, the standard deviations (or uncertainty ranges) are represented symmetrical with respect to the mean curve (they are not one-sided standard deviations). Thus, the uncertainty ranges might include negative loss values, which are not realistic. The most probable explanation is that the losses for some single patients remain large throughout the whole training process. These large losses might have a high impact on the standard deviation value and extended the range. However, we must mention that small negative loss values are technically feasible with our framework. Indeed, the minimum values used for loss normalisation (see eq. (5.6)) are computed by constructing dose loss landscapes on a discretised domain. Indeed, as stated earlier, dose distributions were evaluated at every angle pair with increments of 10 degrees; exact global minima are thus not expected. The networks can in principle converge to regions with values lower than the minimum computed via interpolation for certain patients.

The hyperparameters yielding the results are presented in table 5.1.

Model Mask Type	Full Binary	Inception Binary	Reduced Binary	Full Distance	Inception Distance	Reduced Distance
η	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$
Regularisation	L_1	L_1	L_1	L_1	L_1	L_2
λ_{reg}	$5.0 \cdot 10^{-5}$	$5.0 \cdot 10^{-5}$	$5.0 \cdot 10^{-5}$	$5.0 \cdot 10^{-5}$	$5.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-4}$
λ_{π}	$1.0 \cdot 10^{-5}$	$5.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-4}$	$5.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-4}$
Batch Size	1	1	1	2	1	1

Table 5.1: Best hyperparameters combinations obtained after grid search with the unsupervised learning paradigm. The symmetric patients are not considered.

Despite a slight initial reduction that is apparent for all models and mask types, none of the networks demonstrate convergence in the test loss. In terms of the training loss and the occurrence of overfitting, the Inception model displays superior results. Specifically, both the mean and standard deviations across the training set converge to lower values compared to the other models.

5.4. Discussion

The networks seem to overfit, learning to generalise on the training set only. We investigate why that might be and propose optimisations.

5.4.1. Gradient Based Activation Maps

Similarly to the supervised learning paradigm, gradient-based activation maps are computed to assess whether the networks perceive the input geometries adequately. In that regard, training patient with ID=25 and test patient with ID=0 are used again. The masks are presented in fig. 4.10 and fig. 4.11. The resulting maps are represented for two selected patients in fig. 5.5 fig. 5.6 for binary masks, and in fig. 5.7 and fig. 5.8 for the distance masks. Further results for the Inception model and the Reduced Model are presented in appendix E.

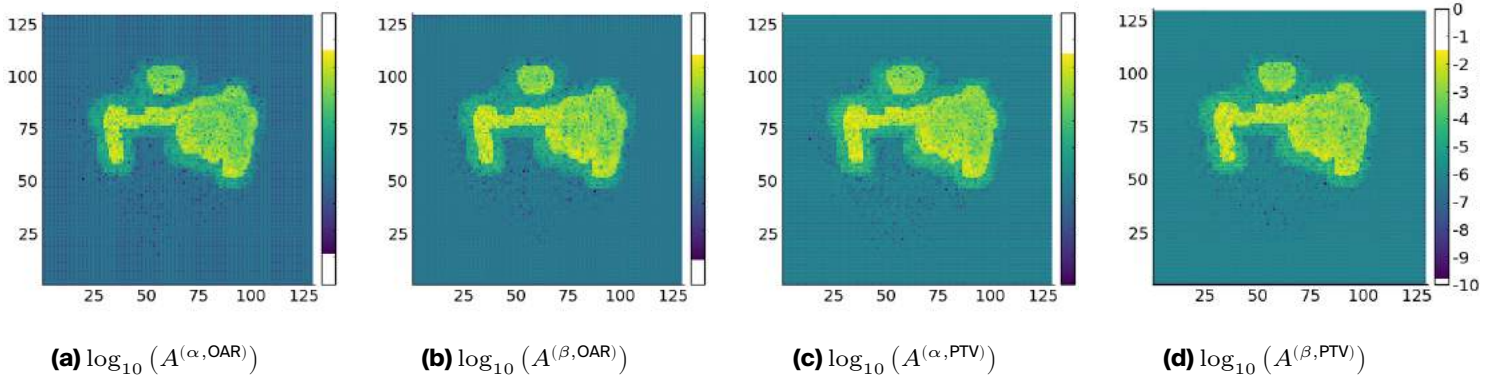


Figure 5.5: Gradient-based activation maps when predicting the angles for training patient with ID=25. The model selected is the unsupervised Full model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs. Slice 50 is represented.

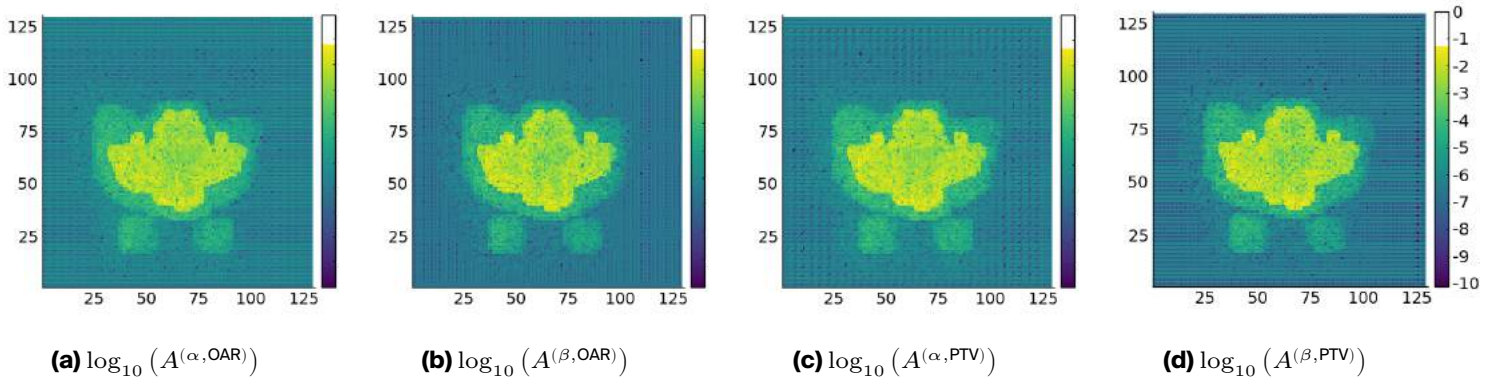


Figure 5.6: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the unsupervised Full model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs. Slice 50 is represented.

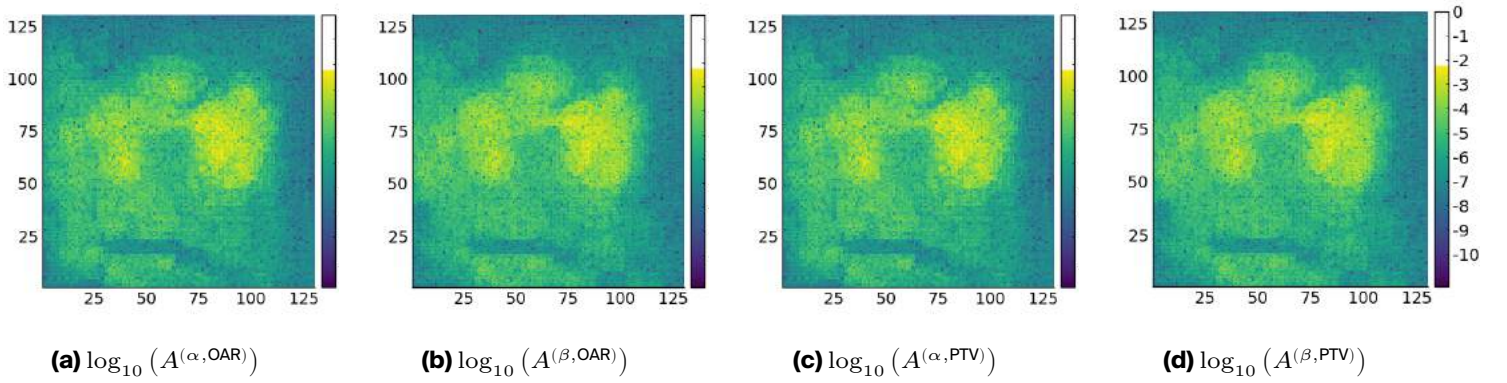


Figure 5.7: Gradient-based activation maps when predicting the angles for training patient with ID=25. The model selected is the unsupervised Full model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs. Slice 50 is represented.

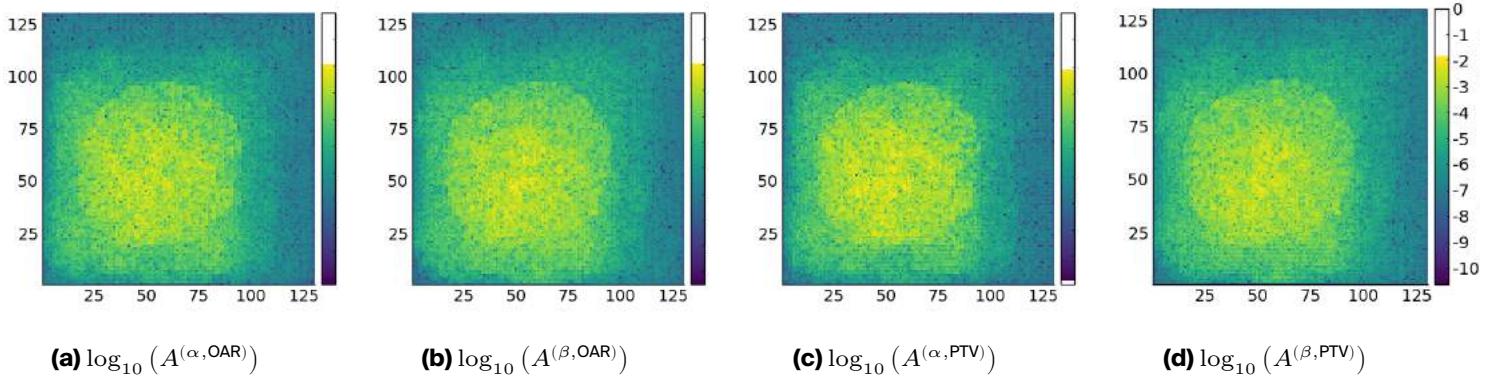


Figure 5.8: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the unsupervised Full model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs. Slice 50 is represented.

The activation maps might again offer valuable insights. Similarly to the supervised approach, the networks relying on binary masks exhibit a clear focus on the areas within the masks that carry the most significant information. There is a distinct boundary outlining the organs, clearly visible in the maps. When analysing the distance masks for the training patient, although discerning this boundary is more challenging than in the supervised case, it remains sharper in comparison. Nevertheless, the networks still tend to allocate their attention to regions that are less relevant. This distinction between informative and less relevant areas is less evident when considering the distance masks for the test patient. Further elaboration on this result is provided later in the discussion.

Similarly to the supervised approach, the networks do not seem to distinguish OARs and PTV mask, further questioning the effectiveness of separating OARs and PTV into distinct channels.

Lastly, for the test patient and all models employing binary masks, the network's focus is directed towards input positions that correspond to typical locations of eyes and temporal lobes. This behaviour indicates that the network has captured the structural characteristics of human cranial organs. In terms of distance masks, the network's attention seems to encompass the entire skull structure rather than specific organs, obscuring the voxels of interest. This observation potentially explains the results seen in Figure 5.4 for distance masks; the networks fail to concentrate on the appropriate regions to formulate their predictions.

This final observation regarding the network's ability to discern certain structures strengthens the proposition of augmenting the dataset to foster a more profound understanding of the pertinent structures across all patients. Such augmentation could enable the networks to establish connections between angles, dosimetric prescriptions, and the nature and position of represented OARs, as well as the characteristics of the PTV (placement and geometry).

5.4.2. Underlying Trends

Similarly to the supervised approach, histograms depicting the overall best mean loss values of each ML experiment across the entire grid searches for each model and input type pairing are presented in fig. 4.16 and fig. 4.17. This analysis aims to provide a broader perspective and identify a potential optimal combination of model and input type.

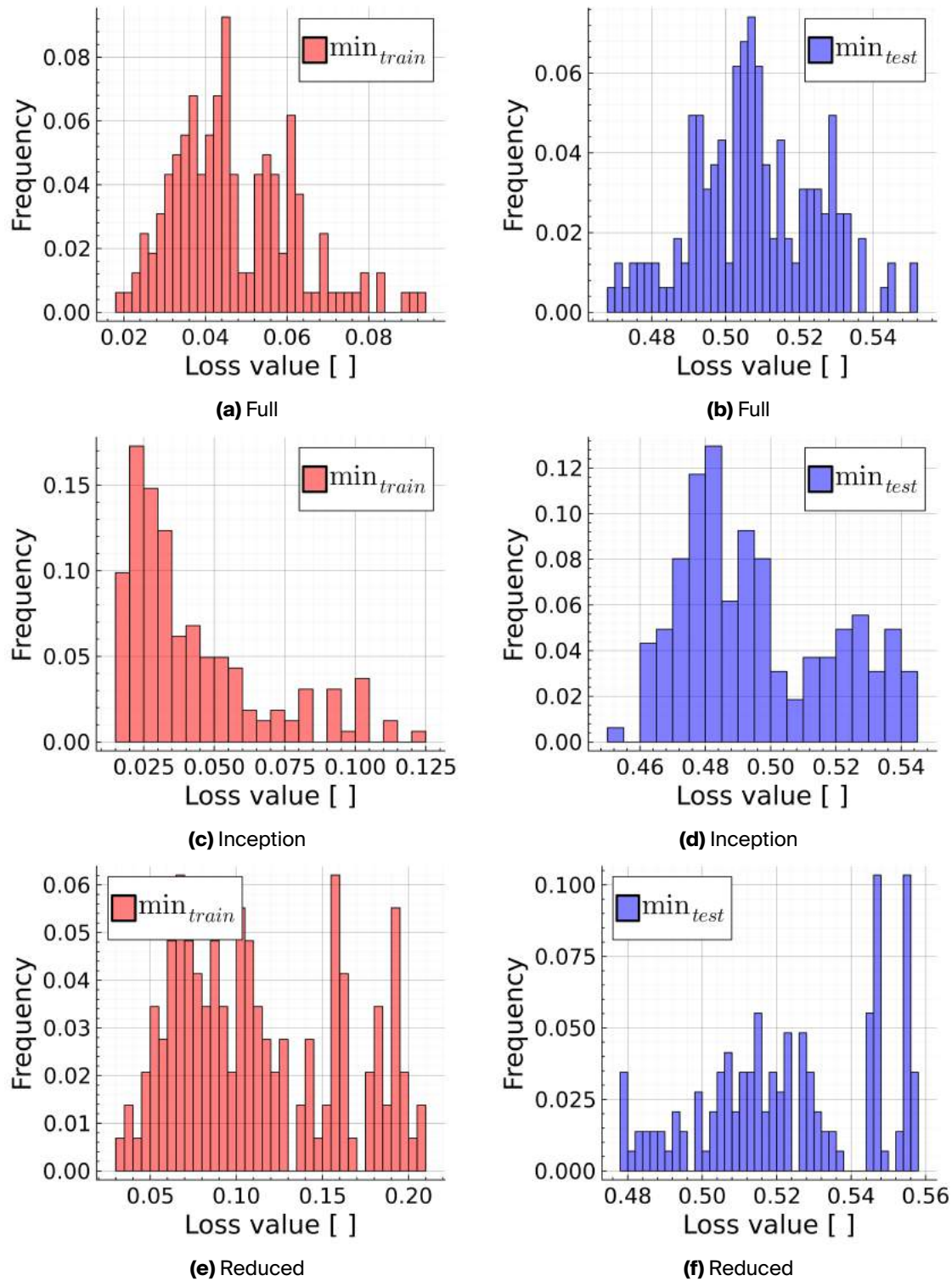


Figure 5.9: Histograms of best mean test loss values of each model trained in an unsupervised fashion. The models were trained with both the initial and augmented dataset. Both results are compared. The input masks are of type **Binary**.

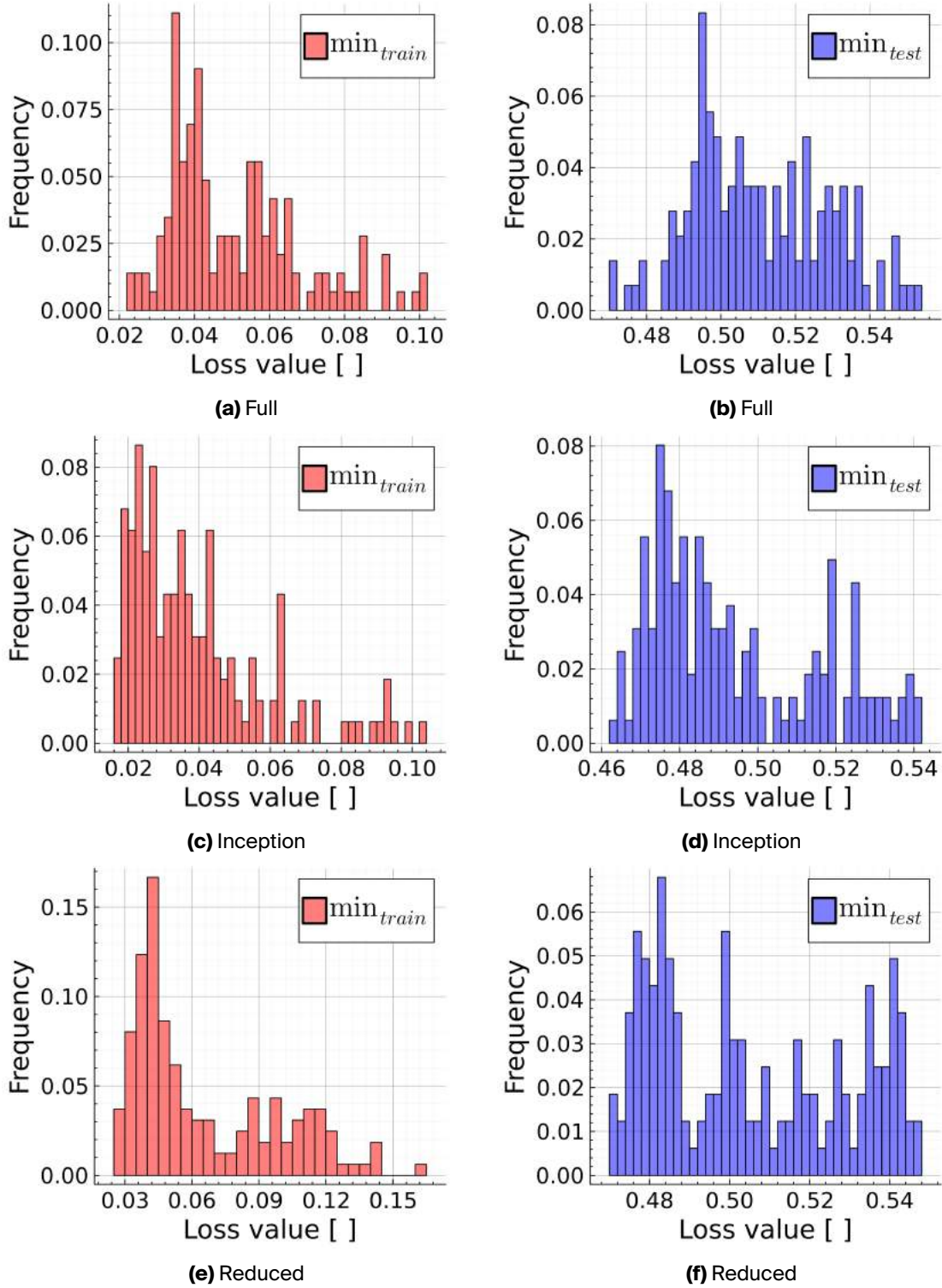


Figure 5.10: Histograms of best mean test loss values of each model trained in an unsupervised fashion. The models were trained with both the initial and augmented dataset. Both results are compared. The input masks are of type **Distance**.

A first observation is that not one or the other mask type yields better results. However, despite not converging in the test losses, the Inception models yield overall better results. The histograms of best values are centred around a lower value than for the Full and Reduced model. Despite

this observations, conclusions regarding overall performances of models would fallacious, as the divergence from the 0.5 value is minimal in any case.

5.4.3. Regularisations

FCN and dropouts were attempted sporadically (i.e for a restricted number of architecture and mask type combinations) but these preliminary results did not yield improvement in performance, similar to the supervised pipeline. Further analysis was not performed but is feasible with AROBASE in further research.

Since this is the first implementation of such an approach for this problem, all the parameters and choices might not be optimal. However, we would expect a minimum performance anyway, which is not reached. Deep learning might not be the right solution. Nevertheless, an extensive framework was developed in the frame of this study, and can be used again if new data becomes available or different inputs, models, losses or optimisers are to be used in further research.

6

Three Fields Dose Descent

6.1. Loss Landscape

Both the supervised and unsupervised Convolutional Neural Network (CNN) approaches did not successfully address the task of predicting a single field's beam angles with the amount of data at hand. This outcome suggests the necessity for a rollback to a less intricate strategy. In pursuit of a method that is not reliant on previous treatment plans, the importance of the dose loss function $\mathcal{L}_D$ persists. To underline this idea, we can map the loss values attributed to all dose distributions generated by the possible combinations of angles. In the case of a single-field dose distribution, a rapid identification of the most optimal angles' arrangement that corresponds to the global minimum of the loss function can be done. This is illustrated in fig. 6.1.

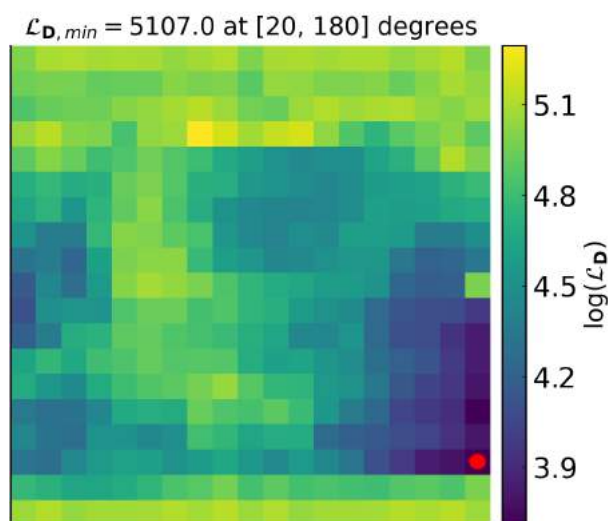


Figure 6.1: Dose loss $\mathcal{L}_D$ landscape (logarithmic scale) for a 1-field treatment delivered to a training patient (ID = 0). The minimum value is located at the red dot : $(\alpha, \beta) = (20^\circ, 180^\circ)$ - which indicates the best range of angles - $\alpha \in [20^\circ, 30^\circ]$ and $\beta = 180^\circ$. The y-axis represents $\alpha \in [0^\circ, 180^\circ]$ and x-axis $\beta \in [0, 180]$. Each coloured tile corresponds to a range of 10° for both α and β .

This strategy could be coined the "brute force" method, where the dose landscape is constructed by computing all possible dose distributions resulting from all angle pairs, along with their corresponding loss function values. From these values, the optimal combination of beam angles and the minimum loss value are deduced. Although this method does not necessitate prior patient

data to generate optimal beam angles tailored to the current patient, it is constrained by the precision with which it is devised (10 degrees increment for each angle in fig. 6.1).

Leveraging the available hardware resources (as outlined in section 3.4), an average duration of approximately 150 minutes¹ is required for sequentially computing all the essential dose distributions for the brute force approach. When considering multiple fields optimisation (involving more than one field for patient treatment), not only the computed dose distribution combinations but also their corresponding loss values must be assessed. This combination process grows exponentially in complexity as the number of fields increases².

Treatment planning is an iterative process, demanding swift time to solution at each iteration. The brute force approach falls short of being optimal despite its ability to reach a global minimum. The ideal algorithm should provide suitable beam angles within a reasonable time frame, necessitating fewer dose distribution calculations. With this goal in mind, a gradient descent method tailored to yield satisfactory beam angles for three fields is introduced – aptly named the Three Fields Dose Descent (3FDD) algorithm.

6.2. Algorithm

A conventional way to automatise the search of a global minimum is to harness convex optimisation algorithms [33]. Our goal is to obtain optimal beam angles from the prescriptions of the clinicians only. As most of the patients in the training and testing set are treated with three fields (see section 3.1.1), we decided to implement an optimisation for this number of fields. A modified gradient descent is proposed, with carefully chosen learning rates and number of steps.

6.2.1. Update Rule and Modified Gradient

The gradient descent update rule adapted to our problem is eq. (6.1)

$$\vec{\alpha}^{(k+1)} = \vec{\alpha}^{(k)} - \eta_k \mathbf{Grad}_\lambda [\mathcal{L}_D(\vec{\alpha}^{(k)})] \quad 0 \leq k \leq N \in \mathbb{N} \quad (6.1)$$

where k the current step of the algorithm, $N \in \mathbb{N}^*$ the number of steps until completion of the gradient descent algorithm, set to 15. $\vec{\alpha}^{(k)} = (\alpha_0, \beta_0, \alpha_1, \beta_1, \alpha_2, \beta_2)^{(k)}$ where $\alpha_i, \beta_i \in [0^\circ, 180^\circ]$ with $i \in \{0, 1, 2\}$ the angles of the 3 different fields at step k , $\eta_k \in \{20, 10, 5\}$ a decaying learning rate. The initial value is $\eta_0 = 20$ and it decays every 5 steps ($\eta_5 = 10$ and $\eta_{10} = 5$). $\mathcal{L}_D$ is the dose loss [23] evaluated at dose distribution $\mathbf{D}(\vec{\alpha})$.

In eq. (6.1), a modified gradient operator $\mathbf{Grad}_\lambda$ is introduced, defined in eq. (6.2)

$$\mathbf{Grad}_\lambda [\mathcal{L}_D(\vec{\alpha}^{(k)})] = \frac{\nabla_{\vec{\alpha}^{(k)}} \mathcal{L}_D(\vec{\alpha}^{(k)}) + \lambda \nabla_{\vec{\alpha}^{(k)}} \mathcal{L}_\pi(\vec{\alpha}^{(k)})}{\|\nabla_{\vec{\alpha}^{(k)}} \mathcal{L}_D(\vec{\alpha}^{(k)}) + \lambda \nabla_{\vec{\alpha}^{(k)}} \mathcal{L}_\pi(\vec{\alpha}^{(k)})\|_2} \quad (6.2)$$

where $\lambda \in \mathbb{R}^+$ is a tunable penalty strength, set to 1, $\nabla_{\vec{\alpha}^{(k)}} \mathcal{L}_\pi(\vec{\alpha}^{(k)})$ is the gradient of a penalty of the angles, as defined in eq. (5.4). It ensures the gradient descent remains in the $[0^\circ, 180^\circ]$ range for every predicted angle.

The gradient calculation is divided into two parts, a gradient of the loss function with respect to the dose distribution and a Jacobian matrix of the dose distribution with respect to the angles:

$$\nabla_{\vec{\alpha}} (\mathcal{L}_D) = \nabla_{\mathbf{D}} [\mathcal{L}_D]^\top \mathbf{J}_{\vec{\alpha}} \mathbf{D} \quad (6.3)$$

¹It's important to note that $\alpha, \beta \in [0^\circ, 180^\circ]$ with fixed increments of 10 degrees, resulting in $19 \cdot 19 = 361$ dose distributions to compute. With an average time of 25 seconds for each dose distribution, the given estimate holds true. The computation time for each patient varies based on their unique geometry.

²For instance, with 3 fields and 19 potential values for each angle (2), the total number of angle combinations equals $(19^2)^{\text{\#fields}} = 47,045,881$.

The modified gradient is normalised, which allows control over the extent each step follows the steepest direction. In other words, the step size is entirely controlled and determined by the learning rate.

6.2.2. Dose Superposition and Single Field Optimisation

The dose distribution $\mathbf{D}(\vec{\alpha})$ passed to the dose loss is a superposition of the SFO distributions of each field; the spots are optimised separately for each field. An element $\mathbf{D}_{ijk}(\vec{\alpha})$ of the total dose distribution is thus defined as eq. (6.4):

$$\mathbf{D}_{ijk}(\vec{\alpha}) = \frac{1}{3} \sum_{n=0}^2 \mathbf{D}_{ijk}(\alpha_n, \beta_n). \quad (6.4)$$

where $\mathbf{D}_{ijk}(\alpha_n, \beta_n)$ is the interpolation defined in eq. (5.1). Again, the spot weight optimisations and dose calculations using CPT's in-house Treatment Planning System FlonA. No OAR sparing is considered in the spot weight optimisation because the FlonA optimiser needs human input to spare OARs. While neglecting OARs completely is unacceptable to generate a clinical plan, it is considered a suitable approximation to find good beam arrangements. A more advanced spot optimisation algorithm can be employed once the final beam arrangement is found.

The individual dose distributions of each field computed for interpolation are saved at each step and shared between fields for later use. Thus, if a dose distribution is required for interpolation by a certain field, it can be used by another field in later steps if necessary. The fact that the global dose distribution is a superposition of independent fields allows for this sharing of dose distributions, drastically reducing the computations required for convergence of the gradient descent.

6.2.3. Educated Initialisation

At first step $k = 0$ of the gradient descent, each angle must be initialised. An educated initialisation is the following :

$$\alpha_i^{(0)}, \beta_i^{(0)} = (90 + 20 \cdot (2X - 1), 90 + 20 \cdot (2X - 1)) \quad (6.5)$$

where $i = 0, 1, 2$, $X \sim U(0, 1)$ a random variable following a continuous uniform distribution on interval $[0, 1] \subset \mathbb{R}$. This means the initial values for each angle pair are constricted to a square of side length $l = 40^\circ$ centred on $(90^\circ, 90^\circ)$ as illustrated in fig. 6.2.

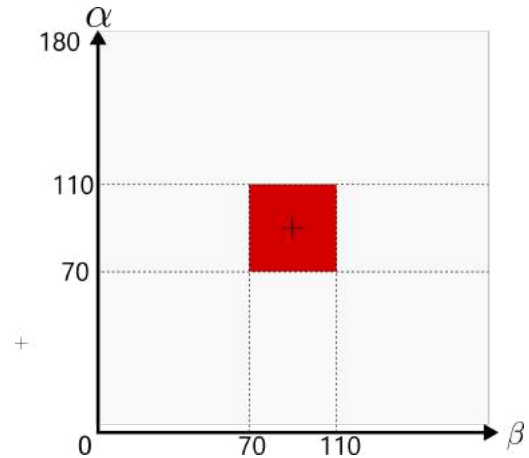


Figure 6.2: Initial conditions of the 3FDD algorithm for each single field. The possible values are delimited by the red square.

We anticipate that this initialisation will assist in the prevention of beam configurations originating from the lower part of the cranium. Nevertheless, it will simultaneously permit the algorithm to investigate alternative sectors within the loss manifold. A field emerging from the lower regions of the patient might traverse either through the oral cavity or along the entire length of the body. In the former scenario, clinicians promptly dismiss it, while in the latter, it becomes nonsensical, given that the proton beam cannot advance to the tumour site.

6.2.4. Compare like with like : SFO and MFO

When considering the final outcomes, one can ask the question: Is it reasonable to compare the predicted dose distributions and the reference clinical plans in the first place? Indeed, many additional optimisations are performed by clinicians once the beam arrangements are found. To circumvent this comparison issue, the reference plan and the predicted plans are (re)calculated in two different ways. The first way computes and optimises the dose distributions of each field with SFO and averages them out in a single distribution. The second approach accounts for each of the three fields and optimises the total resulting dose distribution : MFO. Computing the dose distributions resulting from these two approaches strips the clinical dose distribution from any additional optimisations that are not applied to the predicted fields, thus making them comparable. The dose distributions and Dose Volume Histograms are computed and compared for the five following scenarios (five plans): Clinical Reference dose distribution, 3FDD dose distributions in SFO fashion and in MFO fashion, and SFO and MFO dose distributions computed from the angles of the prescribed fields in the clinical reference plan. We will mainly compare the two SFO computations, as the 3FDD updates the angles based on this paradigm.

6.3. Results

6.3.1. Patients Selection

Six patients are included in this analysis. All are extracted from the dataset used for the previous CNN approach (see section 3.1.1 for more general information). These patients are selected to showcase the method's capacity to accommodate diverse pathologies, patient anatomies, and, consequently, a range of challenges in treatment planning. The particulars of these patients are given in table 6.1. For a comprehensive compilation of the prescriptions of these patients, please refer to appendix F.

type	ID	Condition	Target Dose [Gy]
Train	00	Chordoma - skullbase	74.0
Train	01	Ependymoma	59.4
Train	09	Basalioma (Basal-cell carcinoma)	64.0
Train	10	Glioma low grade	54.0
Train	15	Carcinoma - Head and Neck	74.0
Train	33	Meningioma atypical	62.0

Table 6.1: Patients considered in the Three Fields Dose Descent Approach to beam angles selection. For further information about these cancer types, refer to [34].

6.3.2. Convergence Characteristics

First, we demonstrate the ability of the algorithm to reach minima for all the patients considered. The results are presented in fig. 6.3.

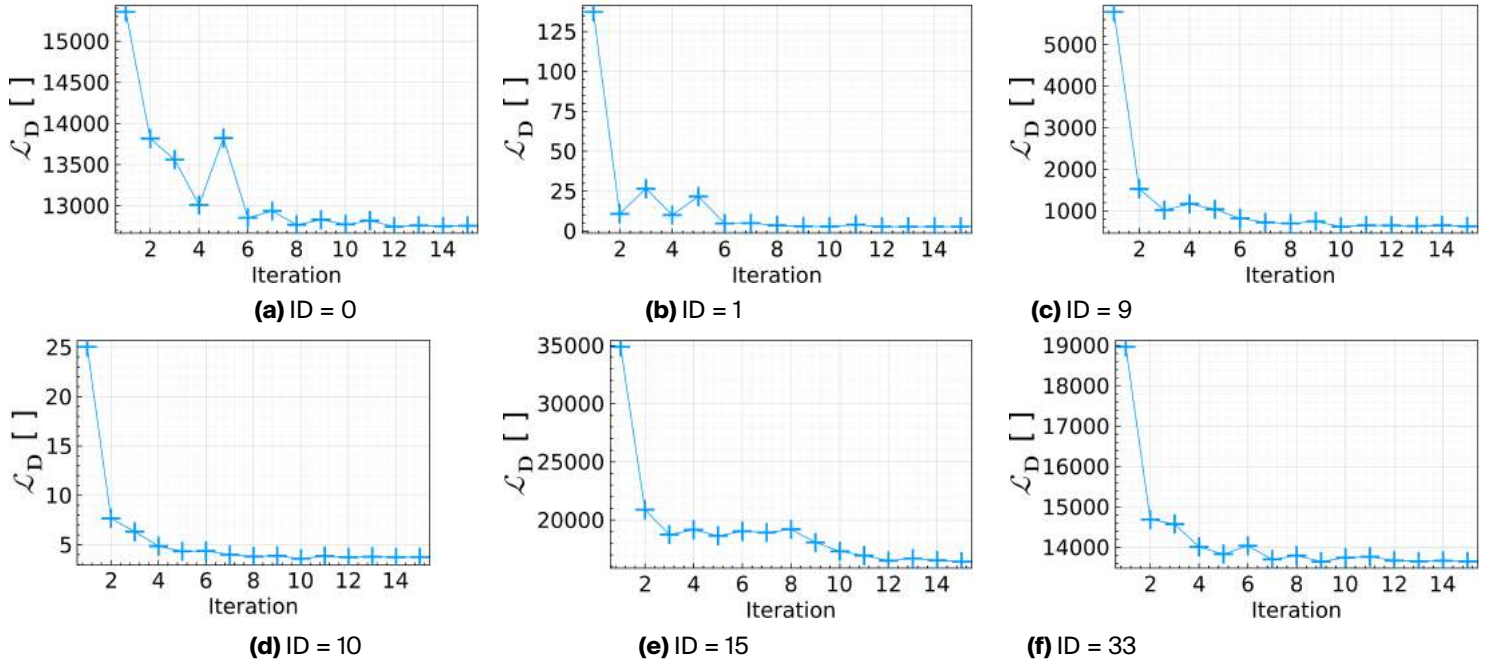


Figure 6.3: Convergence plots of the 3FDD algorithm for six training patients. Their corresponding identifiers are listed. The algorithm is run for 15 iterations for every patient.

Consistency characterises the initialisation of each angle across patients: distinct seeds are employed for the random number generator for each angle, yet these seeds are shared among patients for corresponding angles. For instance, all instances of $\alpha_0^{(0)}$ share the same seed across patients, while $\alpha_0^{(0)}$ and $\beta_0^{(0)}$ possess distinct seeds.

Convergence of the algorithm is typically achieved within approximately 10 steps for each patient. The duration of convergence fluctuates based on the intricacy of the planning objectives. Specifically, as objectives become more intricate, the algorithm might necessitate a greater number of computations for novel dose distributions to thoroughly explore the $\mathcal{L}_D$ manifold. These dose distribution calculations consume the majority of computation time. A breakdown of the time-to-solution durations is provided in table 6.2.

Patient Type	ID	Time to solution
Train	0	11 min. 33 sec.
Train	1	08 min. 55 sec.
Train	9	09 min. 20 sec.
Train	10	16 min. 26 sec.
Train	15	17 min. 38 sec.
Train	33	14 min. 39 sec.

Table 6.2: Time to solution of the Three Fields Dose Descent for six patients. The algorithm was executed for 15 iterations.

It is imperative to emphasise that **there is no guarantee that the final angles attained represent global minima**. A comprehensive evaluation of field placements in a spatial context, dose distributions, and the Dose Volume Histograms (DVHs) associated with the Volumes Of Interest (VOIs) is indispensable before any assertions can be made concerning the predictive quality.

6.3.3. Dose Distributions

The dose distributions for training patient with ID=0 are presented in fig. 6.4 from the axial view, the coronal view, and the sagittal view for each of the scenarios mentioned in section 6.2.4. For clarity, all the results for every other patient, along with the three-dimensional incidences of the reference and predicted beams, are presented in appendix F. These plots are generated using

FlonA. The slices presented are chosen to show interesting profiles of dose distributions³.

6.3.4. Dose Volume Histograms

To further evaluate the quality of the dose distributions, cumulative Dose-Volume Histograms (DVHs) [35] are provided for the six designated patients in each type of plan evoked earlier. Indeed, relying solely on 3D dose distributions for analysis is insufficient. These distributions present a substantial amount of data for human comprehension and for understanding the relationship between dose and three-dimensional anatomy. The sole visual representation of a dose distribution is a challenge in and of itself. DVHs serve to condense the wealth of 3D dose data within a defined volume of interest into simplified one-dimensional curves. They are an essential tool in radiotherapy to depict and analyse the distribution of radiation doses received by distinct volumes within a patient's anatomical structures or target regions. It proves invaluable for treatment planning, assessing treatment quality, and ensuring that radiation doses delivered to OARs adhere to acceptable limits. Nevertheless, it remains crucial to evaluate 3D dose distributions since the reduction of 3D information into DVHs results in their insensitivity to minor 'hot' and 'cold' spots, which correspond to isolated regions of significantly elevated or reduced doses.

In a DVH, the x-axis signifies dose levels in Gray (Gy), a unit measuring absorbed radiation dose. The y-axis represents the volume or percentage of the organ or target exposed to the specified dose. The DVH curve thus illustrates the extent of volume (y-axis) that receives a given dose (x-axis). The curve initiates from the upper left corner (usually near 100% volume at 0 Gy) and progresses toward the lower right corner (0% volume at the highest dose). Crucial points on the DVH curve include the minimum dose to the volume $D_{\min}$; the maximum dose to the volume $D_{\max}$; the mean Dose to the volume D_{mean} ; the median Dose ($D_{50\%}$), which divides the volume into two equal halves. Generally, several values of minimum dose at volume $V\%$ ($D_{V\%}$), denoting the dose received by a specified percentage ($V\%$) of the volume, are assessed (typically the $D_{95\%}$).

DVHs were computed for all VOIs, encompassing the OARs subject to radiation doses alongside the PTV. The comparison of all the DVHs for training patient with ID = 0 is presented in fig. 6.6. The results for other patients are available in appendix F. The plots are generated using JulianA.

³Sometimes, the view is centred on the voxel with maximum dose delivered, but there are cases where this view does not reveal interesting information on the geometries of the patient. A choice was made accordingly for each patient.

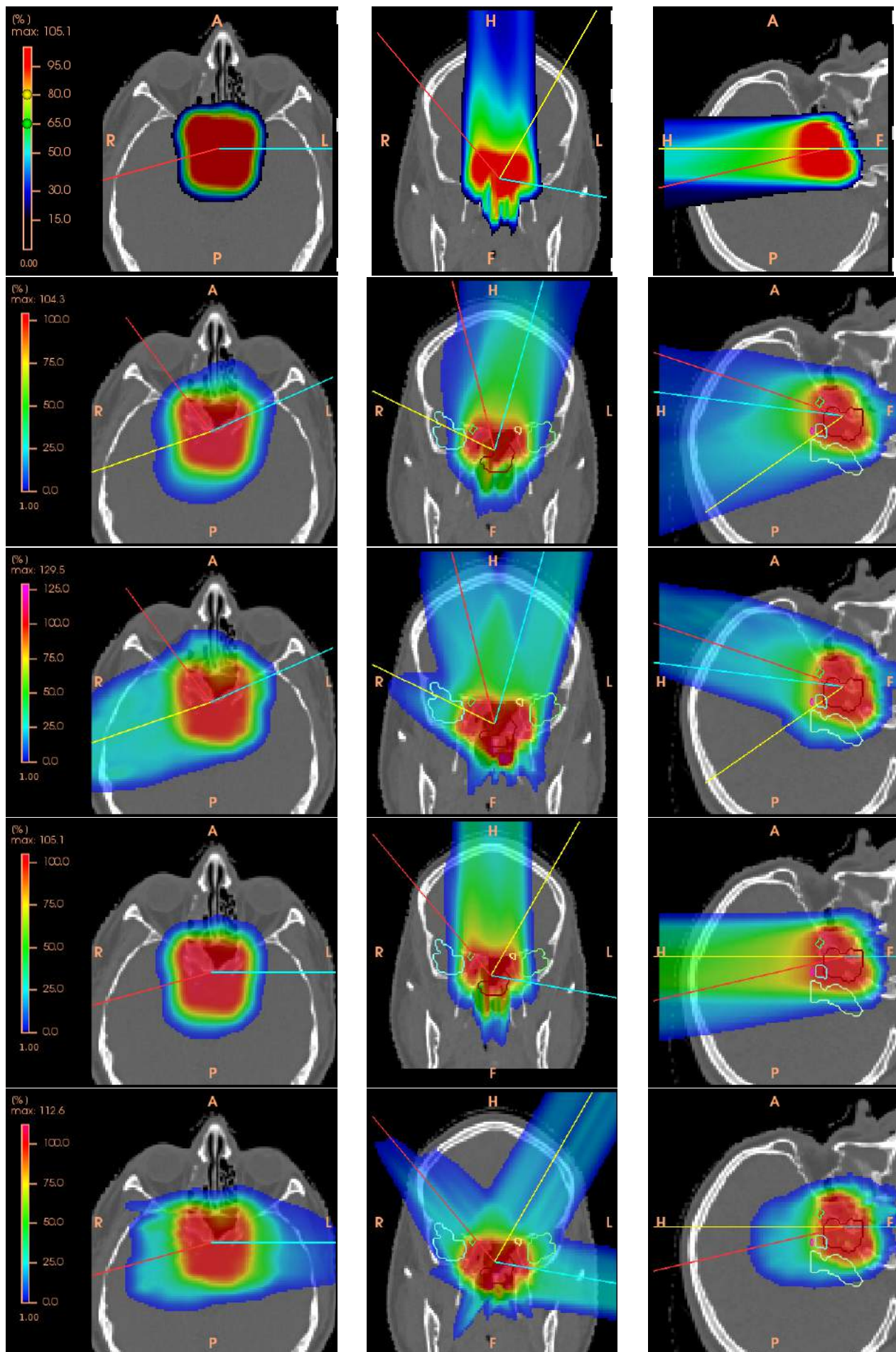


Figure 6.4: Comparison of the dose distributions calculated with 5 methods. The distributions are calculated for training patient with ID = 0 and presented overlapping the patient's CT scan. From views from left to right: axial, coronal, sagittal. The methods used from top to bottom: clinical reference, 3FDD SFO, 3FDD MFO, reference fields recalculated with SFO, and reference fields recalculated with MFO.

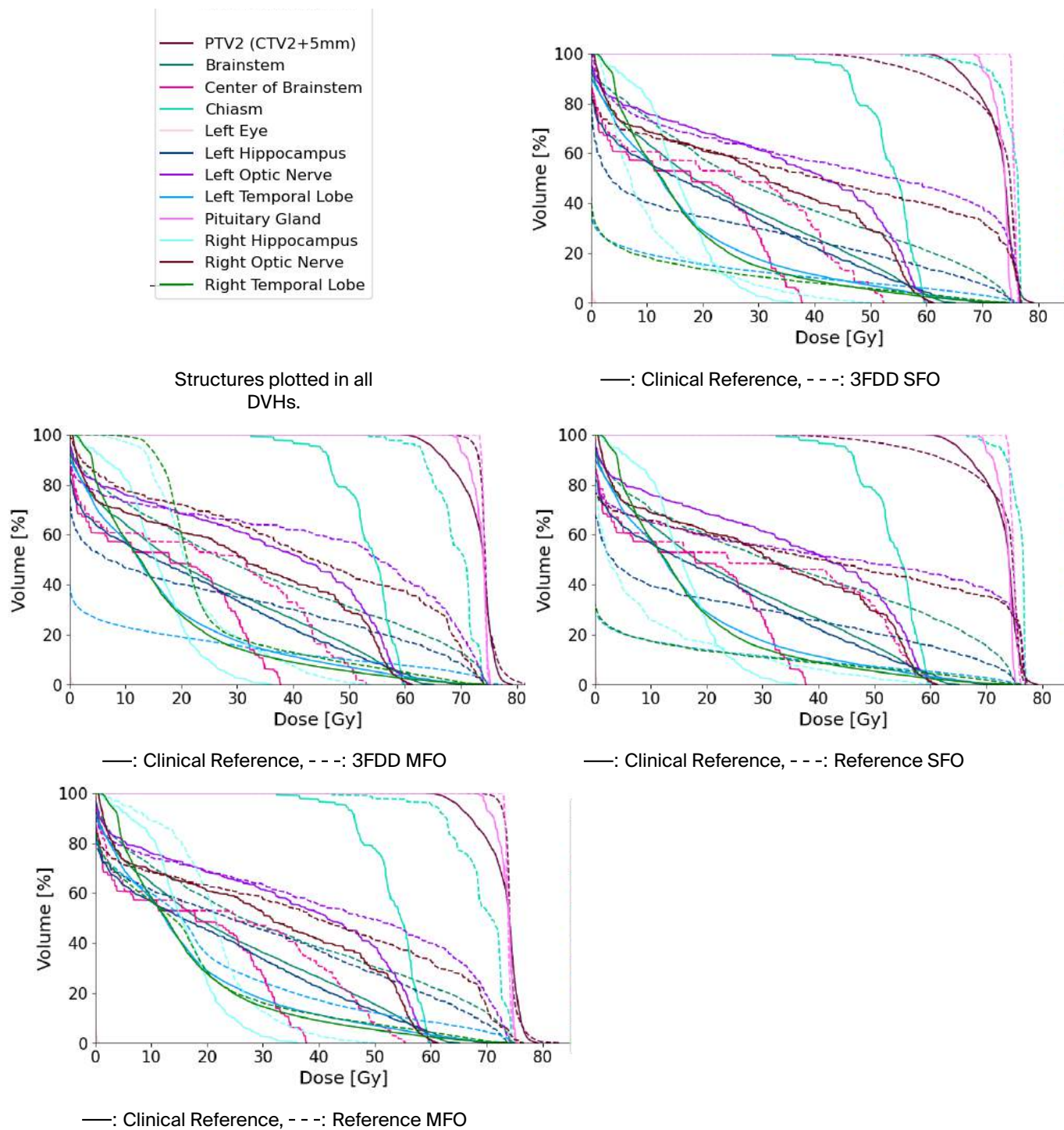


Figure 6.5: Dose volume Histograms comparison of each optimisation strategy to the clinical reference. The patient considered is training patient with ID=0.

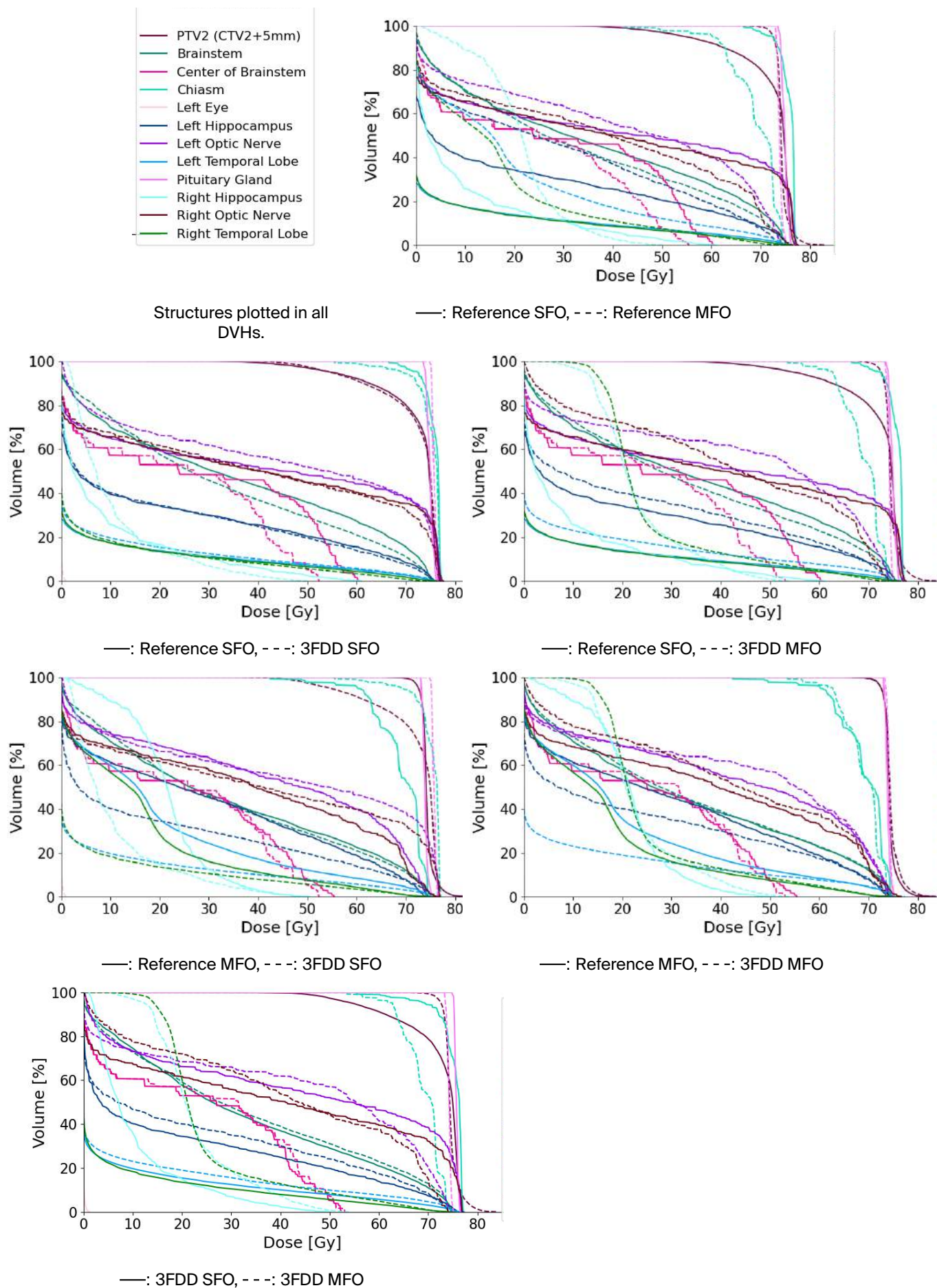


Figure 6.6: Dose volume Histograms comparison of Reference SFO and MFO with 3FDD SFO and MFO. The patient considered is training patient with ID=0.

Another criterion to assess the quality of a plan is the conformity index (CI) of the PTV. The CI is a quantitative measure used to evaluate how well the prescribed treatment volume (target) is covered by the radiation dose while minimising exposure to surrounding healthy tissues. The CI is defined as the volume of the prescription isodose surface divided by the target volume. According to regulations, this ratio shall be between 1.0 and 2.0; and if achieved, there will be no deviation from protocol [36]. A conformity index closest to 1 indicates that the radiation dose closely conforms to the shape and size of the target, leading to better sparing of nearby critical structures and reducing the risk of radiation-related side effects. A high conformity index means that the radiation dose is less conformal to the target, potentially resulting in higher doses to healthy tissues and increasing the likelihood of treatment-related complications. The choice of the optimal conformity index may also depend on specific clinical situations, treatment goals, and the expertise of the radiation oncology team. Since the latter information was not given, we assume that an acceptable CI for the predicted plans shall be closest to 1 or at least as close as possible to the value of the reference plans. The results are presented in table 6.3.

$$CI = \frac{V_{100\%}}{V_{PTV}} \quad (6.6)$$

Patient ID	0	1	9	10	15	33
Clinical Reference	0.562	0.774	1.005	0.745	0.693	0.083
3FDD SFO	0.973	0.572	0.276	0.931	0.678	0.199
3FDD MFO	1.22	0.405	8.982	0.619	1.090	0.482
Reference SFO	1.08	0.358	0.132	0.798	0.775	0.068
Reference MFO	0.95	0.373	8.087	0.533	1.230	0.542

Table 6.3: Conformity indices (CI) of the PTVs of all the patients considered.

A second criterion is the Homogeneity Index (HI). HI has several definitions [37][38], and there is debate around which formulation yields the most expressive and accurate representation of homogeneity⁴. In this work, we select the index that is clinically used at CPT:

$$HI = D_{95\%} - D_{5\%} \quad (6.7)$$

The more homogeneous the dose is delivered, the closer to 0 the HI is. For intuition, it basically assesses how steep the falling part of the curve of the PTV's DVH is. The results for both the 3FDD plans and the reference plans are presented in table 6.4.

Patient ID	0	1	9	10	15	33
Clinical Reference	11.7	5.7	4.31	2.5	16.4	7.4
3FDD SFO	21.7	19.1	18.7	6.9	14.6	17.3
3FDD MFO	5.1	3.1	9.4	2.3	5.9	3.34
Reference SFO	21.4	16.0	18.6	6.9	16.0	15.8
Reference MFO	4.3	3.8	9.0	2.6	4.1	4.2

Table 6.4: Homogeneity indices (HI) of the PTVs of all the patients considered. Five dose distribution optimisations are considered.

⁴The search for expressive features from medical data in radiotherapy is an entire field of study called radiomics.

6.4. Discussion

6.4.1. Quality of the Results

Overall quality

The plans crafted from the predictions display promising potential, all the more so when considering the limited information provided to the algorithm. Keep in mind the method's straightforward nature while evaluating these results. Beyond the patient-specific information encompassing CT scans, organ contours, and field centres, along with the prescriptions, there is no requirement for additional data processing to achieve compelling outcomes. With the educated initialisations, the projected plans consistently avoid suggesting fields originating from caudal directions. Additionally, it appears to successfully circumvent fields traversing through the nasal and oral cavities, which aligns with the preferences of clinicians. This avoidance is particularly significant since the density within the nasal cavity can vary between treatment fractions, potentially affecting proton propagation and subsequently modifying the range of the spread-out Bragg peaks (SOBPs).

Besides avoiding suboptimal regions, the fields predicted are spread out (non-colinear). This behaviour is desirable, as it affects the dosimetric quality. In particular, larger beams spread generally leads to better PTV CI [14]. We can conclude the loss function $\mathcal{L}_D$ from JulianA is well crafted, as it yields these desirable qualities intrinsically.

The importance of the initialisation

The worst plan obtained with 3FDD is for patient with ID=9. However, as suggested earlier, there might be other local minima the algorithm could reach. We test this possibility by starting with a different initialisation (i.e. different seeding⁵). With this new setting, a more acceptable result is obtained as shown in fig. 6.7. Besides, since some dose distributions were already saved from previous descent, running the algorithm a second time only took 5 min. 43 sec.

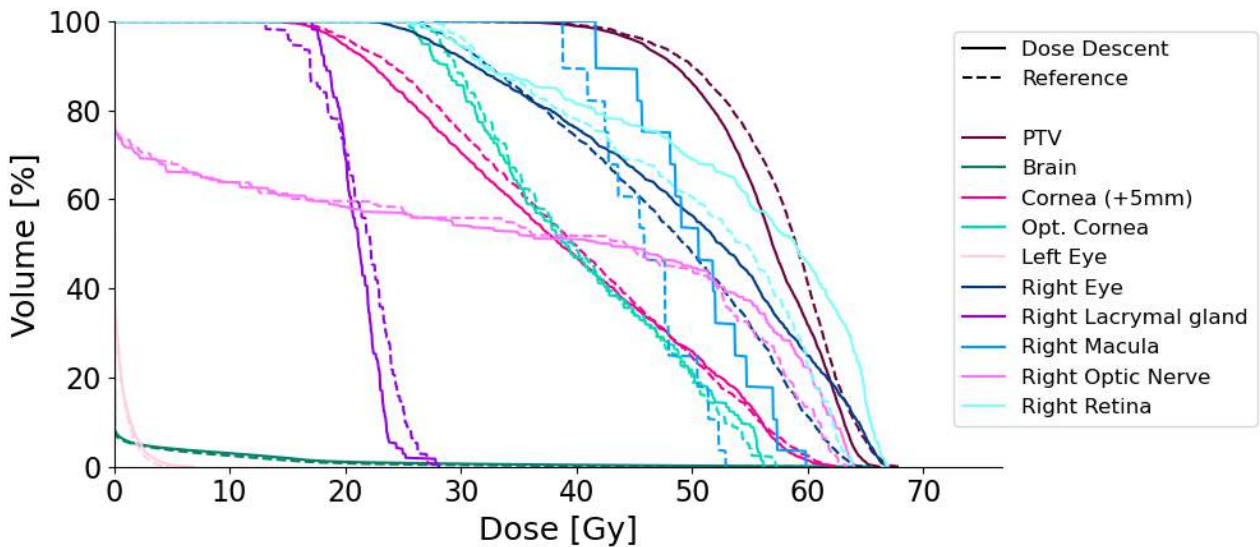


Figure 6.7: Dose Volume Histograms comparison between the predicted plan and the reference plan with **SFO** for training patient with ID=9 with a different initialisation.

These DVHs are already much more comparable for OARs, despite a less optimal target coverage. This manoeuvre confirms the generally recognised sensitivity of the gradient descent approach to initialisation. A different initialisation strategy is detailed further down this document.

⁵We added +100 to the original seeds.

Clinical, SFO and MFO strategies

When considering the results listed in table 6.3, table 6.4 and the DVHs for each of the five methods considered in appendix F, conclusions can be drawn. First, MFO offers better target homogeneity, at the cost of higher hot spots in neighbouring OARs. An analysis regarding conformity is less straightforward, as SFO sometimes yields better results than MFO and conversely.

The clinical reference dose distributions always demonstrate better overall performances compared to the four other strategies. High doses are consistently lower when considering the DVHs of OARs, and the HI and CI of the PTV, if not the best, are always close to optimal. However, discussions with an experienced professional from CPT confirmed that most results from the 3FDD algorithm with SFO were interesting and could use optimisations, mainly in robustness and target coverage. For clinical use, faster time to solution would also be required. We propose optimisations to meet these expectations in the following.

6.4.2. Optimisations

Numerous potential optimisations could enhance the performance of the algorithm, addressing aspects related to result quality and time-to-solution considerations. We outline several recommendations in the subsequent section.

Dose Distributions' Computation Time

The main computational burden of the 3FDD approach is the dose distribution computation. Each dose distribution needed for interpolation takes FlonA about 25 seconds. This is a limitation, not only regarding time to solution, but also regarding the precision of the dose interpolation. Indeed, the current increment in angles between dose distributions is 10° . Decreasing this increment - and thus increasing the precision of the gradient descent - also means a higher chance of needing more dose distributions to proceed to the interpolation on the fly. A faster dose engine would induce less computation time and the possibility for better precision in the final result.

Instead of relying on FlonA for this computation, another possibility could be to rely on JulianA's future dose calculation engine. It follows the same steps as FlonA's but the first benchmarks indicate a faster computation time (order of seconds). This engine was not available at the time of the study. An improvement in the computation time should be the first great improvement of this algorithm.

Minima and Robustness

The 3FDD algorithm does not ensure convergence to a global minimum. From this statement, a wise optimisation lies in further exploration of the dose loss manifold. One means to this end could involve simultaneously executing the algorithm with different initialisations using parallel GPUs (a feasible approach with the computational resources available at PSI). This parallel execution would uphold the sharing of independent dose distributions, potentially yielding comparable computation times across all parallel instances of the algorithm. Additionally, such an approach might provide clinicians with the option to choose from multiple plans that exhibit similar final plan scores.

However, it's important to note that similar scores do not necessarily equate to equivalent plan robustness. In the realm of proton therapy planning, plan robustness implies that the fields produce satisfactory outcomes despite uncertainties in planning variables. Patient positioning is among the most influential variables, where even slight shifts can lead to significant changes. Such shift could cause a corresponding shift in the Bragg peak range of proton beams, particularly if the

beam's path-integrated density is sensitive to such shifts⁶.

Further enhancements aimed at increasing the algorithm's robustness (through selection or real-time adaptation) would constitute a significant step forward in the development of this algorithm.

Fields' Weights

With the current gradient descent approach, all the fields contribute equally to the dose distribution. A new optimisation would be to set weights to each of the fields, which could then be optimised along the angles. Mathematically, this translates into a new expression from eq. (6.8) :

$$\mathbf{D}_{ijk}(\vec{\alpha}) = \sum_{n=0}^2 w_n \cdot \mathbf{D}_{ijk}(\alpha_n, \beta_n). \quad (6.8)$$

with $\mathbf{w} = (w_0, w_1, w_2)$ the weights for each field.

In eq. (6.1), the vector of angles $\vec{\alpha}^{(k)}$ is then replaced by $\vec{\alpha}^{(k)'} = (\vec{\alpha}, w_0^{(k)}, w_1^{(k)}, w_2^{(k)})$, and the same update rule applies. This way, the importance of the fields is adapted to each patient and could yield better performances.

Number of fields

Another optimisation for the 3FDD could involve setting the number of fields as a configurable parameter, rather than restraining to a default of three fields. Indeed, the applicability of a three-field solution might not be universally optimal across all cases. A refined approach would follow this sequence: initialising the problem as a 1-field Dose Descent and selecting and optimising accordingly; then expanding the number of fields and initiating a fresh optimisation; repeating this for three and four fields; and finally selecting the beam configuration that yields the most favourable dose loss (i.e plan score). While this approach could extend the time to solution, it carries the potential for enhanced performance in the dose distribution delivered to the patient, which ultimately remains the primary objective of the treatment planning process.

Fields initialisation

A different "spread-out" initialisation covering more of the phase space could potentially lead to more interesting results. The following one has been tested.

$$\begin{aligned} \alpha_0^{(0)}, \beta_0^{(0)} &= (90 + 20 \cdot (2X - 1), 90 + 20 \cdot (2X - 1)) \\ \alpha_1^{(0)}, \beta_1^{(0)} &= (90 + 20 \cdot (2X - 1), 150 + 20 \cdot (2X - 1)) \\ \alpha_2^{(0)}, \beta_2^{(0)} &= (90 + 20 \cdot (2X - 1), 30 + 20 \cdot (2X - 1)) \end{aligned} \quad (6.9)$$

where $X \sim U(0, 1)$ a random variable following a continuous uniform distribution on interval $[0, 1] \subset \mathbb{R}$. This means the initial values for each angle pair are constricted to a square of side length $l = 40^\circ$ centred on $(90^\circ, 90^\circ)$, $(90^\circ, 150^\circ)$ and $(90^\circ, 30^\circ)$ as illustrated in fig. 6.8.

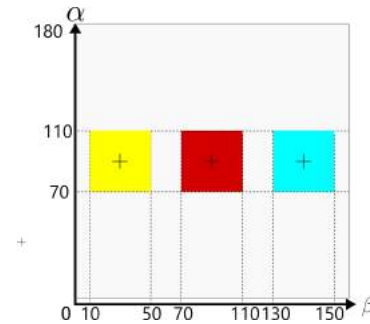


Figure 6.8: Initial "spread-out" conditions of the gradient descent for a single field. The possible values are delimited by the three coloured squares.

Practically, this sets the starting configuration in the neighbourhood of the one illustrated in fig. 6.9.

⁶This situation may arise when the angle between the beam and the skull at the entry point is close to zero or if the beam traverses a complex anatomical region of the skull, such as its base.

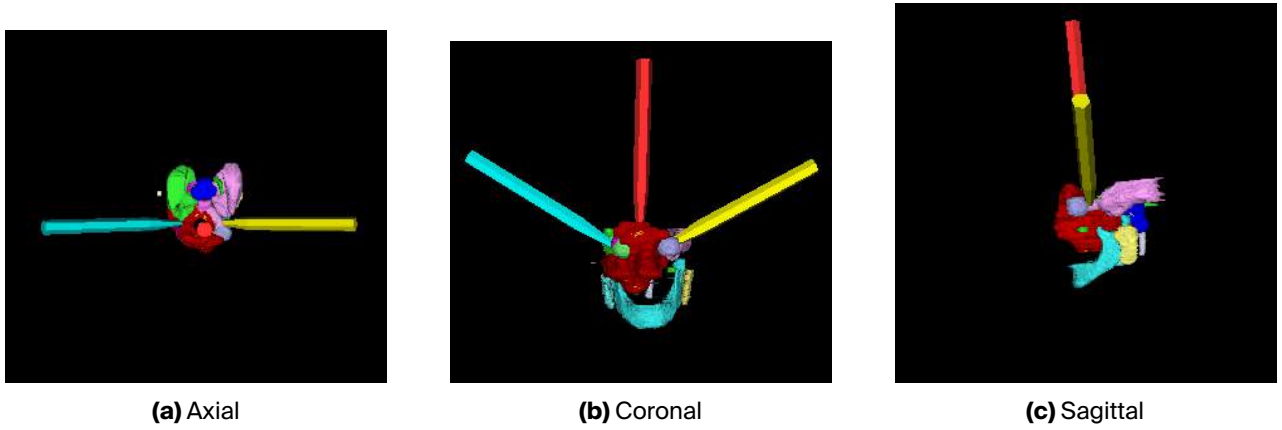


Figure 6.9: Initial "spread-out" conditions for the 3FDD algorithm. Structures of training patient with ID=15 are used for the visualisation. In this representation, the fields are directed to the target, which depends on the patient.

With this new initialisation, the time to solution is increased, since all the starting fields are far away from each other and thus do not share many dose distributions at each interpolation. The results regarding the final loss function are the following (seeds remain unchanged).

Patient ID	0	1	9	10	15	33
Normal Initialisation	12755	2.70	628	3.7	16415.	13647
Spread Initialisation	11662	2.23	734.5	3.22	15396	13552

Table 6.5: Final Dose loss value after 15 steps of the Three Fields Dose Descent algorithm with standard and "spread-out" distribution of initial angles.

The new initialisation seems to yield better final plan scores overall. However, as the random number generator seeds remain unchanged, it is not guaranteed the algorithm reaches the best results for every single patient. A more in-depth statistical analysis should be carried out to draw any conclusion on the balance between plan performance and computation performance of both methods.

The initialisation proposed is general. A different approach is to leave the clinicians to initialise each field according to current practices. Indeed, in the first step of treatment planning, clinicians at CPT are provided with a series of guidelines based on past experience and adapted to specific diseases; they take into account plan robustness, tumour control probability and normal tissue complication probabilities, and the therapeutic ratio (TR). These guidelines encompass fields number and several suggested incidences, with a margin of error. In this application, the process of fine-tuning is left to the algorithm, allowing faster time to the best solution lying in the neighbourhood of the initialisation.

Update rule

As mentioned earlier in chapter 5, $\mathcal{L}_D$ exhibits highly non-convex behaviour concerning the angles. Diverse update strategies can be employed to address such challenges. Despite unsuccessful attempts with Adam and some of its variants, certain refinements remain applicable to the update approach. One notable example is the incorporation of a stopping criterion. This criterion helps to minimise waste of computational resources by identifying scenarios where continuing method execution proves impractical. The restriction of the maximum iteration count to 15 could be removed, with such a criterion introduced in its place. This criterion might be es-

established based on the progression (gradient) of loss function values across iterations or derived from the evolution of the predicted angle values. This would also require a decay rule for the learning rate, such as exponential based on a parameter or linear, which induces additional tuning of the update rule.

On another hand, one could transfer the dose calculation approach from SFO to MFO already in the update rule. This would require a significantly faster dose calculation engine since the algorithm would hardly rely on previously computed dose distributions (instead of 2 parameters defining one distribution, it would have 6 instead). New distributions would be more probably be required for interpolation at each iteration step.

Forbidden spaces

The algorithm always predicted fields outside of the nasal and oral cavity thanks to the initial conditions. However, a selection in these areas is technically not impossible if local or global minima do not lie in between the initialisation and these cavities. An additional penalty added to the loss function could help handle these regions of angles phase space and ensure avoidance. However, the determination of these spaces should be adapted and specific to each patient in order not to restrict the accessible phase space unwisely.

Conclusion

Two methodologies were developed and employed within the scope of this project. The primary approach centred on Convolutional Neural Networks (CNNs), initially in a supervised manner and subsequently in an unsupervised manner. The CNNs were tasked to predict optimal beam angles for a single noncoplanar field. The second strategy involved the application of gradient descent to a heuristic loss function, enabling the prediction of orientations for three fields.

Convolutional Neural Networks

To facilitate the training of neural networks in both the supervised and unsupervised paradigms, we developed an extensive framework. Three distinct models were crafted for evaluation: a traditional 3D CNN, a reduced variant of the former, and a model employing the Inception module to explore varying levels of architectural complexity. The input data for these models was thoughtfully compiled to extract maximal geometrical insights from the available patient data. Two standardised input types were generated – binary masks and distance masks – derived from manually traced contours produced by clinicians. Additionally, data augmentation was performed based on the sagittal symmetry inherent in the human head, expanding the original dataset from 75 training patients to 150. In tandem with the model and input type pairings, a systematic exploration of hyperparameters was conducted via grid searches. Consequently, optimal values are obtained for each pairing, and the most effective networks are constructed. These hyperparameters covered learning rate, regularisation type, strength of the regularisation, and batch size.

In the context of supervised training, the comparison between the output angles and the angles employed in the clinical treatment of the training patients was accomplished with a mean squared error loss function. Despite the comprehensive training efforts, the resulting models fell short of surpassing the median predictor's performance, leading to less than compelling outcomes. To delve deeper into the network's comprehension of input geometries, we computed gradient-based activation maps. The findings validated the network's ability to perceive and comprehend input geometries. A comparative analysis of various model and input type pairings was performed, assessing the minimum mean losses across networks generated via hyperparameter grid searches. This was aimed at offering insights into which combinations might deliver the best outcomes if a larger dataset ever became available. The Inception model yielded the most promising results, and the training set expansion, while only doubling the size of the dataset, improved some of the results. Binary masks exhibited better overall performances, hypothesised to come from the sharper delineations of organs. Several regularisation techniques were attempted, but none improved the results significantly. The conclusion drawn from the supervised approach pointed towards the network requiring more information for effective learning. Although it could discern geometries, it struggled to predict new angles based on them. Improved results would demand more extensive data and comprehensive training. The latter was attempted through an unsupervised learning approach.

In an unsupervised learning strategy, additional task-related information was incorporated through a loss function centred on dosimetric prescriptions. This loss function encapsulates a set of constraints involving minimum dose, maximum dose, and mean dose as well as homogeneity and conformity criteria of the doses delivered to both the tumour and the surrounding organs. Instead of directly incorporating angles into the loss function, they were used for interpolation from

pre-computed dose distributions that were fed into the loss function. Similarly to supervised learning, hyperparameter grid searches were conducted to generate and train the most efficient networks. An additional parameter was added in the form of the strength of a penalty over the angles to ensure the physicality of the outputs. This approach yielded outcomes that were less than convincing. Gradient maps were generated, revealing an increased challenge in recognising the input geometries, particularly for distance maps of a new test patient. In the latter case, the attention of the network was not directed towards the input structures but the entire cranium, potentially resulting in sub-optimal angles. An investigation into the various combinations of architectures and input types was carried out again, using the grid search results to determine which combinations might yield superior results. No conclusions were drawn from this analysis, as the increase in overall test losses for the Inception model was minimal.

The current study produced a framework to train CNNs with standard ingredients and verify if such approaches would be feasible. The conclusion of all our observations is that CNNs, whether trained in a supervised or unsupervised paradigm, would require additional data to yield improved performance. Efforts should be focused on this need in the future, through collaborations with other particle therapy centres. More efficient networks, optimisers, and hyperparameter search methods could then be implemented. Another possibility is the generation of robust synthetic patients. These patients need realistic diseases, organs, CT scans, and adapted plans, based on prior diseases observed at CPT or other facilities. A collaboration between researchers and clinicians is fundamental for such advancement and might require tremendous efforts to implement and use for treatment planning.

Three Fields Dose Descent

Following inconclusive attempts involving neural networks, a more straightforward approach was proposed. This novel approach, based on a modified gradient descent technique, demonstrates the capability to predict beam angles for three fields adapted to the input patient. The objective function minimised to obtain these beam angles is the dose-informed loss function previously employed in the unsupervised CNN approach. For each iteration of the gradient descent, the dose distributions corresponding to the three beam angle pairs predicted are interpolated and averaged. Each beam's dose distribution is spot-optimised independently from the two other ones. The real-time calculation, optimisation, and shared storage of each on-the-fly dose distribution significantly reduce the time to solution, as they can be loaded for later iterations. This algorithm, termed "Three Field Dose Descent" (3FDD), displays an average convergence time of 13 minutes over 15 iterations. This ranges the time to solution to current state-of-the-art algorithms for beam selection in proton therapy. A strategic initialisation technique circumvents sensitive areas for proton therapy, such as the oral and nasal cavities, and avoids cases where the beam could propagate through the body's length.

An exhaustive analysis was performed on resulting dose distributions and their corresponding Dose-Volume Histograms (DVHs). Six patients with pathologies varying in type and severity were reviewed. For each, five dose distributions were analysed and compared. First, the clinical reference dose distribution delivered to the patient is considered, then the three fields derived from the reference optimised under a Single Field Optimisation (SFO) paradigm, and then under a Multiple Field Optimisation (MFO) paradigm, where an additional final step of spot optimisation encompassing all fields was conducted. The three fields predicted via 3FDD were examined under both SFO and MFO paradigms and constituted the fourth and fifth distributions. Additional individual homogeneity and conformity indices were computed to deepen the predictions' quality assessment.

The SFO and MFO plans were comparable or superior to the reference SFO and MFO plans. This

is visible when comparing the DVHs of all four methods. Expert evaluation supported further research for this method after review of the first results. To fully integrate it into clinical practice and potentially publish this work, further developments are necessary. These involve incorporation of robustness during the optimisation, broader statistical investigations across more patients, and faster convergence times. Aside from these recommendations, we suggest additional optimisations to enhance overall performance, such as an adaptive number of fields, learnable field weights, and avoidance of forbidden spaces via patient-adapted penalty on angles. Finally, a new "spread" initialisation was tested. While showing higher times to solutions, it yielded improved final dose loss values. We also suggested using the algorithm as a tuning step after initialising fields to standard clinics recommendations.

A last word

This thesis explored two approaches to beam angles selection in proton therapy. First, Convolutional Neural Networks. Trained first in a supervised manner - relying on past clinical plans - then in an unsupervised manner - relying on dosimetric constraints in the form of a heuristic loss - the networks were employed for prediction of a single non-coplanar field's incidence. Yielding unsatisfactory results, hypothesised to result from data scarcity, a second approach was implemented for three non-coplanar fields : a modified gradient descent exploiting earlier mentioned loss. Tested in both SFO and MFO paradigms against the clinical reference, this method showed results encouraging enough for further research in the future. With additional data, it would be wise to reconsider the CNN approaches.

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Nomenclature

Abbreviations

Abbreviation	Definition
3FDD	Three Fields Dose Descent
AROBASE	Automatic and Rapid tool for Optimal Beam Angles SElection
CPT	Center for Proton Therapy (PSI)
CI	Conformity Index
CNN	Convolutional Neural Network
CT	Computed Tomography
DL	Deep Learning
DVH	Dose Volume Histogram
FCN	Fully Convolutional Network
FlonA	Flexible Treatment Planning Program
G2	Gantry 2
GPU	Graphics Processing Unit
HI	Homogeneity Index
ID	Patient unique identifier
IMPT	Intensity Modulated Proton Therapy
IMRT	Intensity Modulated Radio-Therapy
JulianA	Porte-manteau term combining Julia and FlonA
MFO	Multiple Field Optimisation
ML	Machine Learning
NN	Neural Network
OAR	Organ At Risk
PTV	Planning Target Volume
PSI	Paul Scherrer Institute
PT	Proton Therapy
SFO	Single Field Optimisation
SOBP	Spread Out Bragg Peak
VOI	Volume Of Interest

Symbols

Symbol	Definition	Unit (S.I.)
α	Gantry Angle	[°] or [rad]
β	Couch Angle	[°] or [rad]
η	Learning Rate of the update rule	[]
λ	Regularisation factor	[]
θ	Learnable network parameter	[]
Θ	Set of learnable network parameters θ	[]
σ	Activation function	[]
b	batch size	[]
$\mathcal{L}$	Loss function (general denomination)	[]
$\mathcal{L}_D$	Loss function based on dosimetry prescriptions	[]
Gy	Gray (Absolute dose Unit)	[J/kg]
D	Dose distribution	[Gy]

Symbol	Definition	Unit (S.I.)
s	Structure included in the VOIs	[]
$M^{(s)}$	Binary mask of structure s	[]
$m^{(s)}$	Standardised binary mask of structure s	[]
$D^{(s)}$	Distance mask of structure s	[cm]
$d^{(s)}$	Standardised distance mask of structure s	[]
$A_{ijk}^{(\gamma, I)}$	Gradient based activation map of angle γ w.r.t input I	[°] or [°/cm]
#	Generic symbol for the word "number"	—

A

Supervised Multiple Layers Perceptron

Aside from attempting the convolutional network approach, we tried to implement a simple Multiple Layers Perceptron (MLP) using features used initially in a nearest neighbours model for beam angle selection - an project from CPT. This network is tasked with the prediction of a single field's angles prediction.

The features were crafted in cooperation with clinicians at CPT. They convey information about the dose characteristics of the treatment plan delivered to the patients. For example, the minimum and maximum dose delivered to OARs appearing in the plan or homogeneity criteria.

As all the features are shared by all the patients regardless of their pathology, some features in the dataset were labelled as missing for certain patients. A simple data imputation is performed to replace the missing values with the feature-wise median of all the existing values. Next, the features are standardised on the [0,1] or [-1,1] intervals (depending if there are negative values)

A simple MLP is constructed, composed of three hidden fully connected layers. The whole network is described in table A.1

Layer type	Input Layer	Fully connected	Fully connected	Fully connected	Final layer
Number of nodes	98	64	32	16	2
Activation applied	/	Leaky ReLu	Leaky ReLyu	Leaky ReLu	Identity

Table A.1: Multi-Layer Perceptron architecture used for a supervised single field beam angle prediction

Leaky Relu was chosen to take negative values into account. A L2 regularisation was added to the MSE and batch size can be set. After a hyperparamter search, it is found that learning rate $\eta = 10^{-4}$, regularisation strength 10^{-4} and batch size of 2 yield the best validation loss. The plot of the loss for this specific case is presented in fig. A.1.

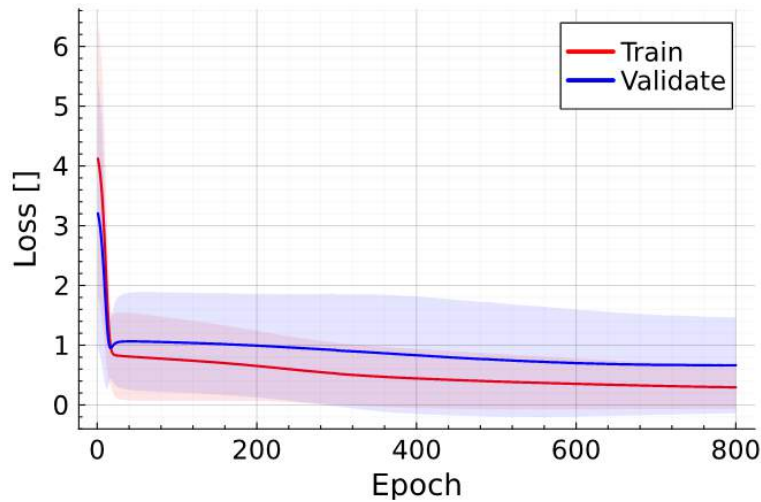


Figure A.1: Convergence of the mean training and validation total loss for the supervised MLP approach to the 1-field beam angle prediction problem.

The predictions are the following

Patient ID	Reference angles	Predicted angles
66	[90,90]	[73.8,108.1]
67	[160,0]	[114.4,97.9]
68	[100,170]	[79.9,183.9]
69	[80,20]	[84.4,41.5]
70	[85,170]	[89.8,204.5]
71	[80,30]	[52.0,22.4]
72	[95,165]	[84.0,109.3]
73	[80,20]	[96.3,139.6]
74	[70,160]	[93.1,141.5]

Table A.2: Beam angle predictions for one field using the supervised Multi-layer perceptron for the validation set.

Despite encouraging results for the gantry angles - which were expected due to the restricted range covered in the training set - the network doesn't provide interesting results for a full prediction. Additional data is necessary to further develop this approach. Indeed, similarly to the CNN approach, this is a high dimensional problem, and a DL approach can only be used at full potential if enough data is at hand. A rule of thumb often stated is 10 times more data points than features are required for DL, which here means 1000 samples. Even more so considering the fact that this attempt is restricted to one field only, which is not sufficient for efficient treatment. A more complete prediction for several fields necessitates additional data.

B

Test Patients

table B.1 details each patient included in the test set. All of the patients were treated in our institute.

Patient ID	Diagnosis and location	Prescribed Dose [Gy RBE]
00	Chordoma, clival	73.8
01	Pilocytic astrocytoma, mesencephalic	50.4
02	Haemangioma, orbital	20.0
03	Melanoma, conjunctival	66.0
04	Meningioma, clival	54.0
05	Meningioma, petroclival	54.0
06	Pilomyxoid astrocytoma, optic nerve	52.2
07	Chordoma, clival	74.0
08	Anaplastic astrocytoma, frontal	59.4
09	Chordoma, clival	74.0
10	Germinoma, XXX	24.0
11	Embryonal tumor with multilayered rosettes (ETMR), frontoparietal	59.4
12	Embryonal rhabdomyosarcoma, intraorbital	45.0
13	Atypical meningioma, XX	60.0
14	Meningioma, optic nerve	50.4
15	Chordoma, clival	74.0
16	Ependymoma, prepontine	59.4
17	Papillary ependymoma, thalamic	59.4
18	Pilocystic astrocytoma, optical nerves	54.0

Table B.1: Patients' characteristics (n=19) for the auto treatment planning analysis utilising JulianA.

C

Supervised Learning - Activation Maps

The activation maps are presented for the Inception and Reduced models trained on the initial dataset only. The colorbars ticks are shared by all the plots and are thus represented only once by group of activation map. The white parts of the colorbar are values that are not covered in the corresponding activation maps. The ticks' values of the colorbar are shared by all the activation maps but the colour gradients are unique to each of them. In all figures, slice 50 of the 3D activation map is represented.

C.1. Inception model

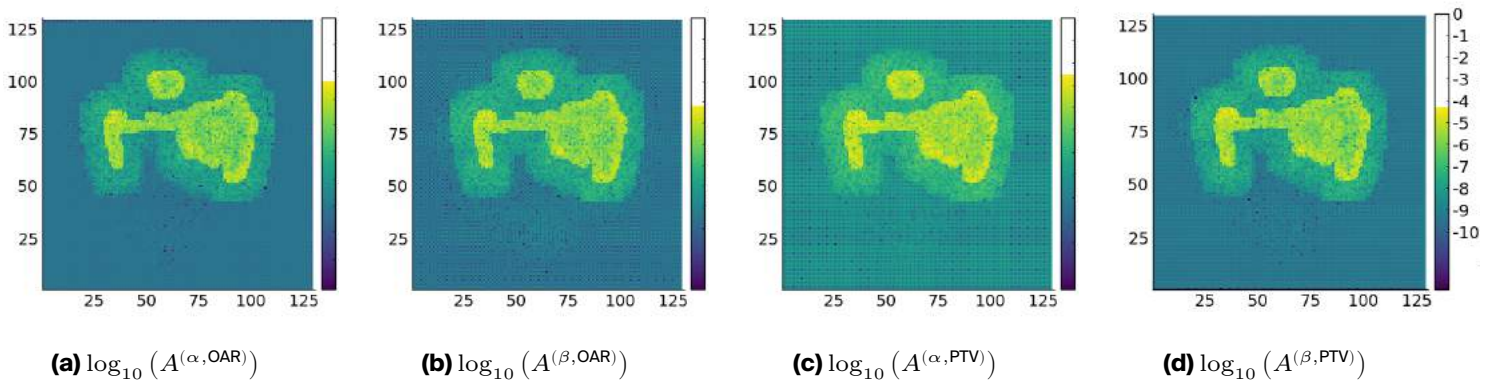


Figure C.1: Gradient-based activation maps when predicting the angles for train patient with ID=25. The model selected is the supervised Inception model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs. Slice 50 is represented.

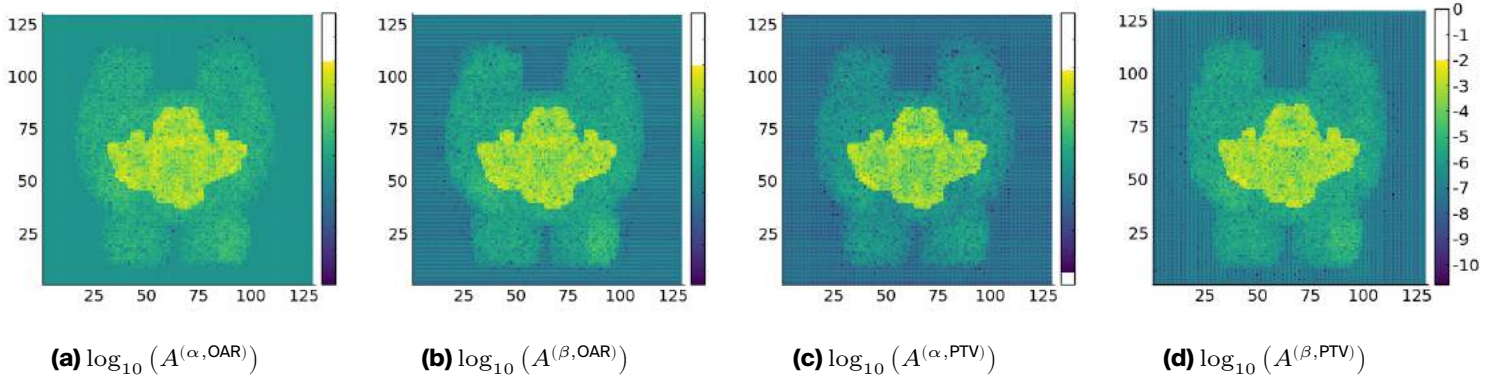


Figure C.2: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the supervised Inception model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs. Slice 50 is represented.

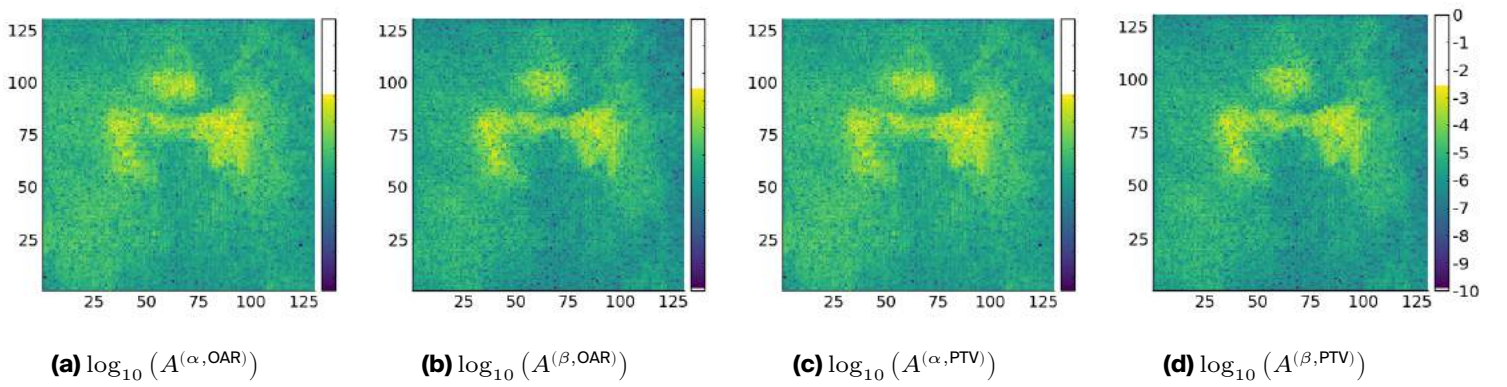


Figure C.3: Gradient-based activation maps when predicting the angles for train patient with ID=25. The model selected is the supervised Inception model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs. Slice 50 is represented.

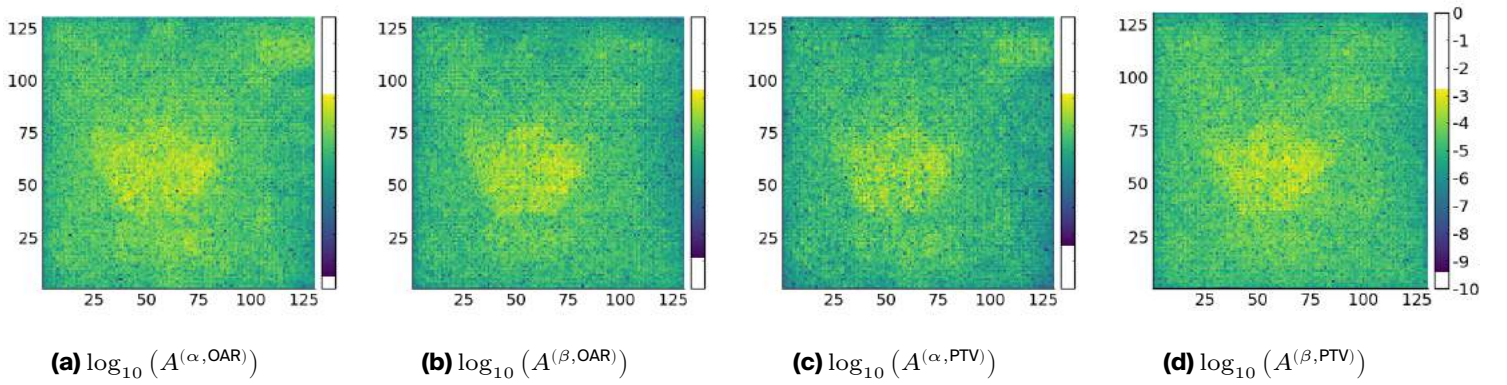


Figure C.4: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the supervised Inception model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs. Slice 50 is represented.

C.2. Reduced Model

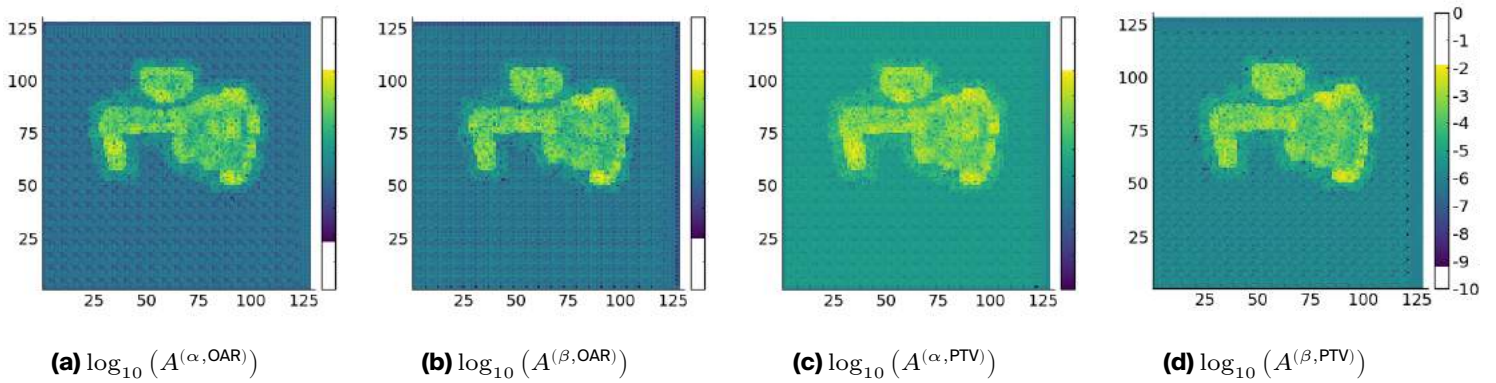


Figure C.5: Gradient-based activation maps when predicting the angles for train patient with ID=25. The model selected is the supervised Reduced model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs. Slice 50 is represented.

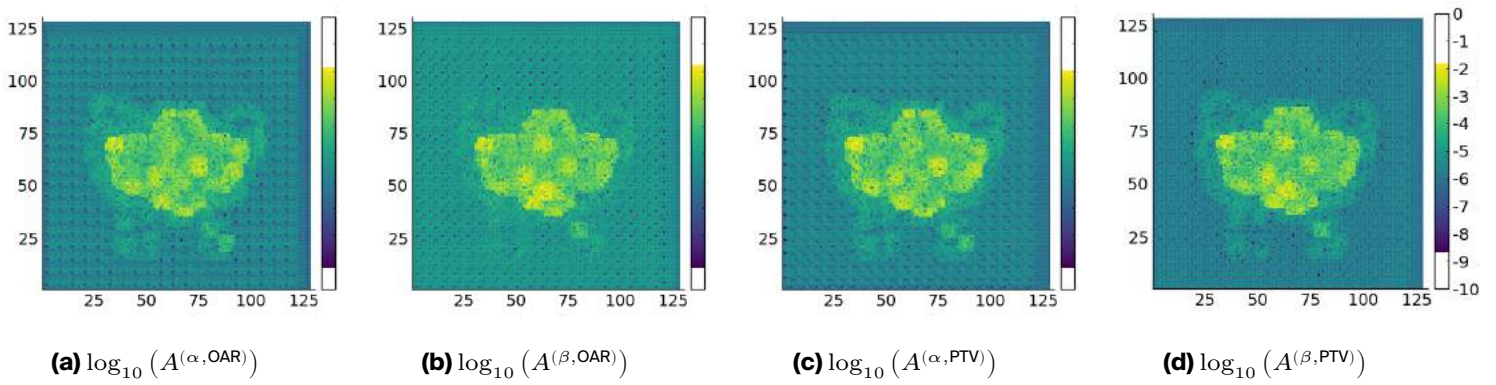


Figure C.6: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the supervised Reduced model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs. Slice 50 is represented.

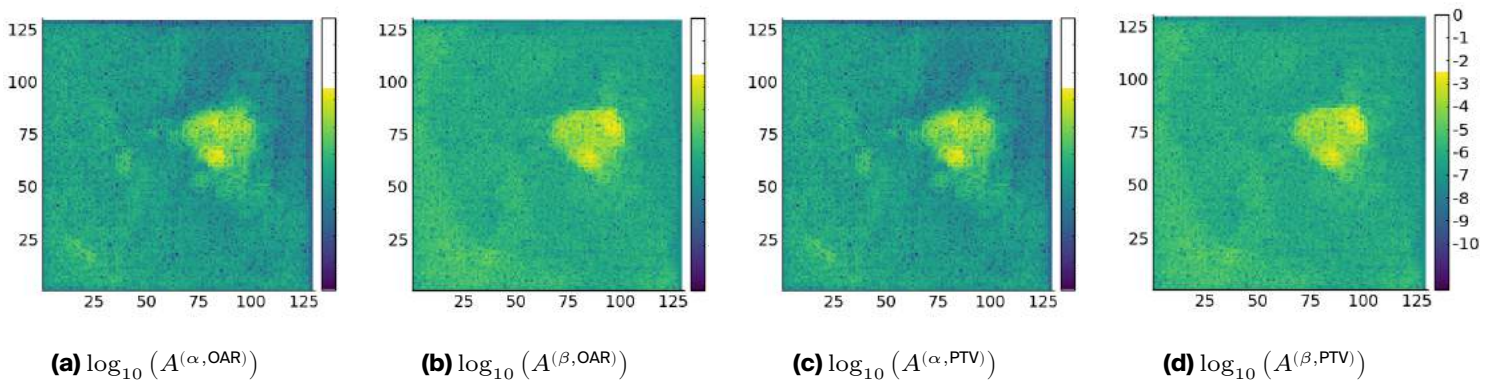


Figure C.7: Gradient-based activation maps when predicting the angles for train patient with ID=25. The model selected is the supervised Reduced model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs. Slice 50 is represented.

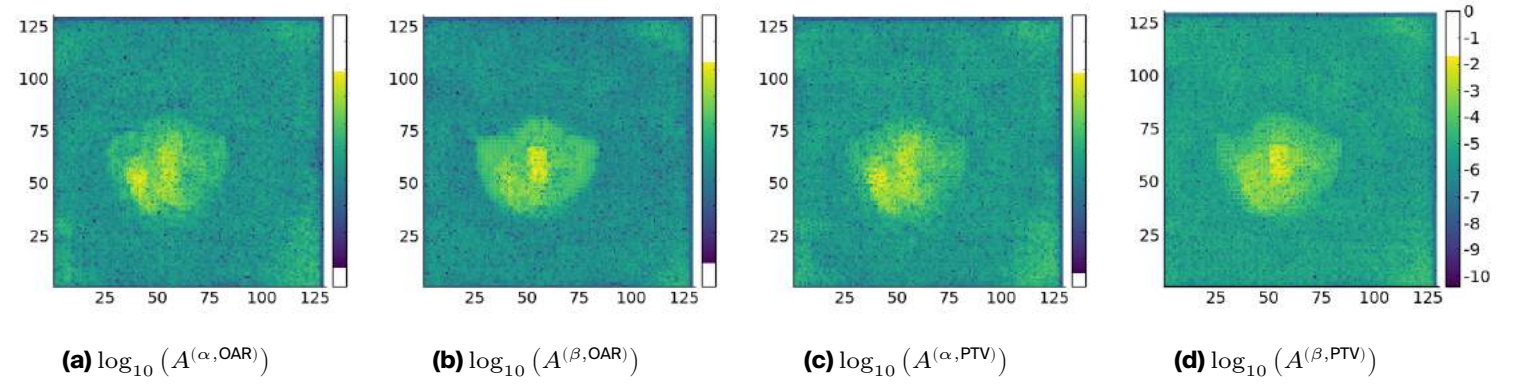


Figure C.8: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the supervised Reduced model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs. Slice 50 is represented.

D

Adaption of $\mathcal{L}_D$ to Automatic Differentiation

Setting

The challenge encountered in the weight update of the unsupervised approach presented in chapter 5 pertains to the loss function calculation. This calculation necessitates the computation of the dose distribution, which in turn relies on retrieving precomputed dose distributions from a file:

$$\nabla_{(\alpha, \beta)} \mathcal{L}_D(\mathbf{D}(\alpha, \beta)) = \underbrace{\nabla_{(\alpha, \beta)} \mathbf{D}(\alpha, \beta)}_{\text{Look-up required}} \cdot \nabla_{\mathbf{D}} \mathcal{L}_D \quad (\text{D.1})$$

The second term can be directly managed by the Flux library. With the incorporation of interpolation, the manual derivation of the first term in the dot product is straightforward:

$$\partial_{\alpha} \mathbf{D}(\alpha, \beta) = \frac{1}{(\beta_2 - \beta_1)(\alpha_2 - \alpha_1)} [(\beta_2 - \beta)(\mathbf{D}_{21} - \mathbf{D}_{11}) + (\beta - \beta_1)(\mathbf{D}_{22} - \mathbf{D}_{12})] \quad (\text{D.2})$$

$$\partial_{\beta} \mathbf{D}(\alpha, \beta) = \frac{1}{(\beta_2 - \beta_1)(\alpha_2 - \alpha_1)} [(\alpha_2 - \alpha)\mathbf{D}_{12} + (\alpha - \alpha_1)\mathbf{D}_{22} - (\alpha_2 - \alpha)\mathbf{D}_{11} - (\alpha - \alpha_1)\mathbf{D}_{21}] \quad (\text{D.3})$$

However, this look-up operation to retrieve dose distributions $\mathbf{D}_{ij}$ is non-differentiable, and Flux.jl is unable to handle it. Consequently, it cannot be integrated as is into an automatic back-propagation step during the neural network training process.

Neural network and automatic differentiation

Where exactly does this issue manifest in the back-propagation? To elucidate, let's focus on the final layers of the neural networks' predictor, composed of two fully connected layers which supply the two outputs to the loss function. fig. D.1 is a generic representation of such architecture.

We designate the value at node j in layer l as z_j^l :

$$z_j^l = \left(\sum_{k=1}^m w_{jk}^l a_k^{l-1} \right) + b_j^l \quad (\text{D.4})$$

where w_{jk}^l represents the weight linking node k of layer $l-1$ with node j of layer l , and b_j^l denotes the weight applied to node j of layer l . The weight update process employing the back-propagation

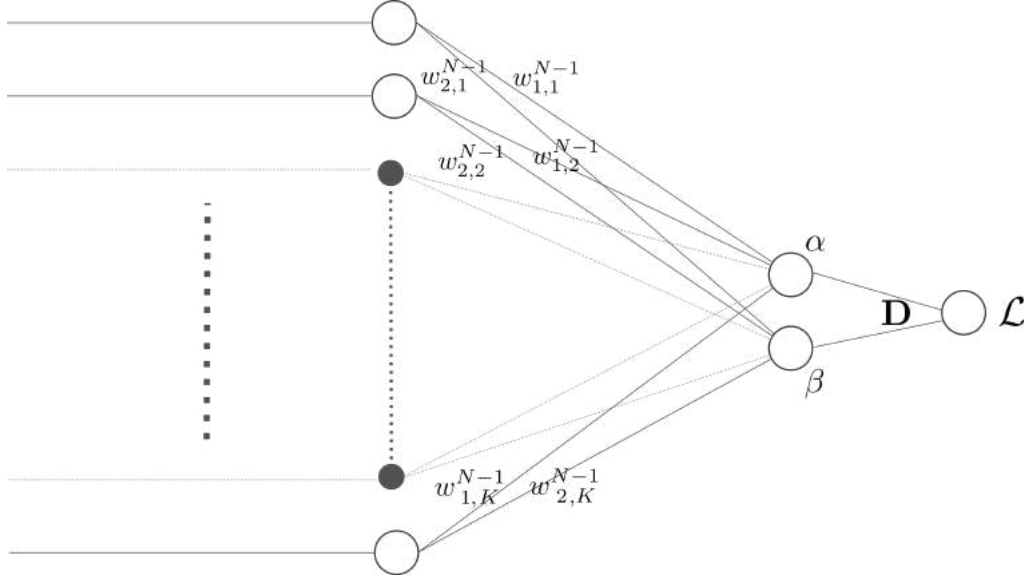


Figure D.1: Predictor of the CNNs crafted for the single field beam angles selection problem.

algorithm incorporates the following values within the selected update scheme (gradient descent update here).

$$\frac{\partial \mathcal{L}}{\partial w_{jk}^l} = \frac{\partial \mathcal{L}}{\partial z_j^l} \frac{\partial z_j^l}{\partial w_{jk}^l} = \frac{\partial \mathcal{L}}{\partial z_j^l} a_k^{l-1} \quad (\text{D.5})$$

$$\frac{\partial \mathcal{L}}{\partial b_j^l} = \frac{\partial \mathcal{L}}{\partial z_j^l} \frac{\partial z_j^l}{\partial b_j^l} = \frac{\partial \mathcal{L}}{\partial z_j^l} \quad (\text{D.6})$$

The derivatives $\frac{\partial \mathcal{L}}{\partial z_j^l}$ are computed using reverse mode automatic differentiation. That is, it requires derivatives of the last layers in the neural network to compute the derivatives of earlier layers. The update of the weights connecting the penultimate and final layers is thus

$$w_{jk}^{N-1} = w_{jk}^{N-1} - \eta \frac{\partial \mathcal{L}}{\partial w_{jk}^{N-1}} = w_{jk}^{N-1} - \eta \frac{\partial \mathcal{L}}{\partial z_j^{N-1}} a_k^{N-2}$$

Using the **adjoints** notation of reverse mode automatic differentiation [39]

$$\frac{\partial \mathcal{L}}{\partial z_j^{N-1}} = \overline{z_j^{N-1}} = \left(\frac{\partial \mathcal{L}}{\partial z_j^{N-1}} z_1^N \right) \overline{z_1^N} + \left(\frac{\partial \mathcal{L}}{\partial z_j^{N-1}} z_2^N \right) \overline{z_2^N}$$

However,

$$\overline{z_1^N} = \frac{\partial \mathcal{L}_{\mathbf{D}}(\alpha, \beta)}{\partial z_1^N} = \frac{\partial \mathcal{L}_{\mathbf{D}}}{\partial \alpha} \quad (\text{D.7})$$

$$\overline{z_2^N} = \frac{\partial \mathcal{L}_{\mathbf{D}}(\alpha, \beta)}{\partial z_2^N} = \frac{\partial \mathcal{L}_{\mathbf{D}}}{\partial \beta} \quad (\text{D.8})$$

Here lies the problem described in the beginning. The gradient of the loss function with respect to the angles is required for all the derivatives of all the weights computed using the reverse mode automatic differentiation.

A solution

To bypass this problem, we replace the final loss function to which the outputs of the network are fed: a linear composition of the derivatives :

$$\mathcal{L}_{\mathbf{D}}'(\alpha, \beta) = \partial_{\alpha} \mathcal{L}_{\mathbf{D}} \cdot \alpha + \partial_{\beta} \mathcal{L}_{\mathbf{D}} \cdot \beta$$

where $\partial_{\alpha} \mathcal{L}_{\mathbf{D}}$ and $\partial_{\beta} \mathcal{L}_{\mathbf{D}}$ are treated as constants that can be computed prior the back-propagation step. This way, the correct gradients can be used for the derivatives of prior layers :

$$\overline{z_1^N} = \frac{\partial \mathcal{L}_{\mathbf{D}}'(\alpha, \beta)}{\partial z_1^N} = \frac{\partial \mathcal{L}_{\mathbf{D}}'}{\partial \alpha} = \partial_{\alpha} \mathcal{L}_{\mathbf{D}} \quad (\text{D.9})$$

$$\overline{z_2^N} = \frac{\partial \mathcal{L}_{\mathbf{D}}'(\alpha, \beta)}{\partial z_2^N} = \frac{\partial \mathcal{L}_{\mathbf{D}}'}{\partial \beta} = \partial_{\beta} \mathcal{L}_{\mathbf{D}} \quad (\text{D.10})$$

An alternative approach to tackle the gradient issue is to employ a dose distribution calculation method that supports differentiability. This feature is still an on-going project as part of JulianA, and its validation is still ongoing¹. Flux.jl provides the flexibility to redefine gradient computations for specific components. This adaptation could enable the integration of the aforementioned differentiable dose calculation into the training pipeline, albeit necessitating the complete computation of the distribution at each training iteration.

¹Developing a differentiable fluence matrix GPU kernel utilising Automatic Differentiation within the Julia programming language constituted my first project at PSI, prior to this master's thesis.

E

Unsupervised Learning - Activation Maps

The activation maps are presented for the Inception and Reduced models trained on the initial dataset only. The colorbars ticks are shared by all the plots and are thus represented only once by group of activation map. The white parts of the colorbar are values that are not covered in the corresponding activation maps. The ticks' values of the colorbar are shared by all the activation maps but the colour gradients are unique to each of them. In all figures, slice 50 of the 3D activation map is represented.

E.1. Inception model

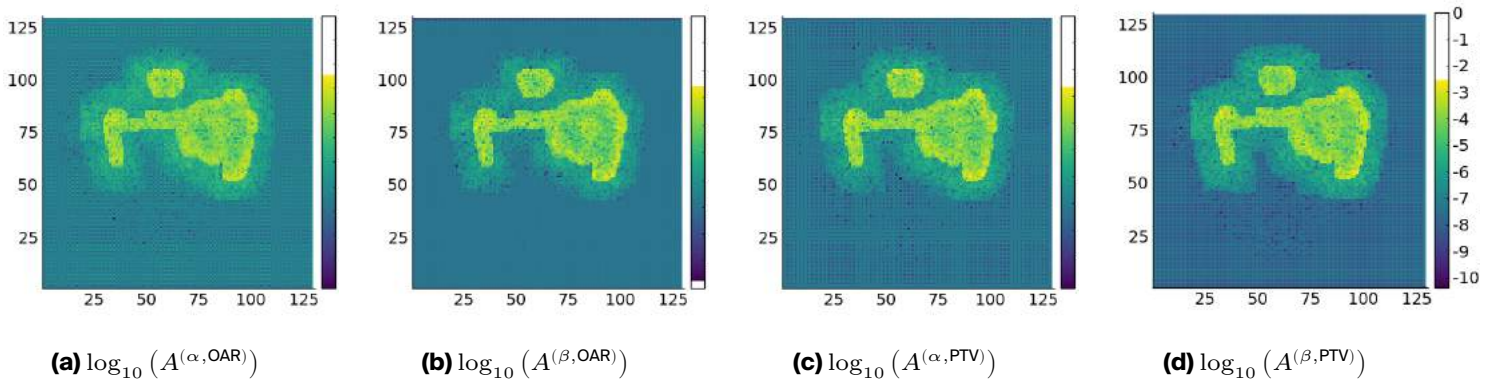


Figure E.1: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the unsupervised Inception model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs.

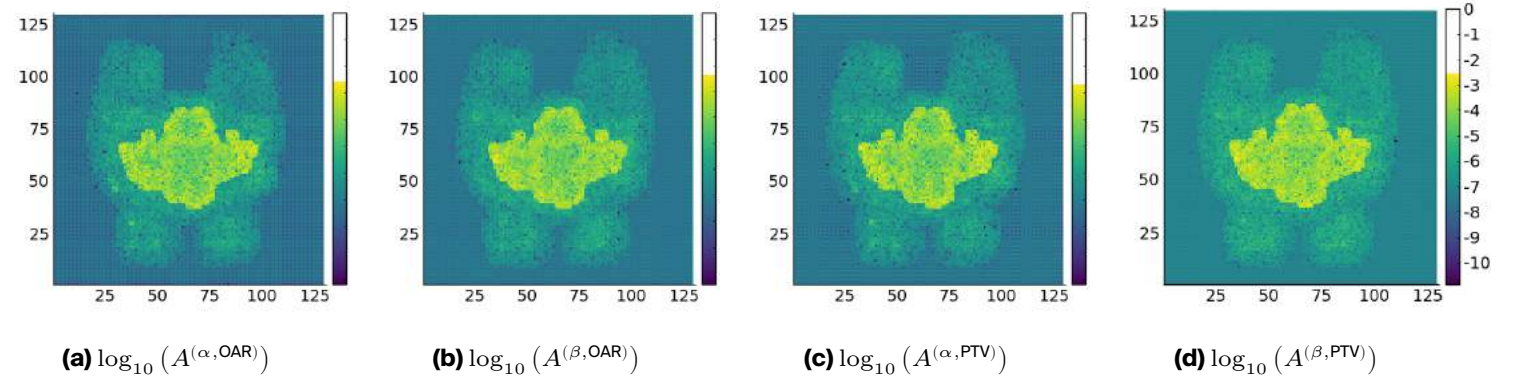


Figure E.2: Gradient-based activation maps when predicting the angles for train patient with ID=25. The model selected is the unsupervised Inception model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs.

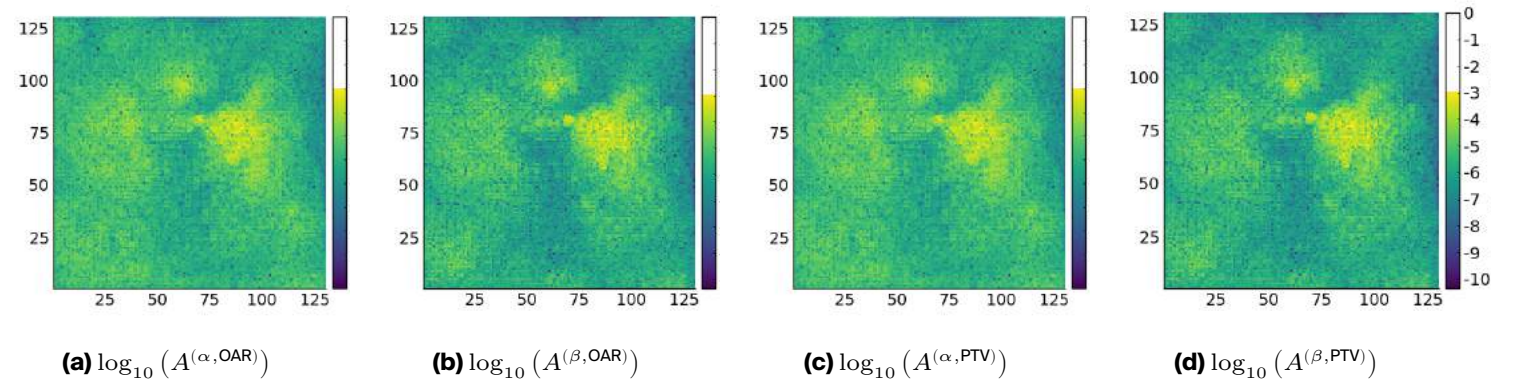


Figure E.3: Gradient-based activation maps when predicting the angles for train patient with ID=25. The model selected is the unsupervised Inception model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs.

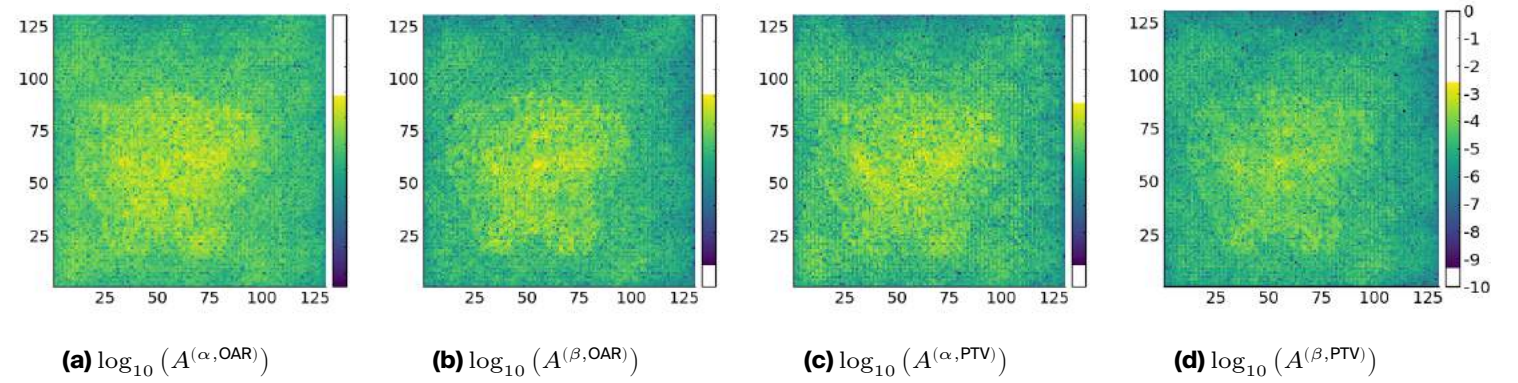


Figure E.4: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the unsupervised Inception model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs.

E.2. Reduced Model

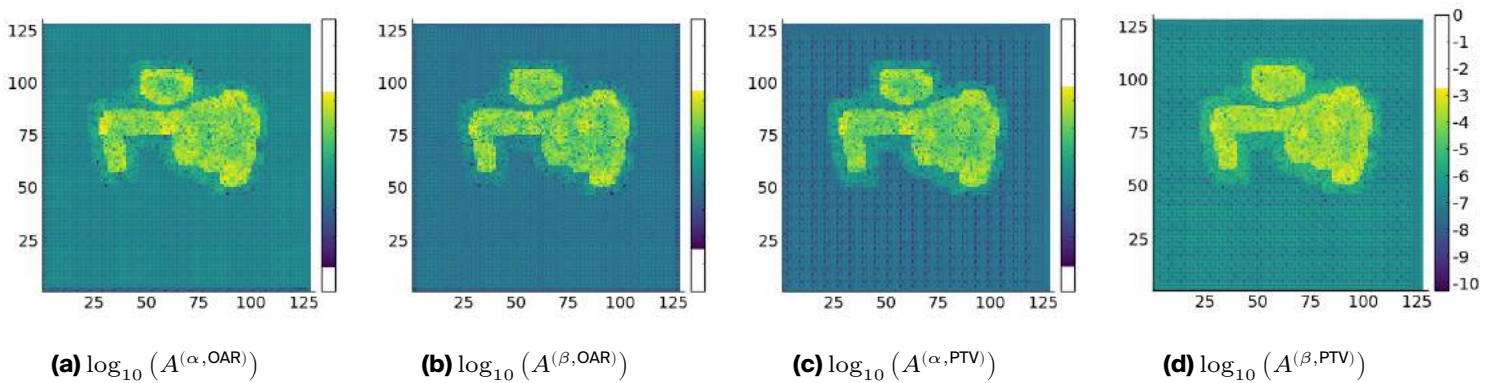


Figure E.5: Gradient-based activation maps when predicting the angles for train patient with ID=25. The model selected is the unsupervised Reduced model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs.

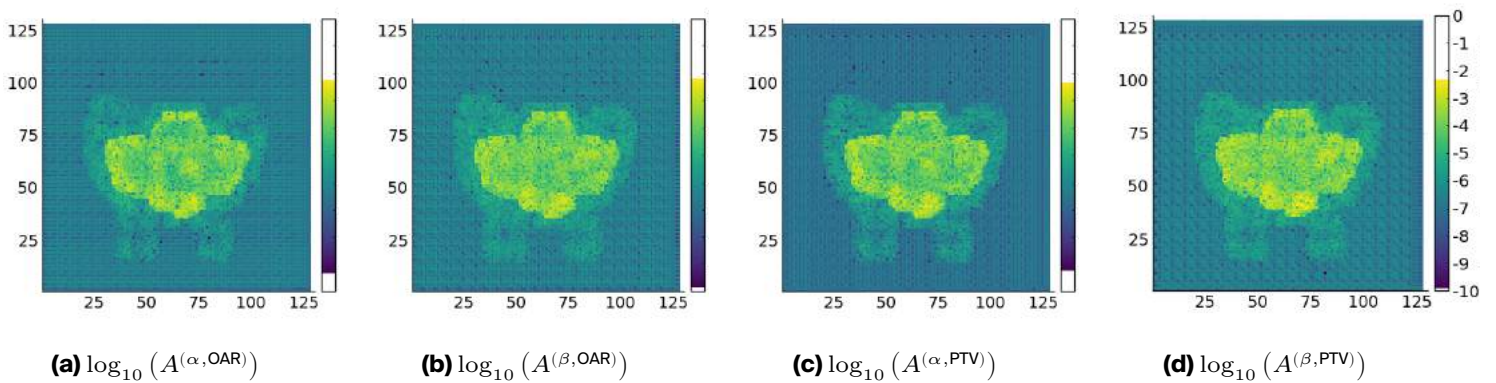


Figure E.6: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the unsupervised Reduced model with the initial dataset only, selected with the best mean test loss across all patients. Binary masks are inputs.

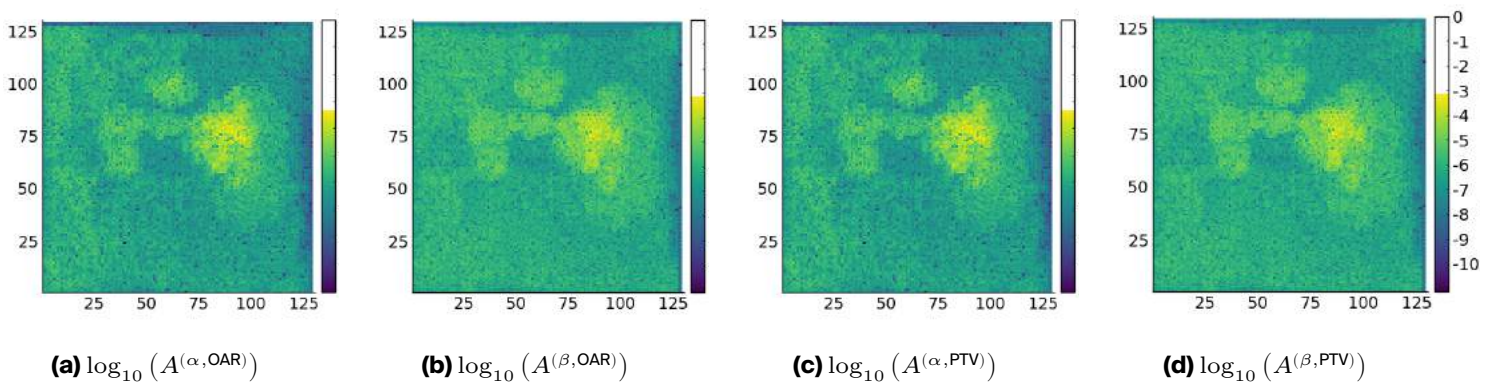


Figure E.7: Gradient-based activation maps when predicting the angles for train patient with ID=25. The model selected is the unsupervised Reduced model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs.

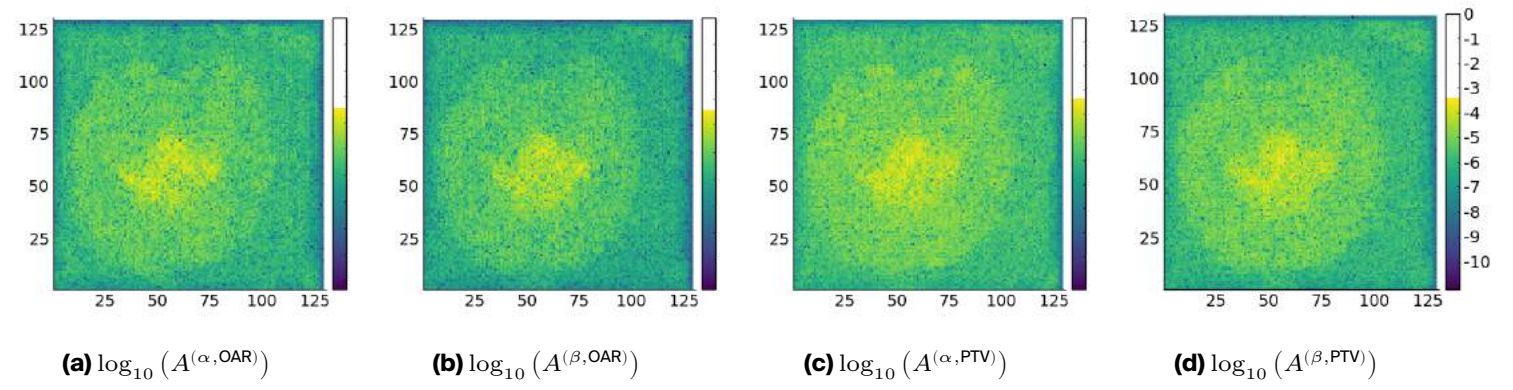


Figure E.8: Gradient-based activation maps when predicting the angles for test patient with ID=0. The model selected is the unsupervised Reduced model with the initial dataset only, selected with the best mean test loss across all patients. Distance masks are inputs.

F

Dose Descent Patients

F.1. Prescriptions

Definitions

- D_{2cc} : dose to maximally exposed 2 cm³ of the OAR / $D_{90\%}$ Gross Tumor Volume / High Risk CTV
- $D_{X\%}$: percentage of the minimum dose that covered $X\%$ of the target volume
- D_{max} : Maximum dose delivered
- D_{mean} : Mean dose delivered
- $V_{95\%}$: percentage of PTV volume receiving $\geq 95\%$ of its prescribed dose.

structure name	Soft or Hard constraint	constraint quantity	constraint threshold
Brainstem	-	$D_{2\%}$	63 Gy
CBS	-	$D_{2\%}$	54 Gy
Chiasm	-	D_{mean}	54 Gy
Chiasm	-	$D_{2\%}$	60 Gy
Left Optic Nerve	-	$D_{2\%}$	60 Gy
Right Optic Nerve	-	$D_{2\%}$	60 Gy
Left Temporal Lobe	-	D_{max}	103.09%
Right Temporal Lobe	-	D_{max}	103.09%
Left Lens	-	D_{mean}	10 Gy
Right Lens	-	D_{mean}	10 Gy
Left Lacrimal Gland	-	D_{mean}	25 Gy
Right Lacrimal Gland	-	D_{mean}	25 Gy
Left Cochlea	-	D_{mean}	25 Gy
Right Cochlea	-	D_{mean}	25 Gy

Table F.1: Prescriptions of train patient, ID = 0

structure name	Soft or Hard constraint	constraint quantity	constraint threshold
Hippocampus Left	unknown	$D_{40\%}$	7.3 Gy
OpticTract >80% (H)	hard	D_{max}	59.4 Gy
OpticTract >80% (H)	hard	$D_{30\%}$	30%

Table F.2: Prescriptions of train patient, ID = 01

structure name	Soft or Hard constraint	constraint quantity	constraint threshold
unknown	hard	$D_{2\%}$	60.8Gy
Chiasm	hard	$D_{2\%}$	10Gy
Cornea Left	hard	D_{mean}	30 Gy
Cornea Left	hard	$D_{2\%}$	50Gy
Cornea Right	hard	$D_{2\%}$	60.8Gy
Eye Left	hard	$D_{2\%}$	50Gy
Eye Right	hard	$D_{2\%}$	103%
Lacrymal Gland Left	soft	D_{mean}	15 Gy
Lacrymal Gland Right	soft	D_{mean}	35 Gy
Lens Left	soft	D_{mean}	7 Gy
Lens Right	hard	$D_{2\%}$	60Gy
Macula Left	hard	$D_{2\%}$	25Gy
Macula Right	hard	D_{mean}	55 Gy
Optic Nerve Left	hard	$D_{2\%}$	50Gy
Optic Nerve Right	hard	$D_{2\%}$	58Gy
Pituitary	soft	D_{mean}	10 Gy
Retina Left	hard	D_{mean}	20 Gy
Retina Left	hard	$D_{2\%}$	40Gy
Retina Right	hard	D_{mean}	50 Gy
Retina Right	hard	$D_{2\%}$	103%

Table F.3: Prescriptions of train patient, ID = 09

here

structure name	Soft or Hard constraint	constraint quantity	constraint threshold
Brainstem	hard	D_{max}	57.78 Gy
Chiasm	hard	D_{max}	56 Gy
Eyes	soft	D_{max}	45 Gy
Hippocampi	soft	$D_{40\%}$	7.3 Gy
Hypothalamus	soft	D_{max}	54 Gy
Lenses	hard	D_{max}	5 Gy
Cochleae	hard	D_{max}	45 Gy
Pituitary Gland	soft	D_{mean}	45 Gy
Temporal Lobes	soft	D_{max}	57.8 Gy

Table F.4: Prescriptions of train patient, ID = 10

structure name	Soft or Hard constraint	constraint quantity	constraint threshold
Eye Left	-	D _{2%}	54 Gy
Eye Right	-	D _{2%}	64 Gy
Lacrimal Gland Left	-	D _{mean}	30 Gy
Lacrimal Gland Right	soft	D _{mean}	40 Gy
Optic Nerve Right	-	D _{2%}	60 Gy
Optic Nerve Left	-	D _{2%}	56 Gy
Chiasm	soft	D _{mean}	50 Gy
Chiasm	soft	D _{2%}	60 Gy
Brain stem	-	D _{2%}	64 Gy
CBS	-	D _{2%}	54 Gy
Spinal cord surface	-	D _{2%}	54 Gy
Cochlea Right	soft	D _{mean}	45 Gy
Cochlea Left	-	D _{mean}	30 Gy
Temporal Lobe Left	-	D _{2cc}	74 Gy
Temporal Lobe Right	-	D _{2cc}	74 Gy
Hippocampi	-	D _{40%}	7.3 Gy
Hippocampi	-	D _{40%}	7.3 Gy
TMJ's Left	-	D _{2%}	70 Gy
TMJ's Right	-	D _{2%}	70 Gy
Parotis Left	-	D _{mean}	21 Gy
Parotis Right	-	D _{mean}	20 Gy

Table F.5: Prescriptions of train patient, ID = 15

structure name	Soft or Hard constraint	constraint quantity	constraint threshold
Lens Left	-	D _{2%}	7 Gy
Lens Right	-	D _{2%}	0 Gy
Eye Left	-	D _{mean}	35 Gy
Eye Left	-	D _{2%}	45 Gy
Eye Right	-	D _{2%}	0 Gy
Optic Nerve Left	-	D _{mean}	50 Gy
Optic Nerve Left	-	D _{2%}	54 Gy
Optic Nerve Right	-	D _{2%}	0 Gy
Chiasm	-	D _{mean}	50 Gy
Chiasm	-	D _{2%}	54 Gy
Lacrimal Gland Left	-	D _{mean}	25 Gy
Lacrimal Gland Left	-	D _{2%}	35 Gy
Lacrimal Gland Right	-	D _{2%}	0 Gy
Brainstem, surface	-	D _{2%}	54 Gy
Brainstem, center	-	D _{2%}	50 Gy
Cochlea Left	-	D _{mean}	25 Gy
Cochlea Left	-	D _{2%}	36 Gy
Cochlea Right	-	D _{2%}	0 Gy
Pituitary	-	D _{mean}	40 Gy
Pituitary	-	D _{2%}	45 Gy
Hippocampus Left	soft	D _{40%}	7.3 Gy
Hippocampus Right	-	D _{40%}	7.3 Gy
Temporal Lobe Left	-	D _{2%}	103.09%
Temporal Lobe Right	-	D _{2%}	103.09%

Table F.6: Prescriptions of train patient, ID = 33

F.2. Fields Incidences

The beams from the Dose Descent plans and from the Reference plans are illustrated and compared in the following figures. They take the form of red, blue and yellow pencils. The views are always presented in the same order, from left to right : axial view from above, frontal coronal view, sagittal view (eyes heading to the left to the page, except for patient 33). The structures represented are the ones selected for the DVHs except the eyes which are always both represented for visualisation purposes.

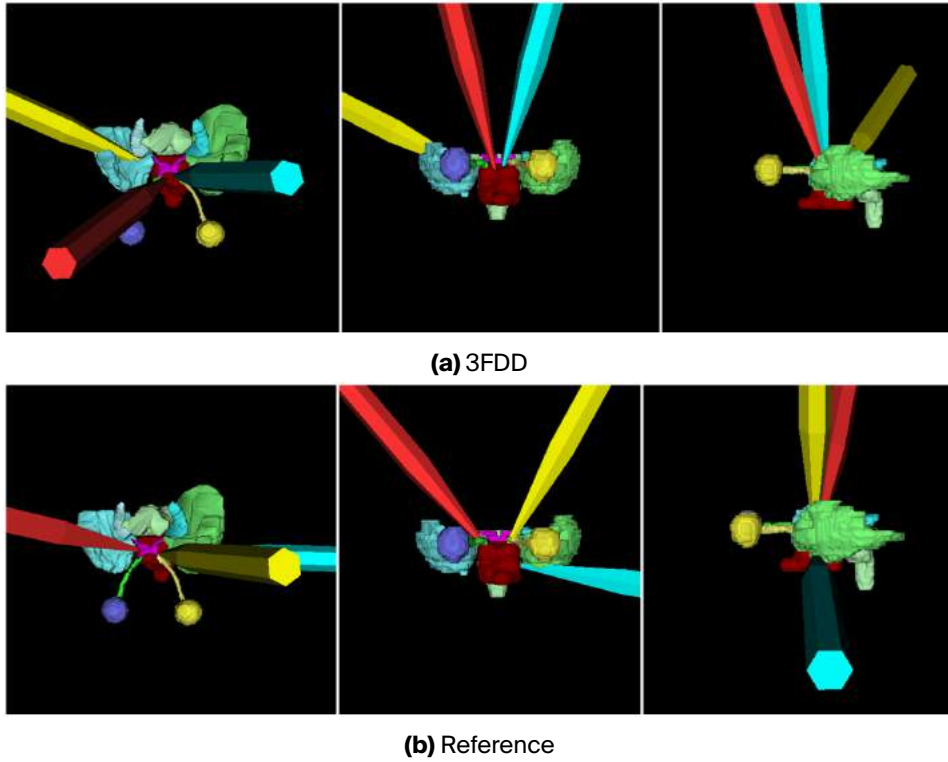


Figure F.1: Fields directions comparison between predicted and reference plans. Training patient with ID = 0 is considered.

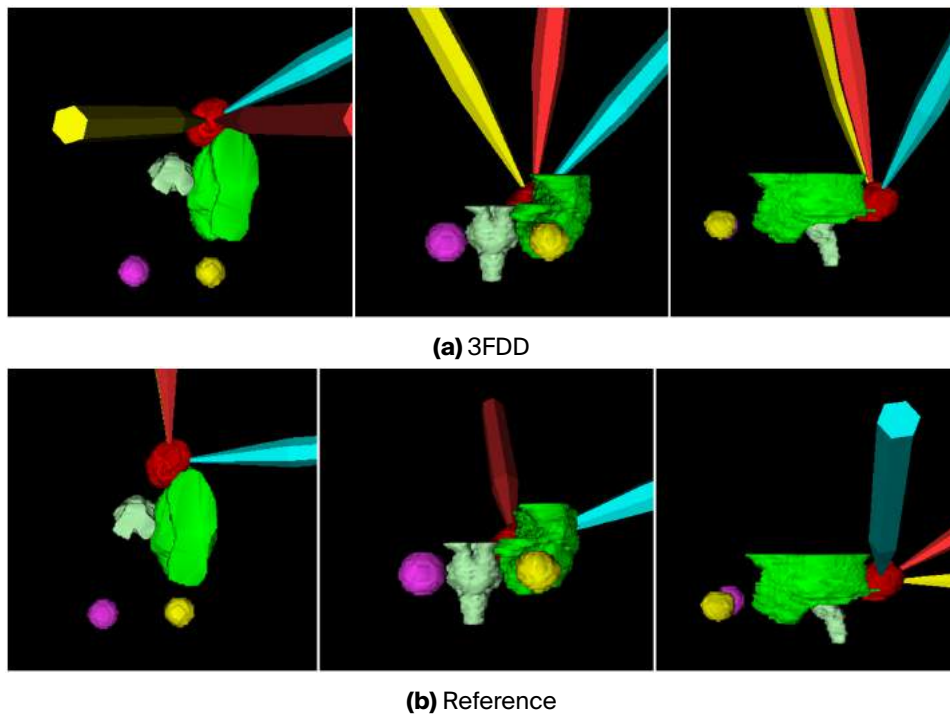


Figure F.2: Fields directions comparison between predicted and reference plans. Training patient with ID = 1 is considered.

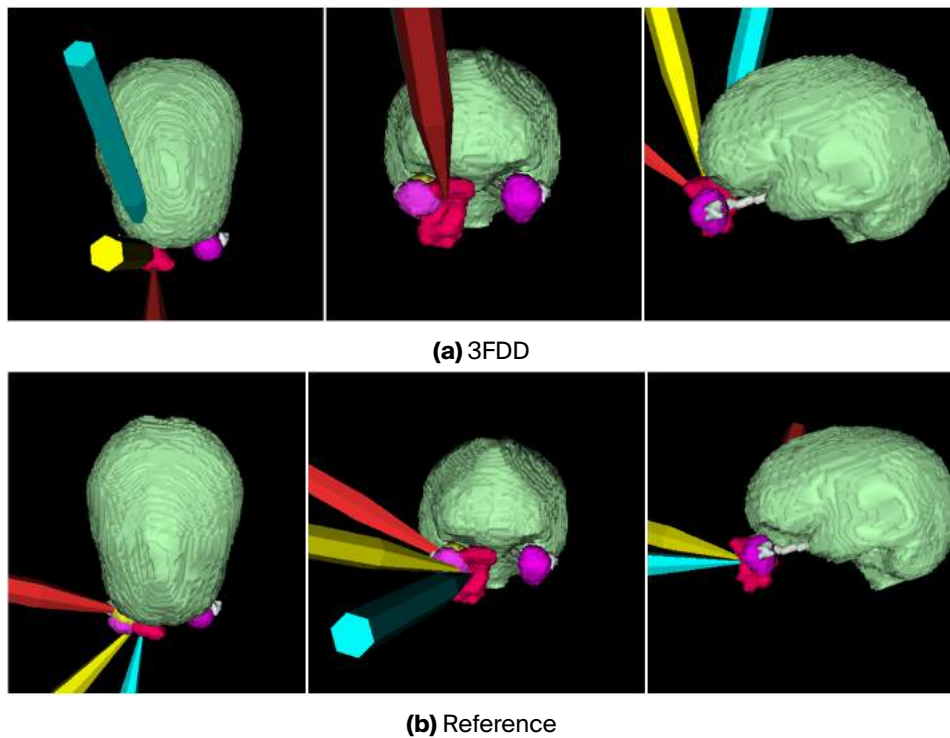


Figure F.3: Fields directions comparison between predicted and reference plans. Training patient with ID = 9 is considered.

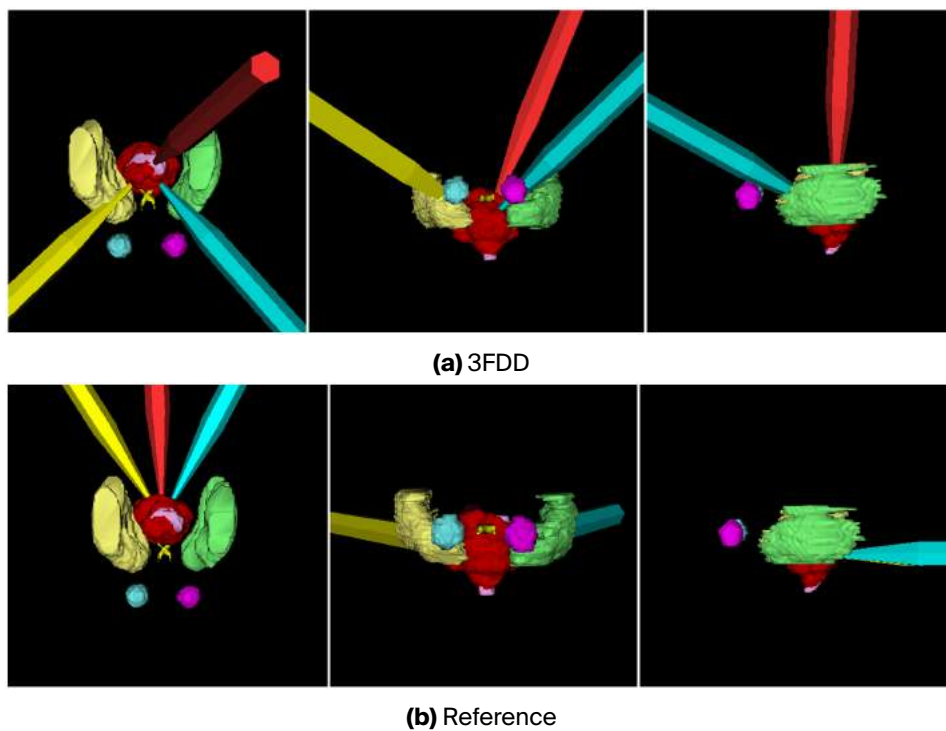


Figure F.4: Fields directions comparison between predicted and reference plans. Training patient with ID = 10 is considered.

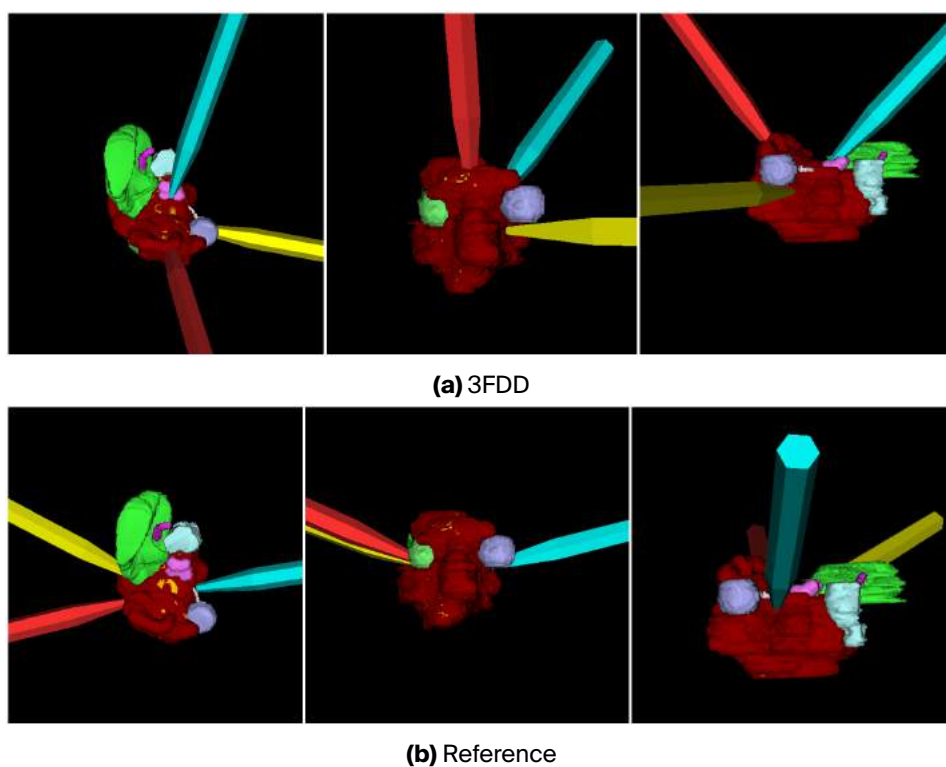


Figure F.5: Fields directions comparison between predicted and reference plans. Training patient with ID = 15 is considered.

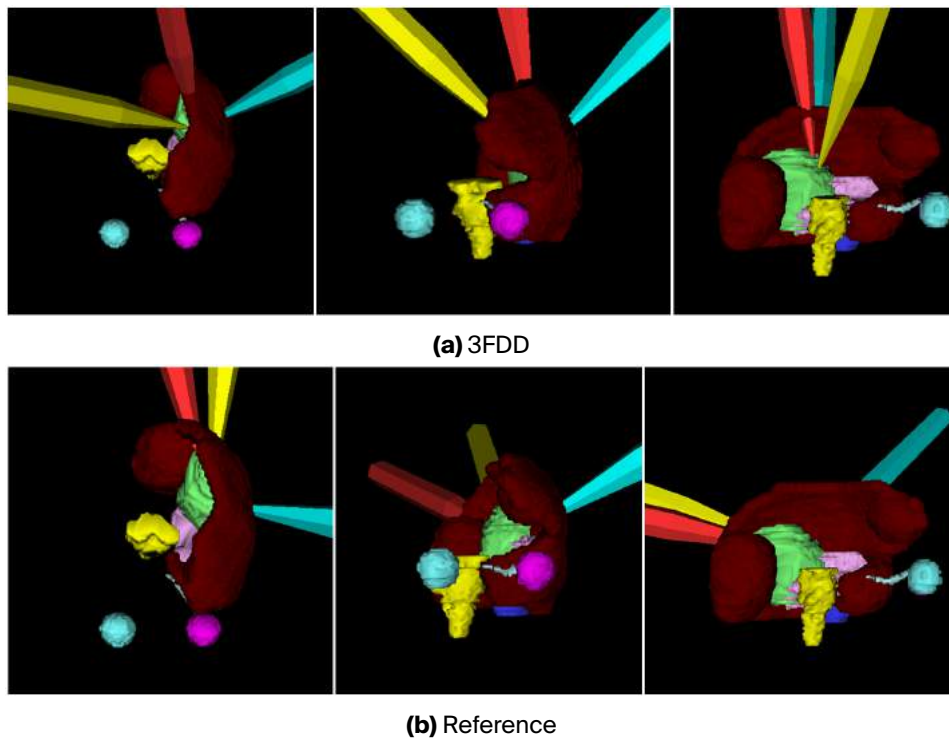


Figure F.6: Fields directions comparison between predicted and reference plans. Training patient with ID = 33 is considered.

F.3. Dose Distributions Comparison

The dose distributions are always presented in the following format. From left to right : axial view, coronal view, sagittal view. The methods used from top to bottom : clinical reference, 3FDD SFO, 3FDD MFO, reference fields recalculated with SFO and reference fields recalculated with MFO.

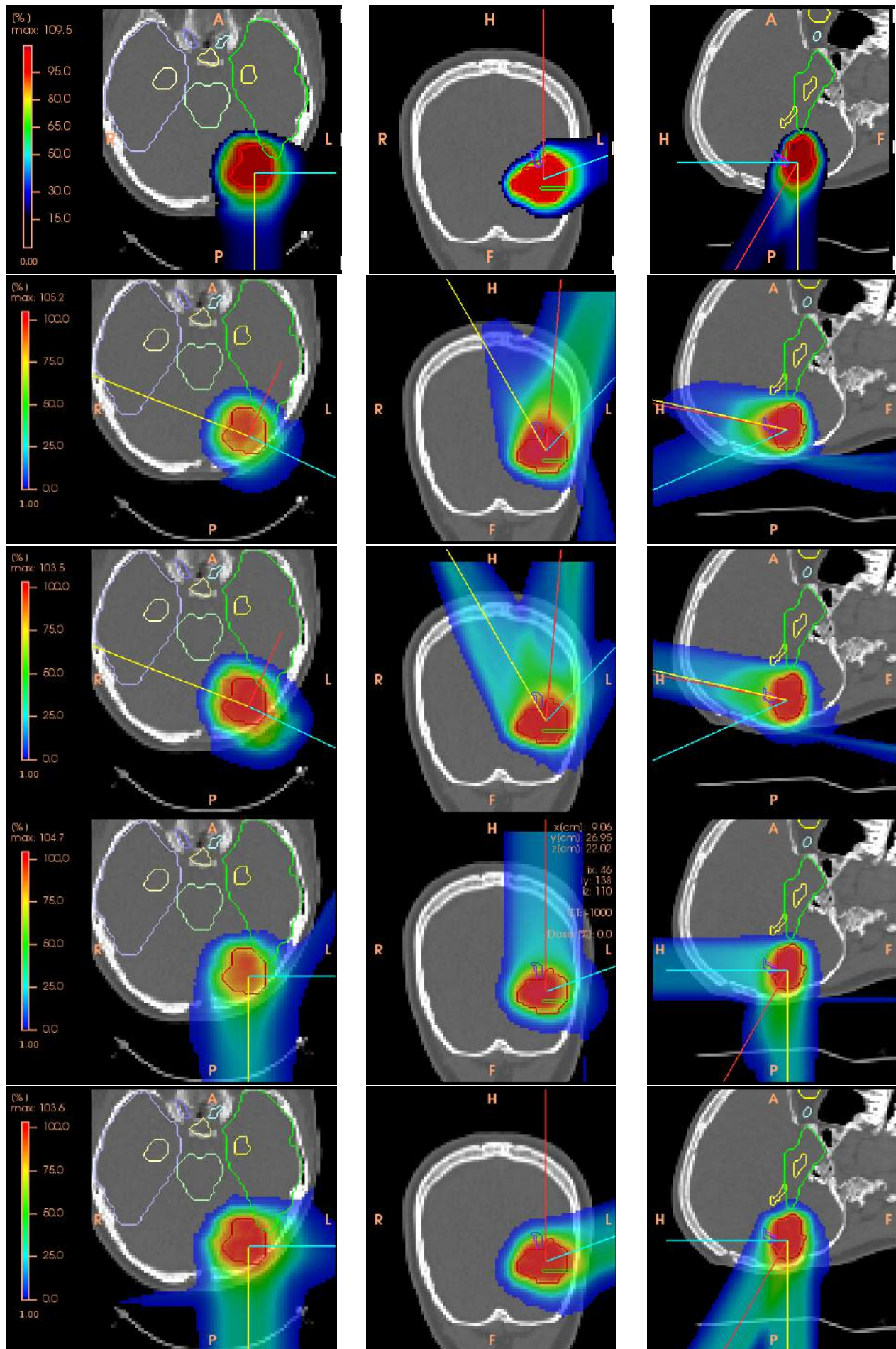


Figure F.7: Comparison of the dose distributions calculated with 5 methods. The dose distributions are calculated for training patient with ID = 1 and presented overlapping the patient's CT scan.

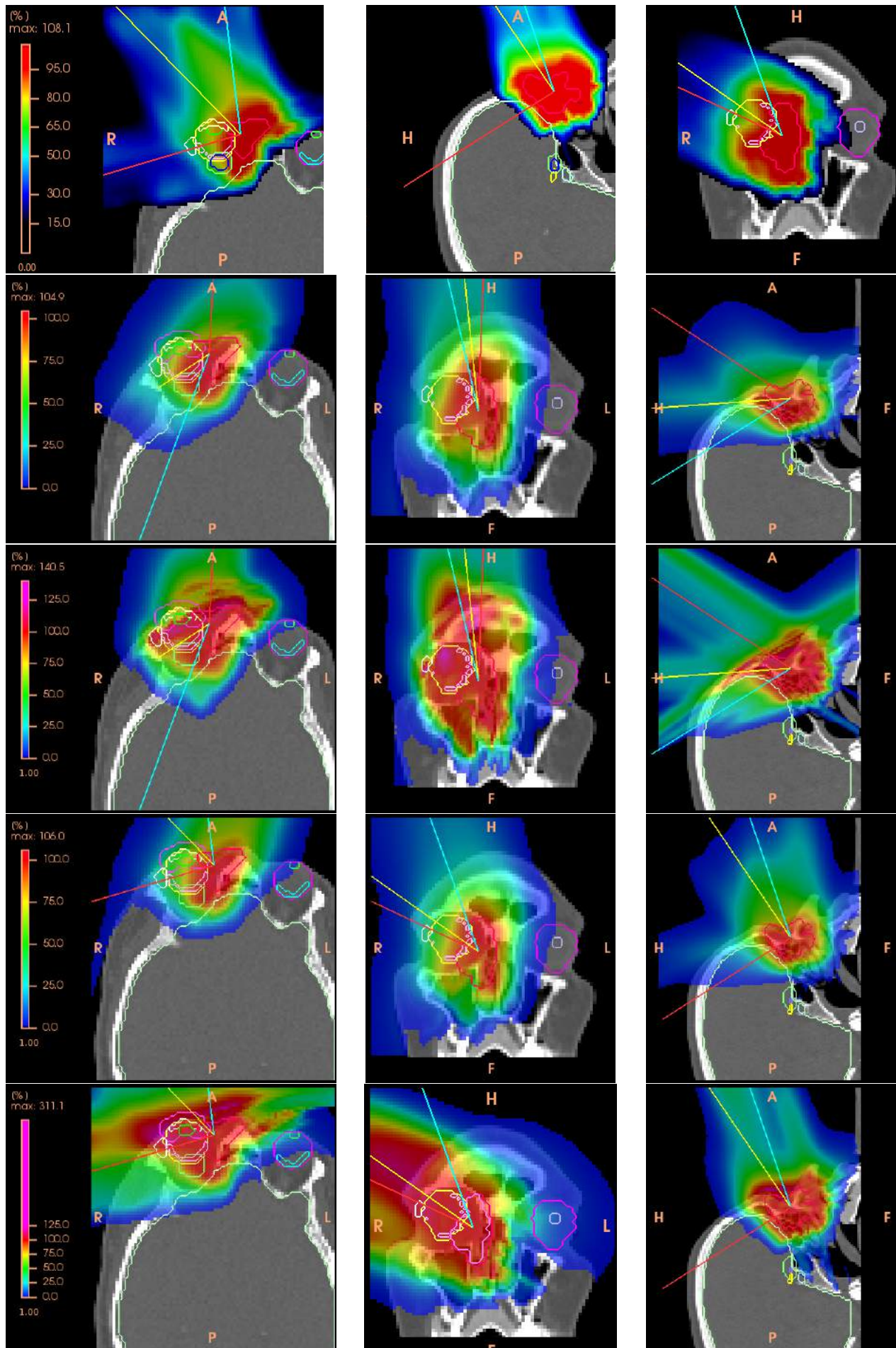


Figure F.8: Comparison of the dose distributions calculated with 5 methods. The dose distributions are calculated for training patient with ID = 09 and presented overlapping the patient's CT scan.

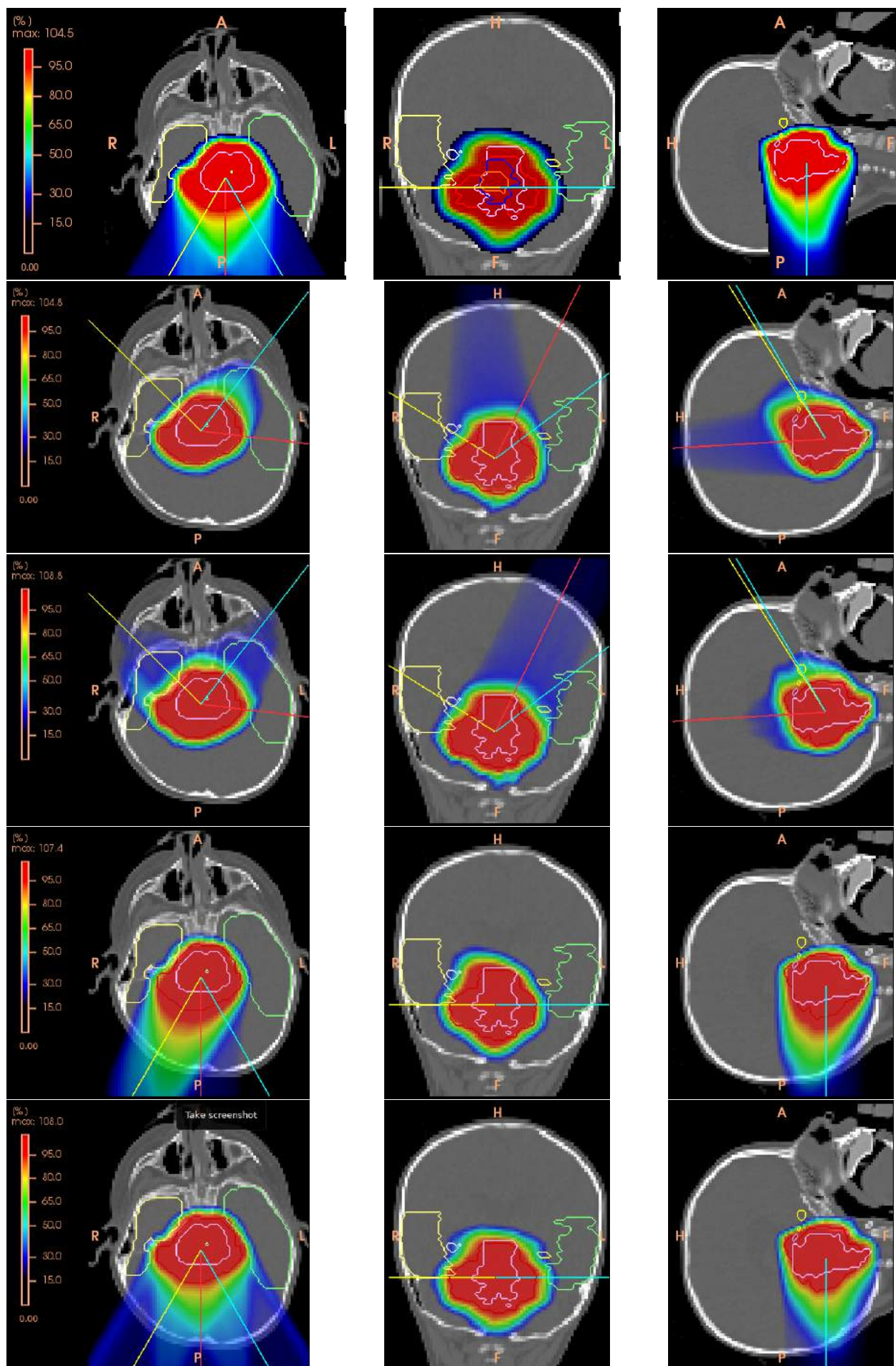


Figure F.9: Comparison of the dose distributions calculated with 5 methods. The dose distributions are calculated for training patient with ID = 10 and presented overlapping the patient's CT scan.

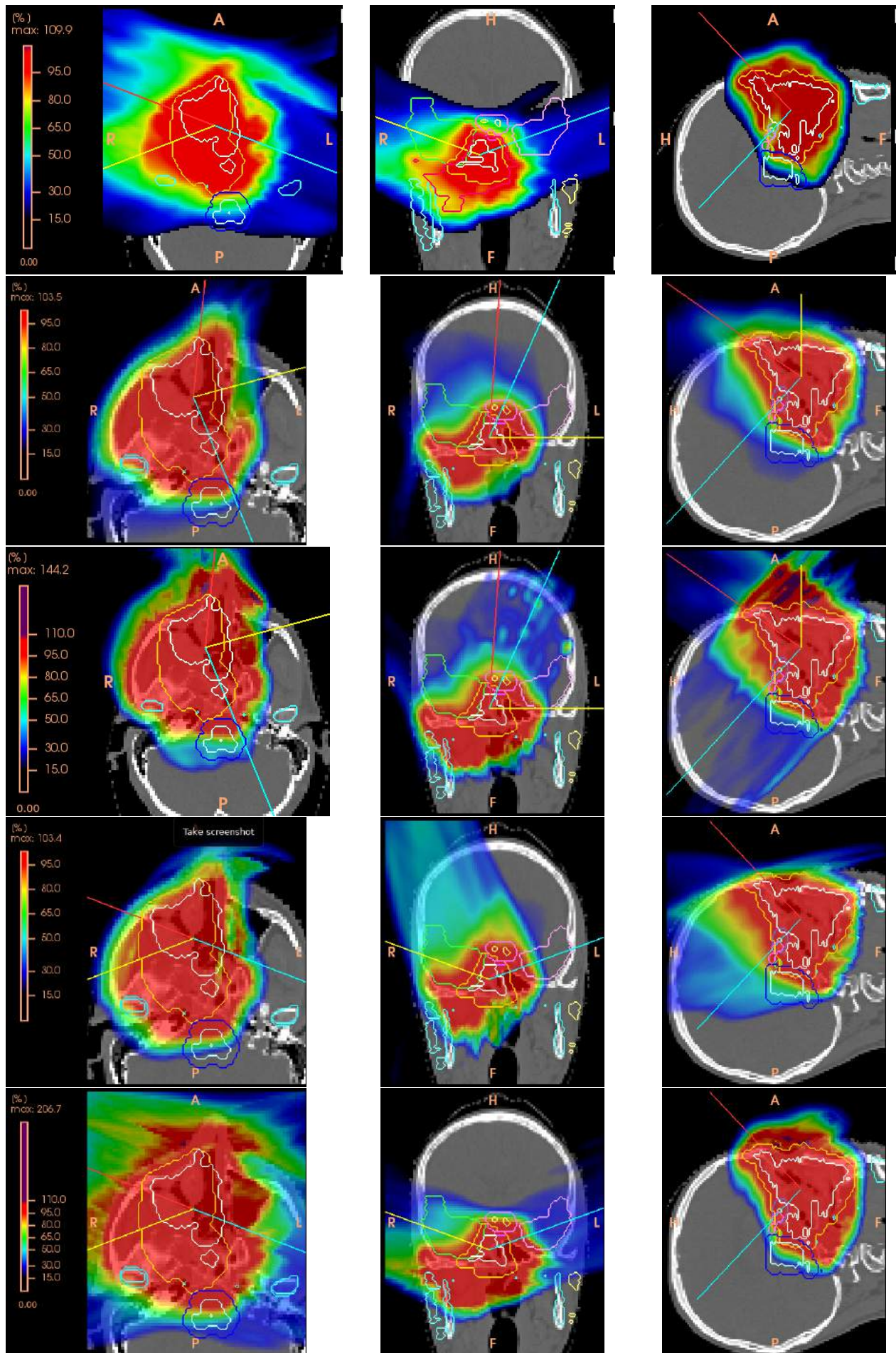


Figure F.10: Comparison of the dose distributions calculated with 5 methods. The dose distributions are calculated for training patient with ID = 15 and presented overlapping the patient's CT scan.

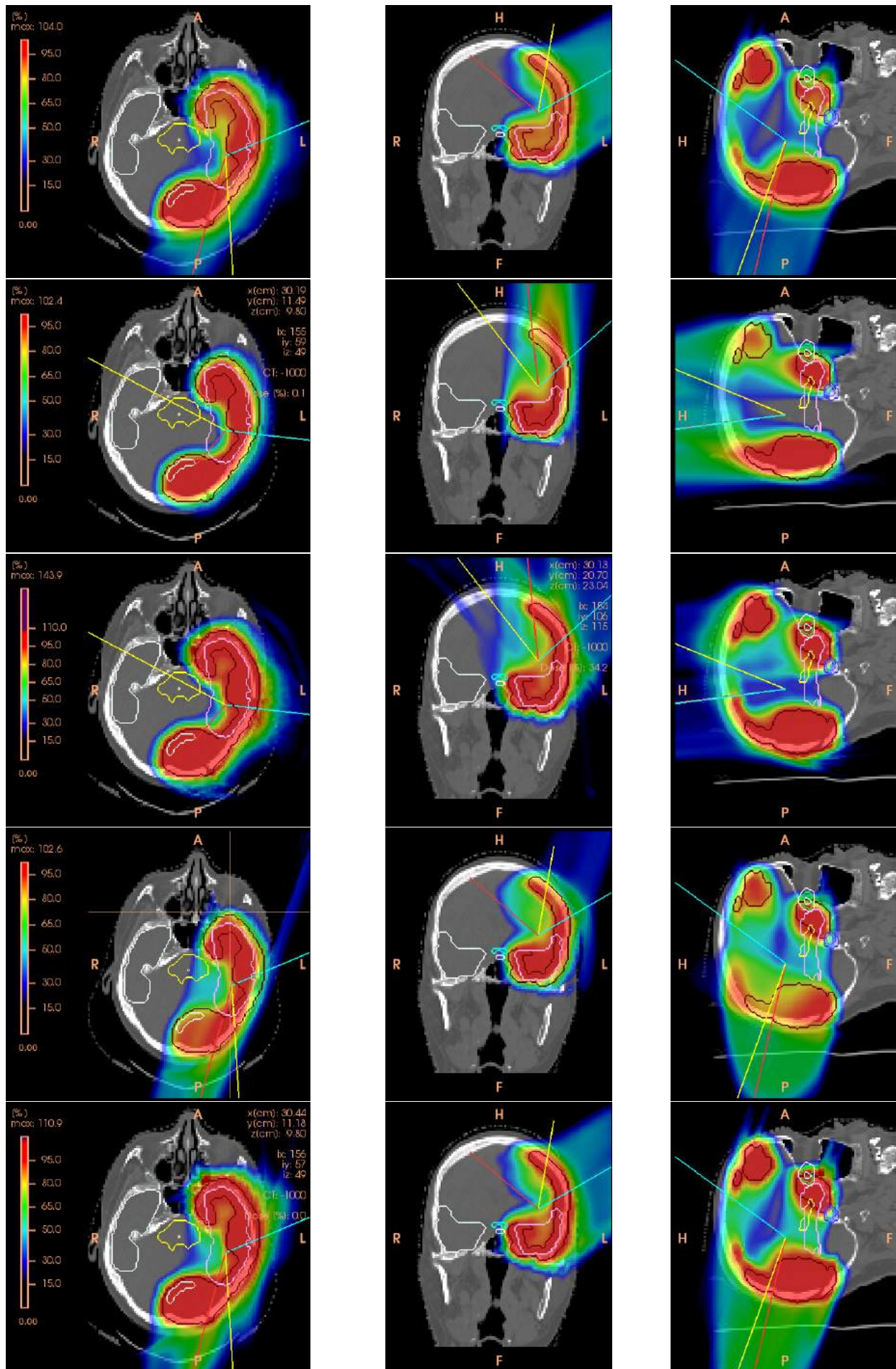


Figure F.11: Comparison of the dose distributions calculated with 5 methods. The dose distributions are calculated for training patient with ID = 33 and presented overlapping the patient's CT scan.

F.4. Dose Volume Histograms

The Dose Volume histograms of training patients with $ID \in \{1, 9, 10, 11, 15, 33\}$ are presented in the following.

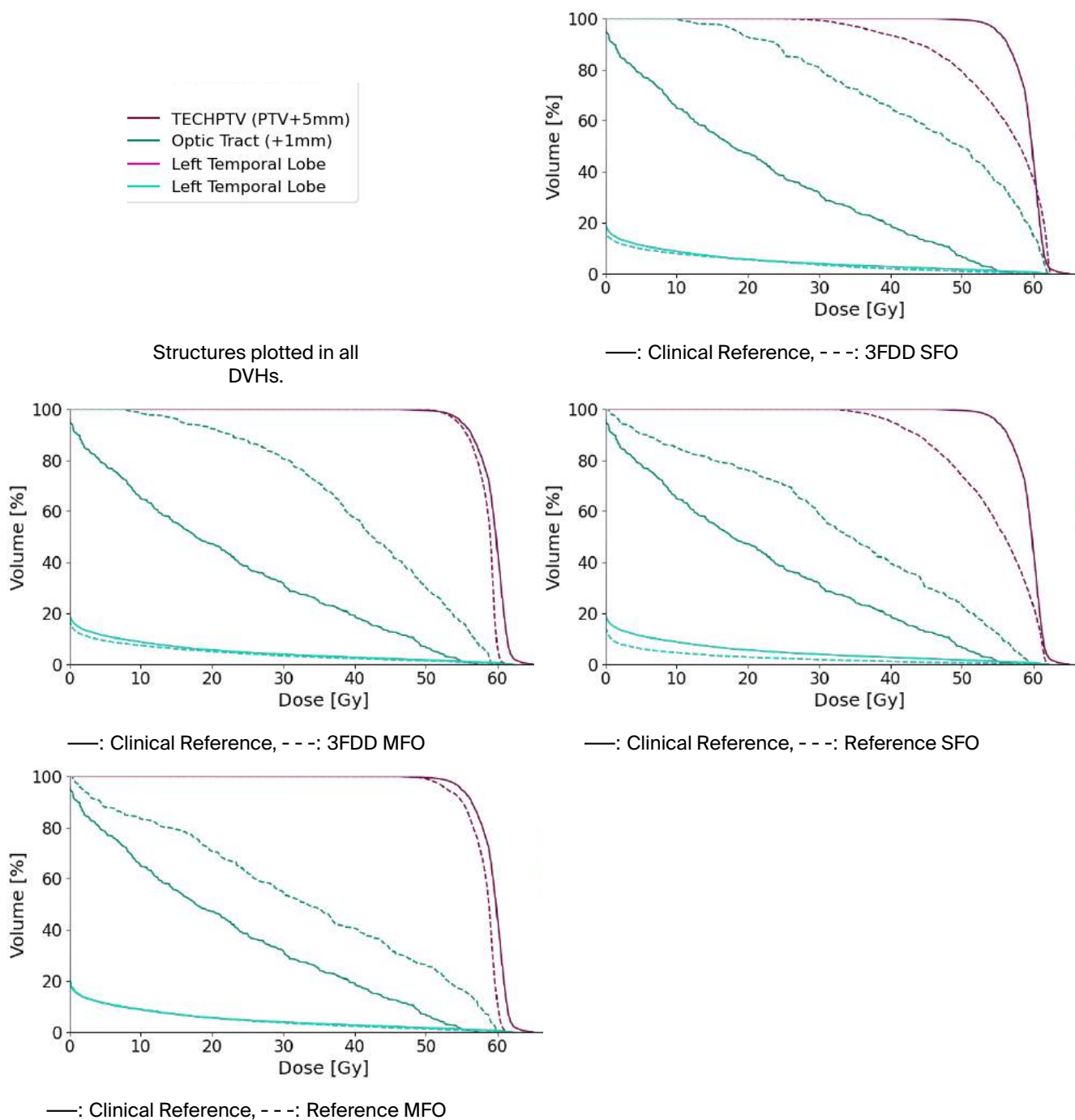


Figure F.12: Dose volume Histograms comparison of each plan strategy to the clinical reference. The patient considered is training patient with ID=1.

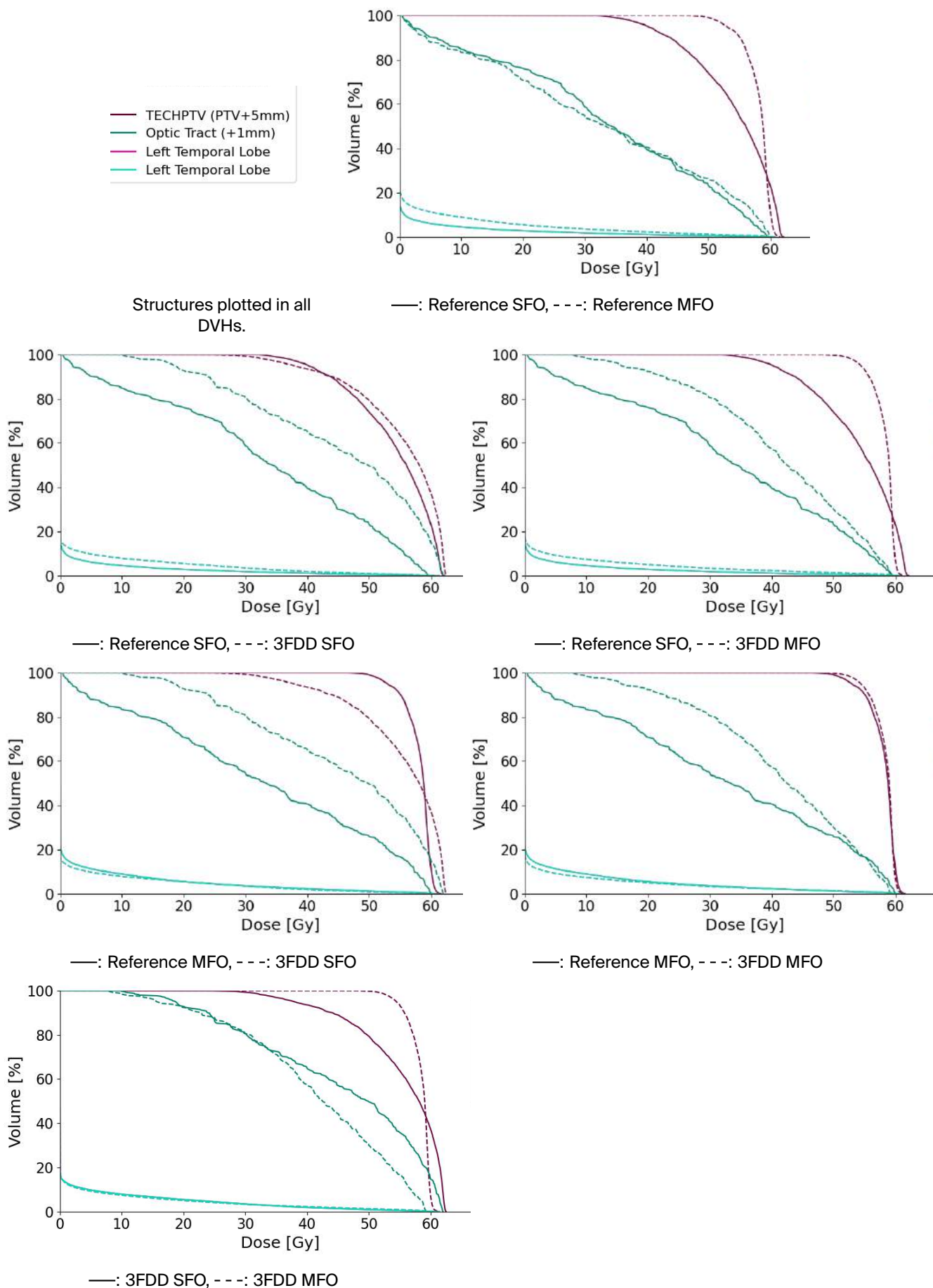


Figure F.13: Dose volume Histograms comparison of Reference SFO and MFO with 3FDD SFO and MFO. The patient considered is training patient with ID=1.

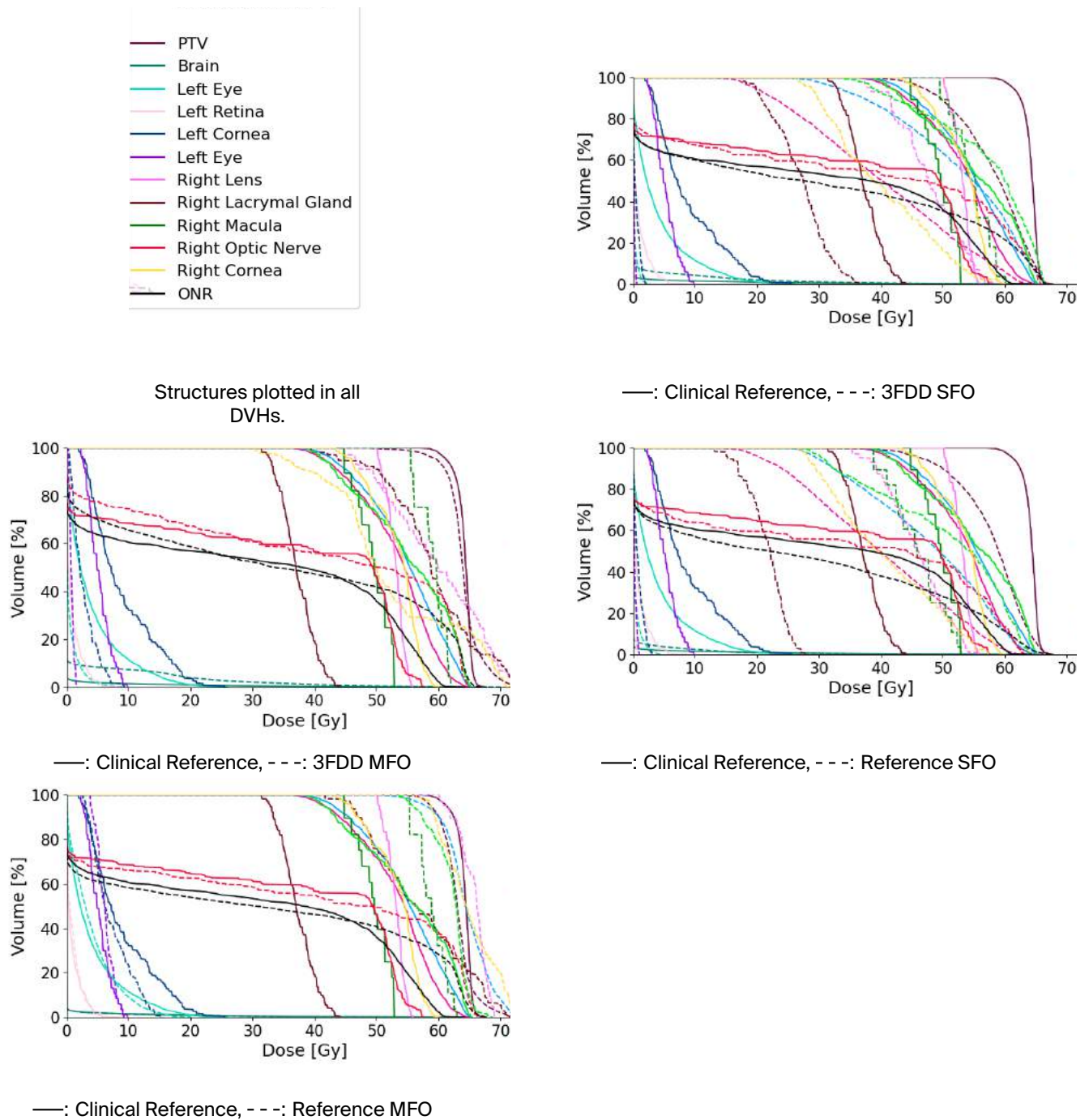


Figure F.14: Dose volume Histograms comparison of each plan strategy to the clinical reference. The patient considered is training patient with ID=9.

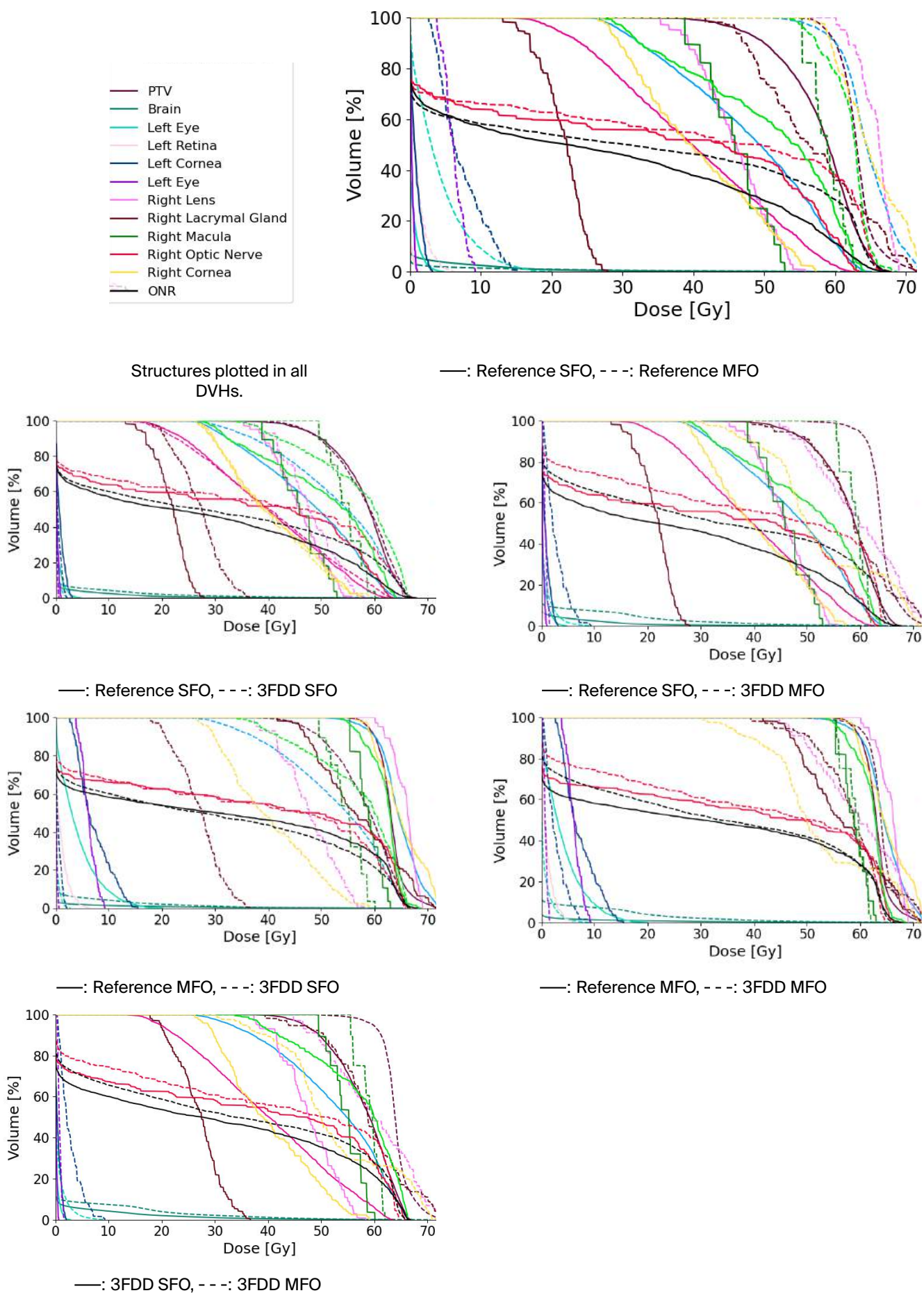


Figure F.15: Dose volume Histograms comparison of Reference SFO and MFO with 3FDD SFO and MFO. The patient considered is training patient with ID=9.

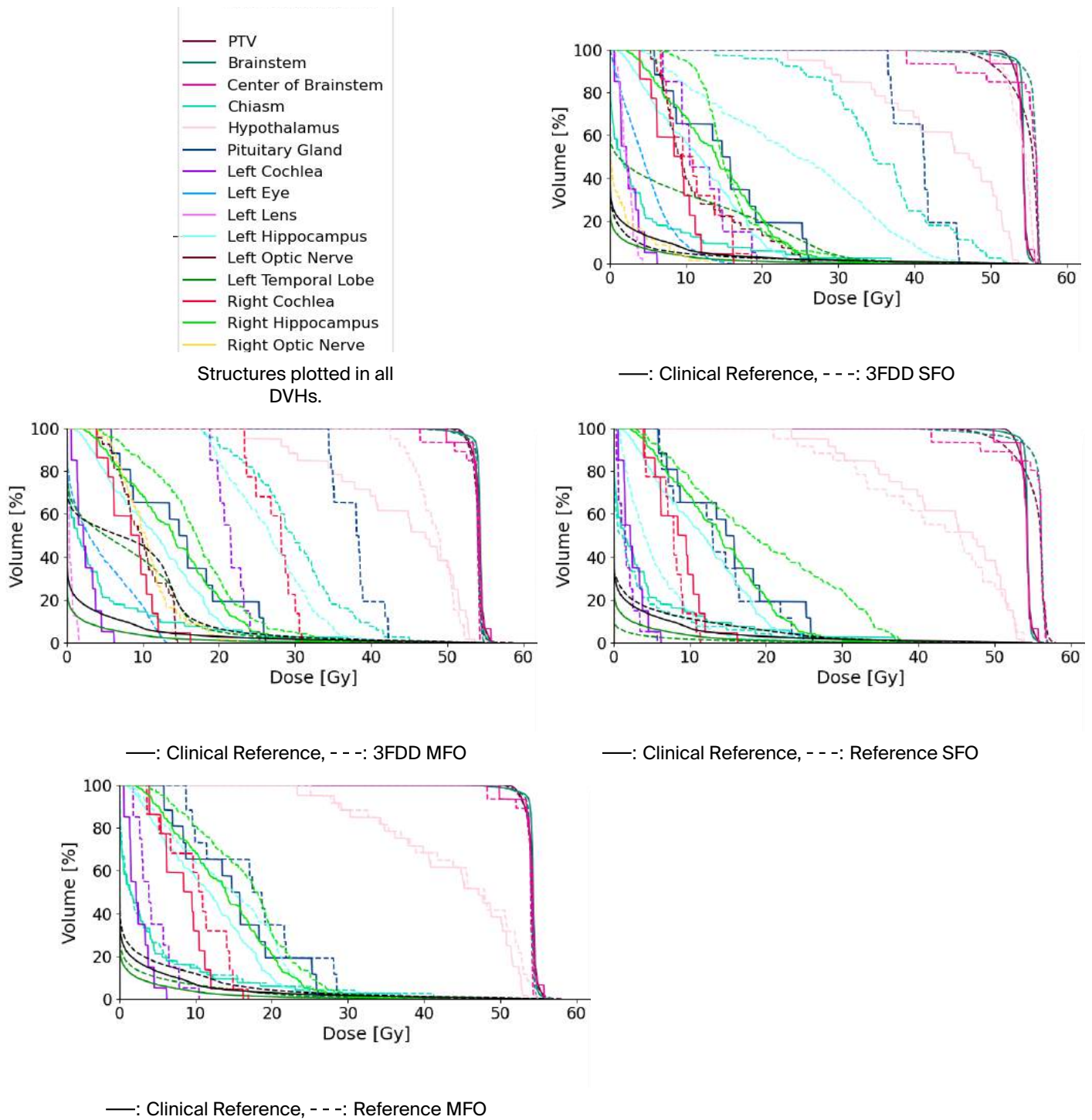


Figure F.16: Dose volume Histograms comparison of each plan strategy to the clinical reference. The patient considered is training patient with ID=10.

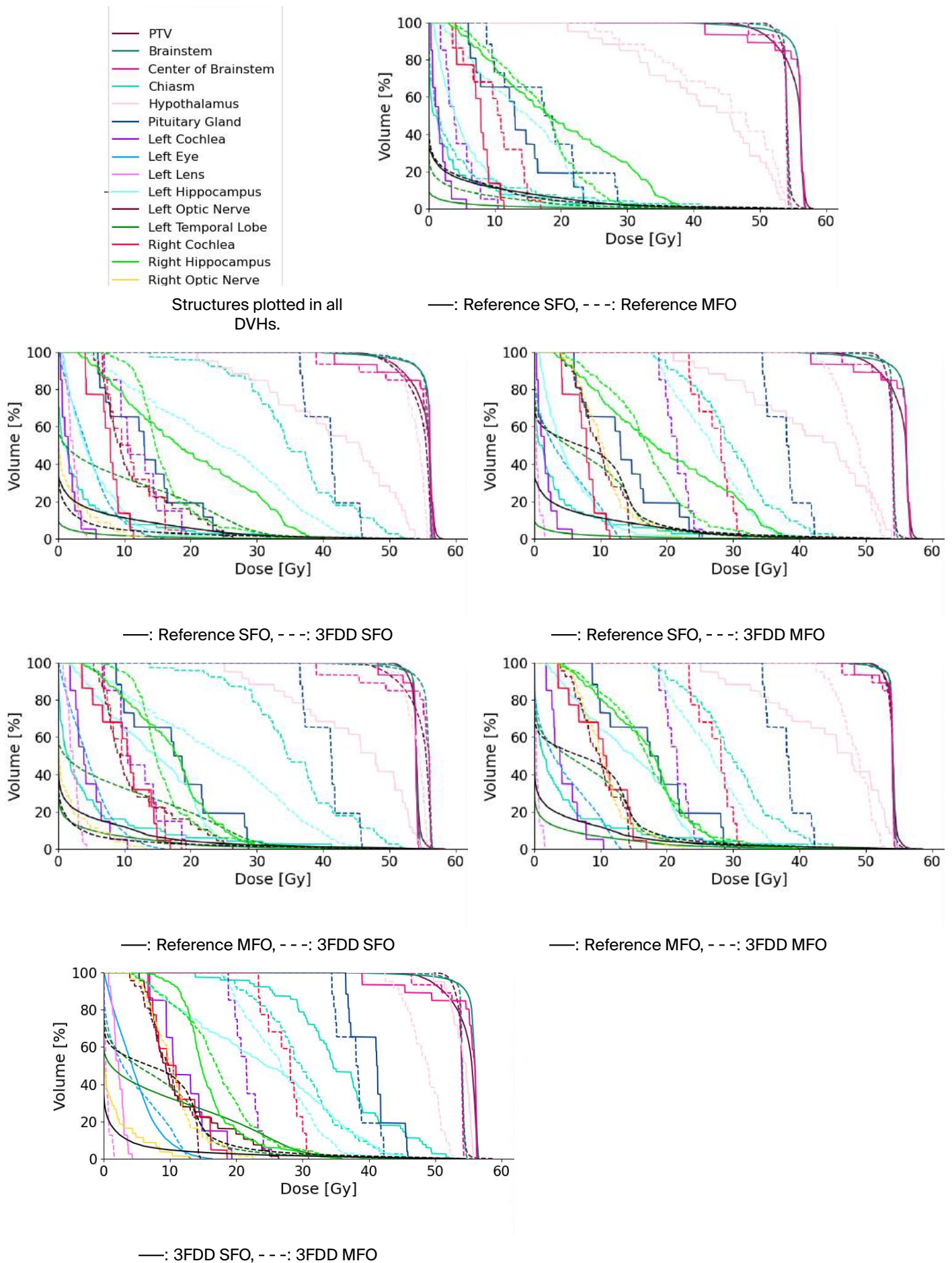


Figure F.17: Dose volume Histograms comparison of Reference SFO and MFO with 3FDD SFO and MFO. The patient considered is training patient with ID=10.

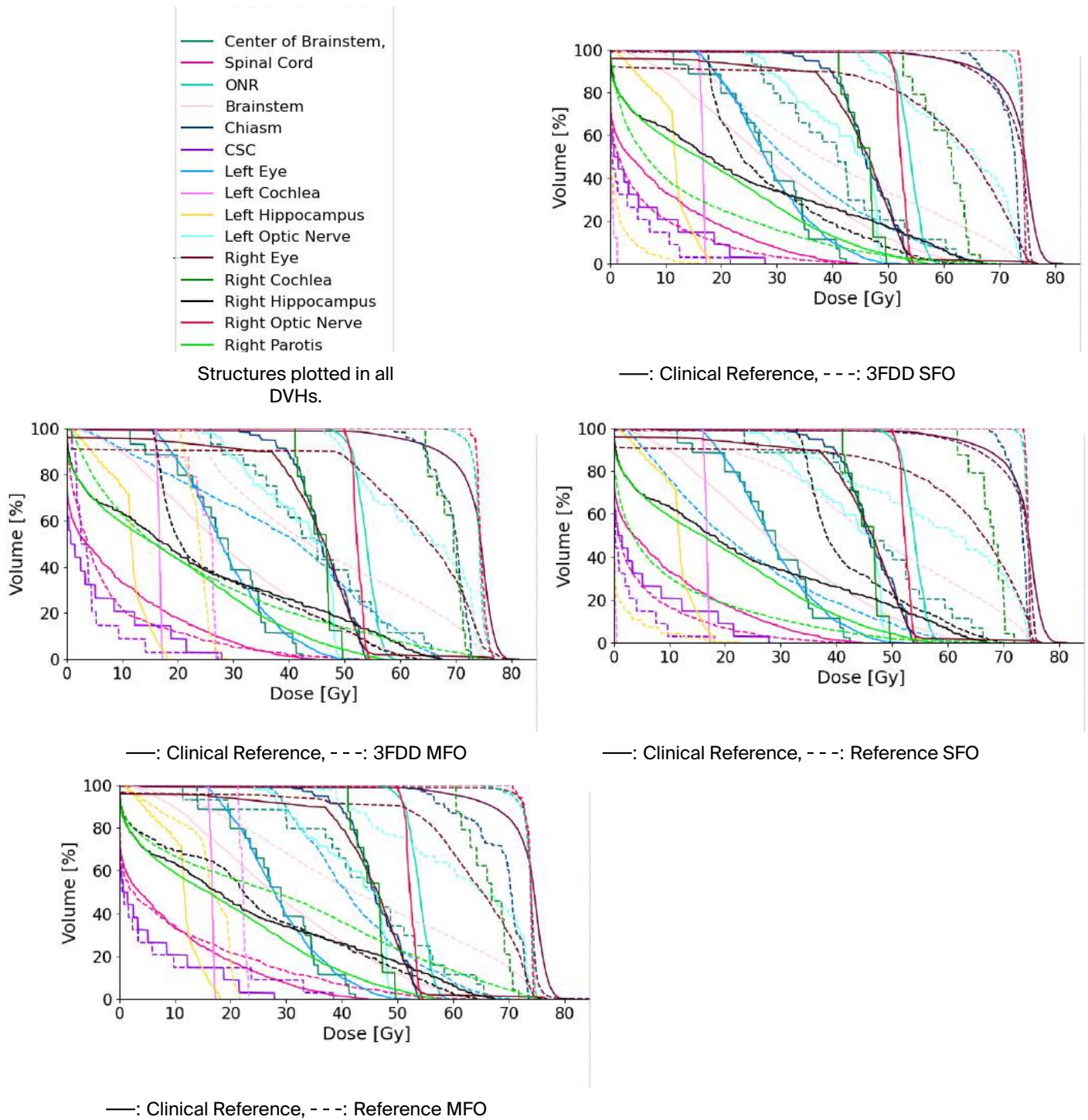


Figure F.18: Dose volume Histograms comparison of each plan strategy to the clinical reference. The patient considered is training patient with ID=15.

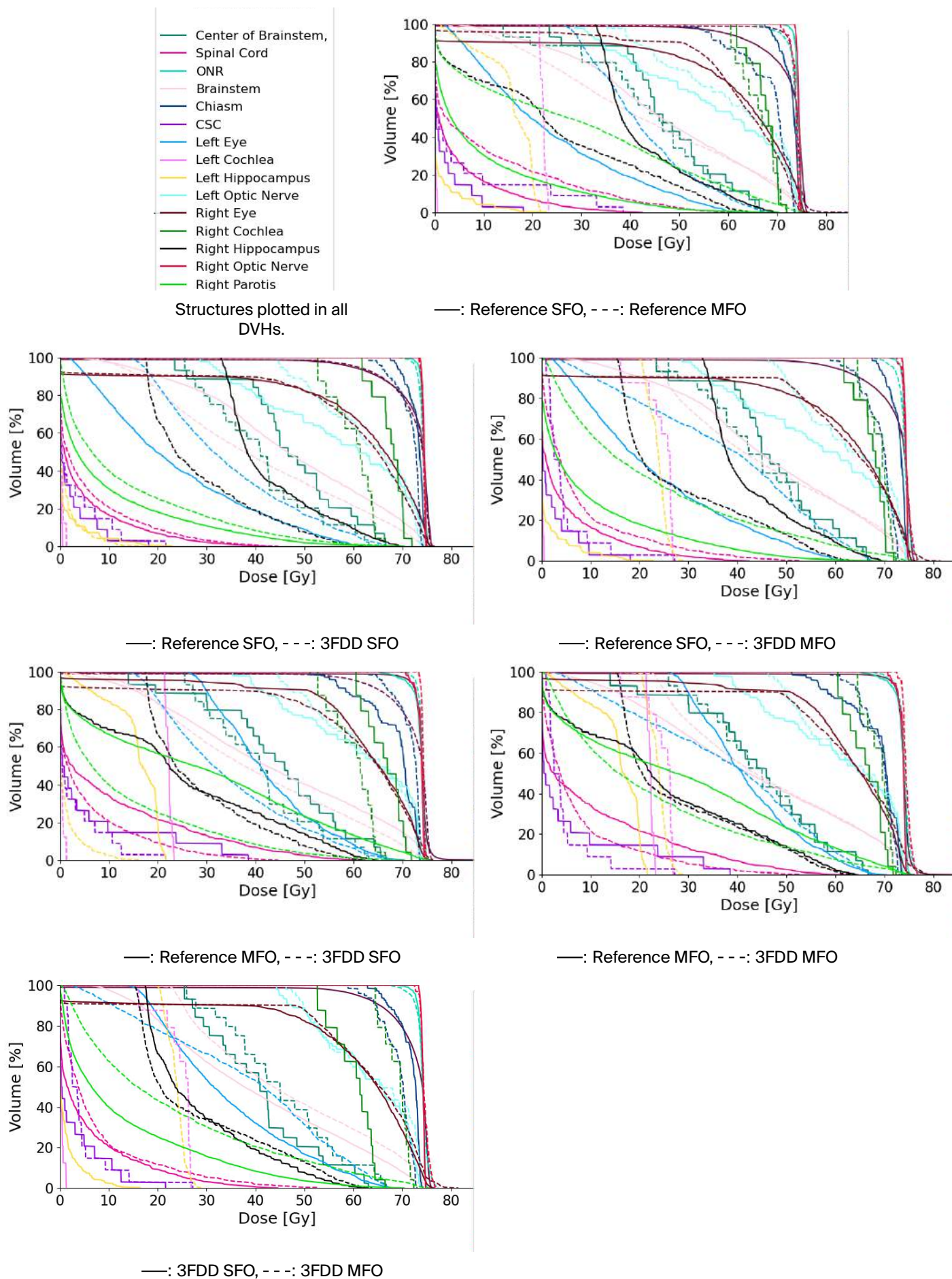


Figure F.19: Dose volume Histograms comparison of Reference SFO and MFO with 3FDD SFO and MFO. The patient considered is training patient with ID=15.

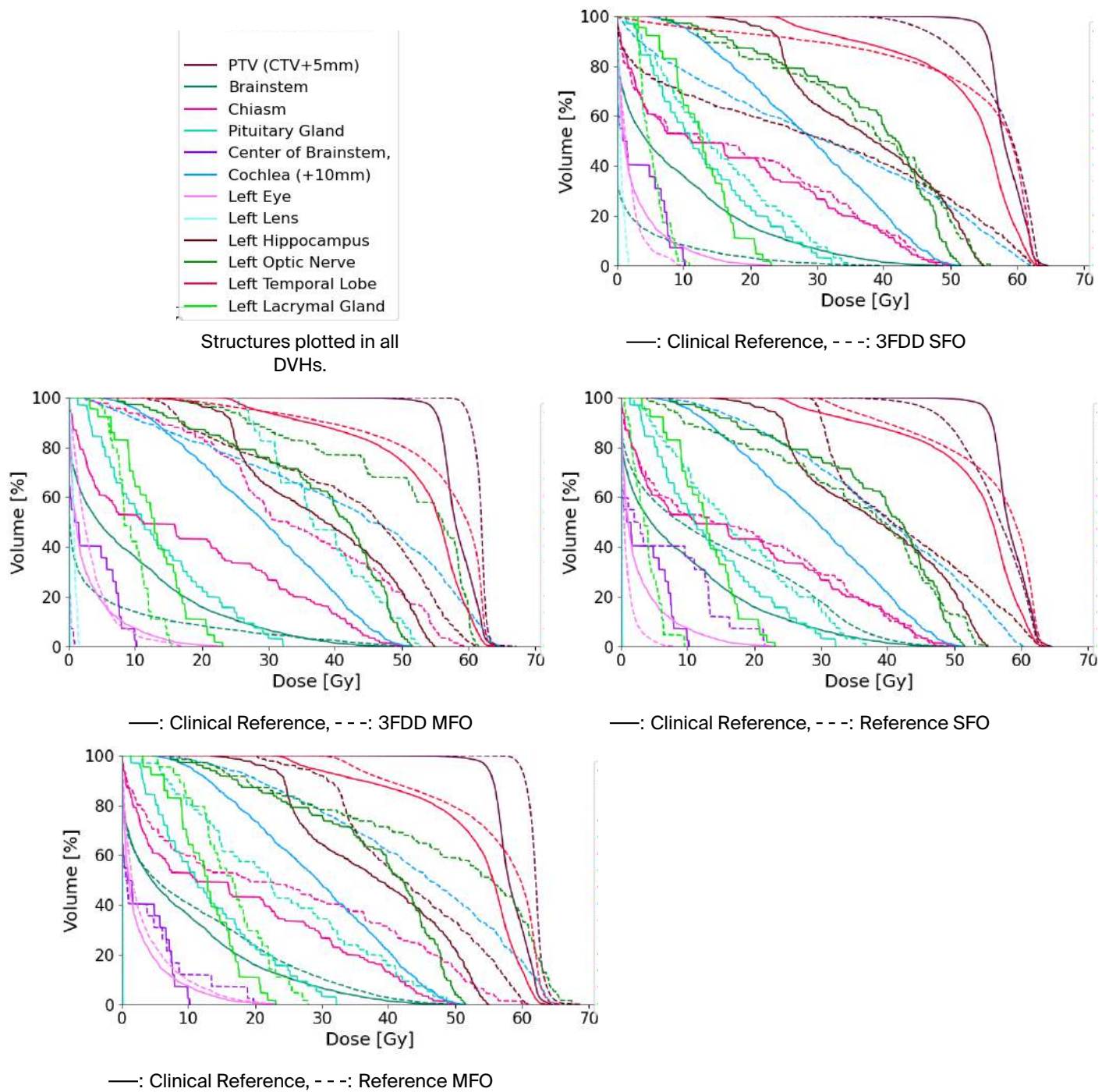


Figure F.20: Dose volume Histograms comparison of each plan strategy to the clinical reference. The patient considered is training patient with ID=33.

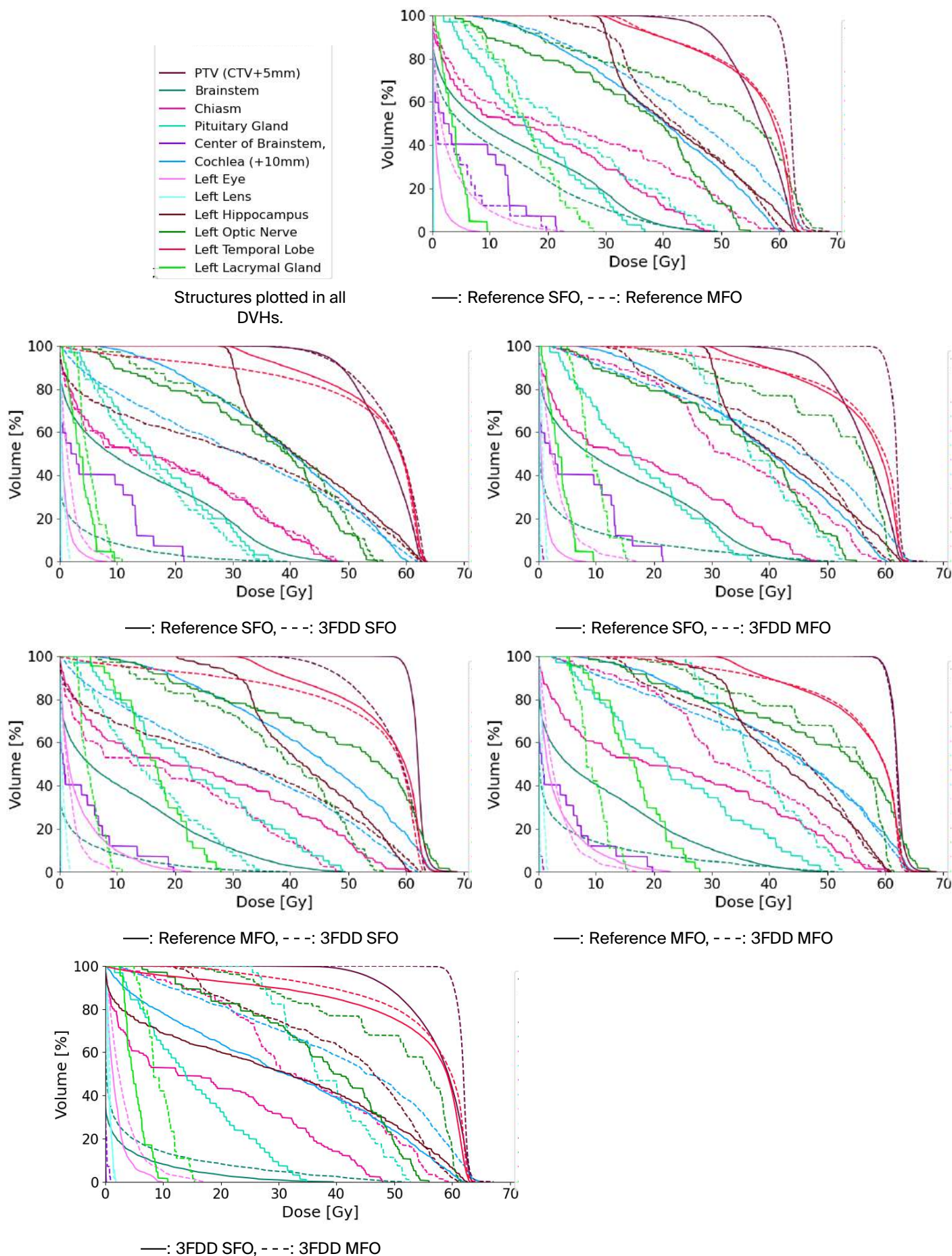


Figure F.21: Dose volume Histograms comparison of Reference SFO and MFO with 3FDD SFO and MFO. The patient considered is training patient with ID=33.